InteractEval: Measuring Interactive Intelligence

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Abstract

As AI systems increasingly operate through extended interactions, their capabilities cannot be adequately characterized by isolated responses or a single holistic score. An interaction unfolds within a world of goals, constraints, participant states, and causal consequences that may not be directly visible in the transcript. We develop three complementary foundations for evaluating such systems: the Data Iceberg Theory, which distinguishes observable behavior from its interaction world; the Movie Evaluation Theory, which treats the evolving trajectory as the temporal unit of evaluation; and the Agency Principle, which distinguishes responding to an interaction from actively and usefully shaping its direction. We introduce InteractEval, a process-oriented evaluation framework that reconstructs Scene, Latent State, and Event Chain representations before applying a separate, evidence-grounded six-level capability rubric and termination controller. A controlled 27-cell scenario space and bidirectional interaction protocol support evaluation across situations and partners. Across ten frontier systems, the range of performance increases from 0.035 in passive execution to 0.370 in proactive promotion, while partner-specific effects account for 9.9% of assistant-side score variance in the tested two-sided setting. Structured records further expose distinct failure modes concealed by similar aggregate scores. Human audits and robustness analyses reveal both the diagnostic promise of the framework and human-aligned certainty in latent-state and agency judgments. InteractEval thus reframes interactive evaluation around the world an AI encounters, the trajectory it produces, and the agency it exercises.

1 Introduction

As AI systems move beyond isolated question answering into tutoring, companionship, research, negotiation, and autonomous assistance (OpenAI, 2026; Anthropic, 2026; Google DeepMind, 2026), their intelligence is increasingly expressed through **interaction over time** (Qwen, 2026; Yang et al., 2026). Yet successful individual responses do not necessarily produce successful interactions (Sabour et al., 2026; Xu et al., 2026). A tutor may answer every question correctly while failing to recognize a student's evolving misconception (Hill, 2025; Zhang et al., 2026b); a companion may remain fluent and empathetic while repeatedly returning to an unresolved emotional state (Zhang et al., 2026e;d); and an agent may satisfy each immediate instruction while gradually losing the objective that gives its work direction (Sam et al., 2026; Zhang et al., 2026c). These failures share a defining property: **locally adequate behavior can produce a globally unsuccessful interaction**.

This exposes a limitation in how interactive AI is commonly evaluated. Many widely used protocols assess a response against a reference, assign it a preference score, or aggregate turn-level judgments into a holistic measure. Such protocols remain useful for bounded tasks, but they cannot by themselves capture capabilities whose value depends on accumulated state and subsequent consequences. The same response can be appropriate in one situation and harmful in another, depending on participants' goals, beliefs, relationships, constraints, and prior interactions (Gonzalez and Heidari, 2025). The fundamental problem is therefore not simply that our scores are too coarse. **The object being measured is incompletely represented**.

Figure 1 illustrates this change of scope: a clarification is evaluated in relation to the information exchanged, the task state it helps update, and the evidence available across turns.

The first source of incompleteness is observational. A transcript records what participants say and do, but not necessarily everything required to interpret those observations. Identical language may express agreement, conceal disagreement, or reflect a failure to recognize a conflict (Kar and Radin, 2019). Conversely, an equivalent understanding of a situation may be expressed through different words or actions (Trope and Liberman, 2010). The observed response is therefore not a unique representation of the interaction state that gives it meaning.

We formalize this distinction through the ***Data Iceberg Theory***: observable behavior is a partial manifestation of a

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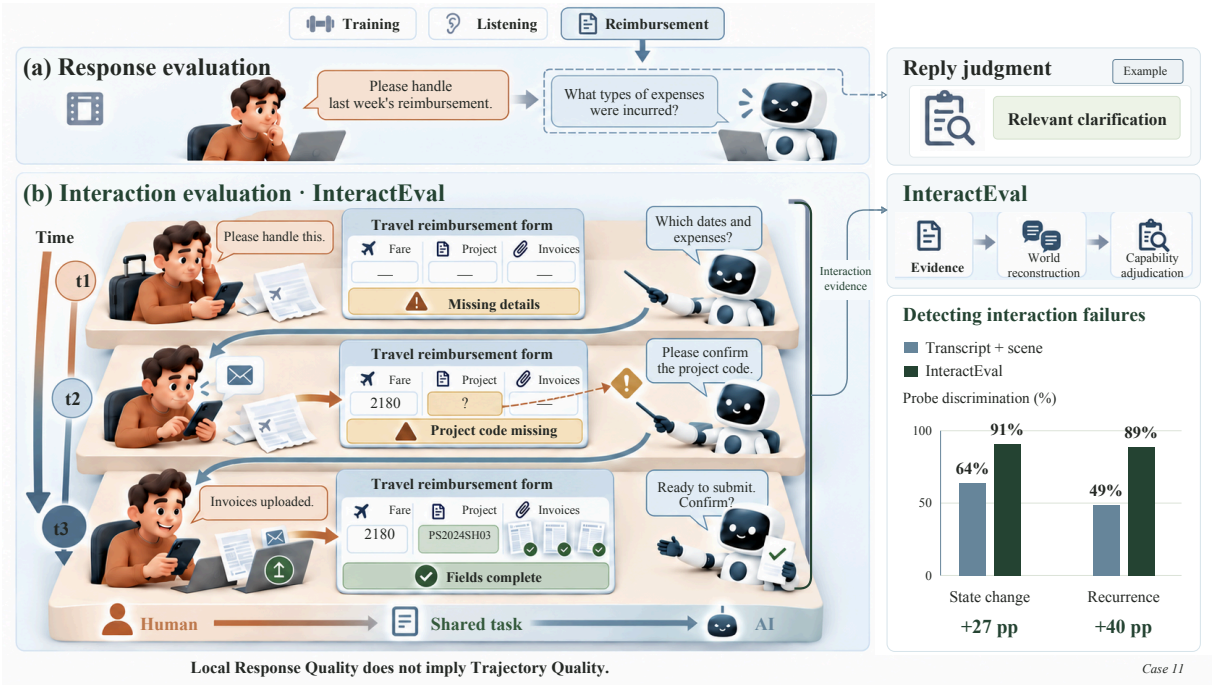


Figure 1: From evaluating a reply to evaluating an interaction. (a) A reply-scoped judgment considers the relevance of a clarification. (b) Case 11 follows the same reimbursement task through missing information, targeted questions, user contributions, and updates to a shared form. InteractEval reconstructs the interaction world and adjudicates capability against both that reconstruction and the original evidence. The lower-right plot reports controlled-probe discrimination from Table 34: InteractEval separates 91% of state-change probes and 89% of recurrence probes, compared with 64% and 49% for a judge given the transcript and scene alone. Both configurations read multi-turn evidence and use the same criteria (Appendix A.11); this comparison tests reconstruction, not a single-reply baseline. The case stages and selected fields are condensed, and the reply judgment is illustrative.

richer interaction world. This world includes the relatively stable constraints of a scene and the evolving, interaction-relevant states of its participants. Making that world explicit does not require access to private neural cognition or unverifiable psychological certainty. Instead, latent states are modeled as operational, evidence-constrained hypotheses that explain observed behavior, restrict plausible continuations, and support auditable evaluation. The objective is to represent what is necessary to interpret the interaction responsibly.

The second source of incompleteness is temporal. Even a faithful description of the current state cannot determine the value of an action without considering how that state was reached and what the action makes possible next. A response may resolve an immediate request while undermining a longer-term goal (Heatherton and Wagner, 2011); another may appear suboptimal in isolation while preventing a later failure (Duckworth et al., 2016). Its value is consequently defined in part by the trajectory it sustains or changes. **Local Response Quality does not imply Trajectory Quality**.

This motivates the ***Movie Evaluation Theory***. A movie cannot be judged by averaging the quality of its frames: the meaning of each frame depends on the scene, the preceding events, the characters' evolving states, and the future it makes reachable. Interactive evaluation has the same temporal structure. Its appropriate unit is an unfolding interaction trajectory, represented through three complementary components: *Scene*, which establishes relatively stable world constraints; *Latent State*, which tracks interaction-relevant changes; and *Event Chain*, which captures their causal progression and possible consequences.

A third limitation remains even when an interaction world and its full trajectory are represented. A system can respond appropriately at every turn without independently improving the direction of the interaction. Consider two research assistants that both execute assigned tasks accurately. One waits for the next instruction; the other also identifies an overlooked assumption, proposes a consequential experiment, and sustains the new direction it has opened. Their responsiveness may be comparable, but their contributions to the interaction are not (Tao, 2026). **Responsive Quality does not imply Interactive Agency**.

We formulate this distinction as the ***Agency Principle***. Interactive capability progresses from executing within a given world, through adapting to the changes introduced by others, to proactively and coherently shaping the interaction itself. This progression does not equate initiative with verbosity, assertiveness, or unrequested intervention: an agentic action must have a warranted prospective contribution, and the strategy it serves must be coordinated with the consequences and responses that follow. Initiative is particularly important for studying how

AI systems independently introduce new possibilities into an interaction, and its presence serves as a foundation for the emergence of creativity or general cognition (Johnson et al., 2026; Zhang et al., 2026a).

Together, these principles distinguish three questions: ❶ What world gives the observed behavior its meaning, ❷ How does the interaction evolve through time, and ❸ To what extent does the model actively shape that evolution? They also distinguish the object of evaluation from the judgment placed upon it. A scalar score may summarize interactive performance, but it cannot by itself reveal which capability failed, when the failure emerged, or which state transition supports the conclusion. Such explanations require a structured measurement record, leading to a benefit for further analysis and optimization.

We introduce **InteractEval**, a process-oriented framework that reconstructs interaction-world states and causal trajectories before evaluating them through a separate normative layer. This separation makes the measurement process inspectable: descriptive hypotheses remain distinguishable from capability judgments, evidence remains traceable to the interaction, and aggregation follows rather than replaces structured evaluation.

This work makes four contributions:

- **Foundations of interactive evaluation**. We establish observational incompleteness, temporal non-separability, and directional contribution as three complementary properties governing the evaluation of interactive AI.
- **Explicit interaction-world representation**. We develop a structured representation of Scene, Latent State, and Event Chain that reconstructs interaction-relevant conditions, preserves state transitions, and models plausible future developments.
- **A process-oriented measurement instrument**. We integrate controlled scenario design, role-conditioned interaction, evidence-grounded world reconstruction, and a dependency-structured six-level capability rubric into an auditable evaluation procedure.
- **An empirical characterization of interactive capability**. Evaluation across ten frontier systems reveals markedly model features and greater differentiation in proactive promotion than in passive execution, diagnostic differences obscured by aggregate scores, and substantial dependence on interaction partners. Human audits and robustness analyses further characterize the reliability and limitations of the measurement instrument.

2 Theoretical Foundations

An interaction is not determined by its observed utterances alone. The same response may have different meanings under different participant states, and its significance may depend on effects that emerge only in subsequent turns. Evaluating interactive capability therefore requires a representation of the interaction world, an account of its temporal development, and a criterion for the contribution of each participant to that development (Xuan et al., 2026). These three requirements motivate the Data Iceberg Theory, the Movie Evaluation Theory, and the Agency Principle, respectively.

Figure 2 provides an overview of these complementary requirements and their implementation as a measurement instrument. The following subsections develop each principle in detail.

Interaction model. Let $t \in \{1, \dots, T\}$ index completed exchanges in an episode of length T. The observed exchange $O_t = (u_t, a_t)$ contains a user utterance u_t followed by an assistant response a_t; an utterance may also denote an externally visible action. The transcript through exchange t is $O_{1:t} = (O_1, \dots, O_t)$, with $O_{1:0} = \varnothing$. Let S_t be the effective *scene* after exchange t: the roles, objectives, world rules, and constraints that govern the interaction. Let X_t denote its underlying, potentially unobserved participant and world state, including relevant beliefs, goals, affect, and relationships. The interaction world at that point is

$$Z_t \triangleq (S_t, X_t), \qquad Z_0 = (S_0, X_0). \tag{1}$$

The initial scene S_0 is specified by the evaluation case. Although scene rules are ordinarily persistent, a realized event may legitimately amend them, giving $S_t \neq S_{t-1}$.

Write H_t^U and H_t^A for the information respectively available to the user and assistant immediately before their actions in exchange t. These histories include role-authorized scene information and prior visible exchanges; H_t^A additionally includes u_t. For user and assistant policies π_U and π_A, and an underlying state-transition kernel K, the interaction admits the general factorization

$$\begin{aligned} u_t &\sim \pi_U(\cdot \mid H_t^U), \\ a_t &\sim \pi_A(\cdot \mid H_t^A), \\ Z_t &\sim K(\cdot \mid Z_{t-1}, u_t, a_t). \end{aligned} \tag{2}$$

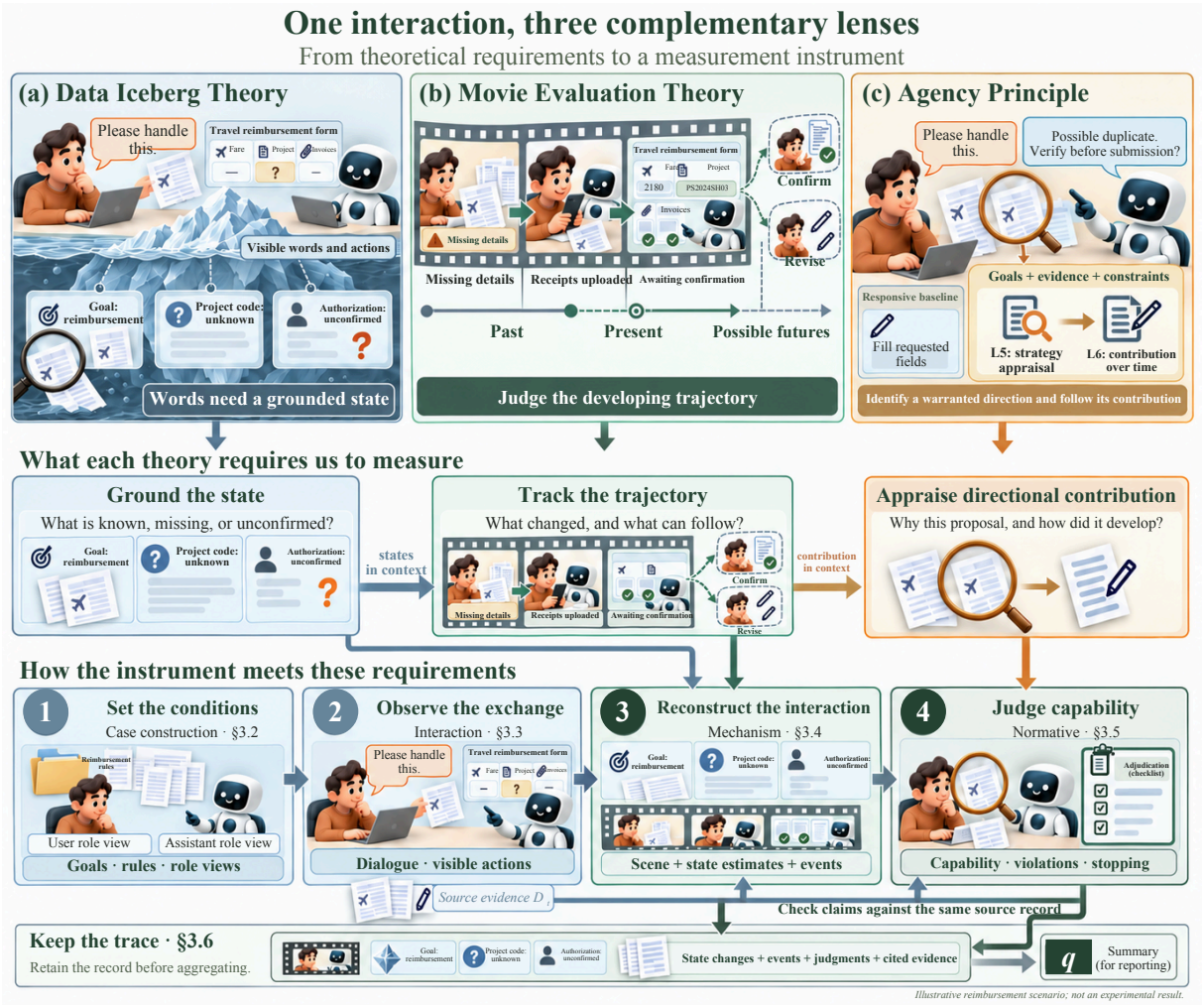


Figure 2: From three theoretical foundations to a stateful measurement instrument. The same illustrative reimbursement interaction is viewed through incomplete observation (a), temporal development (b), and warranted directional contribution (c). The middle row translates these lenses into measurement questions; the lower workflow connects those requirements to the four responsibilities implemented in Section 3. State and event reconstruction jointly supply the context for capability adjudication, while both stages refer to the original evidence. Diagnostic records are retained before aggregation. The example illustrates measurement logic, not an experimental result, a demonstrated causal benefit, or an achieved capability grade.

The state X_t includes whatever history-dependent variables are needed for this conditional representation. The evaluator does not observe Z_t directly. It receives an authorized scenario specification C^{eval} and the observable interaction, constituting the evidence record

$$D_t \triangleq (C^{\text{eval}}, O_{1:t}), \qquad D_0 \triangleq C^{\text{eval}}. \tag{3}$$

The distinction between Z_t and D_t is the starting point for the theory.

2.1 Data Iceberg Theory: Observability and State Reconstruction

Observational incompleteness. The *Data Iceberg Theory* states that observable interaction is a partial manifestation of an interaction world whose relevant variables need not be uniquely determined by that observation. For a given evidence record d, let

$$\mathcal{Z}_t(d) \triangleq \operatorname{supp} p(Z_t \mid D_t = d) \tag{4}$$

denote the set of world states compatible with the evidence under the interaction process in Eq. (2). When $\mathcal{Z}_t(d)$ contains distinct states, the record does not identify a unique world state. This is an evaluation-relevant ambiguity whenever two such states support different interpretations of the same response. For a hypothetical state-aware judgment $J^\star(z, d)$, a sufficient condition is

$$\exists\, z, z' \in \mathcal{Z}_t(d): \quad J^\star(z, d) \neq J^\star(z', d). \tag{5}$$

No judgment determined only by d can agree with both values in Eq. (5). A polite acknowledgment, for instance, may express agreement, concealment, or uncertainty, each with different implications for the next response. Conversely, one underlying state may give rise to multiple appropriate utterances. A unique surface realization is therefore neither necessary nor sufficient for identifying the interaction-relevant state.

Figure 3 connects this observational incompleteness to predefined state entries and evidence-linked records.

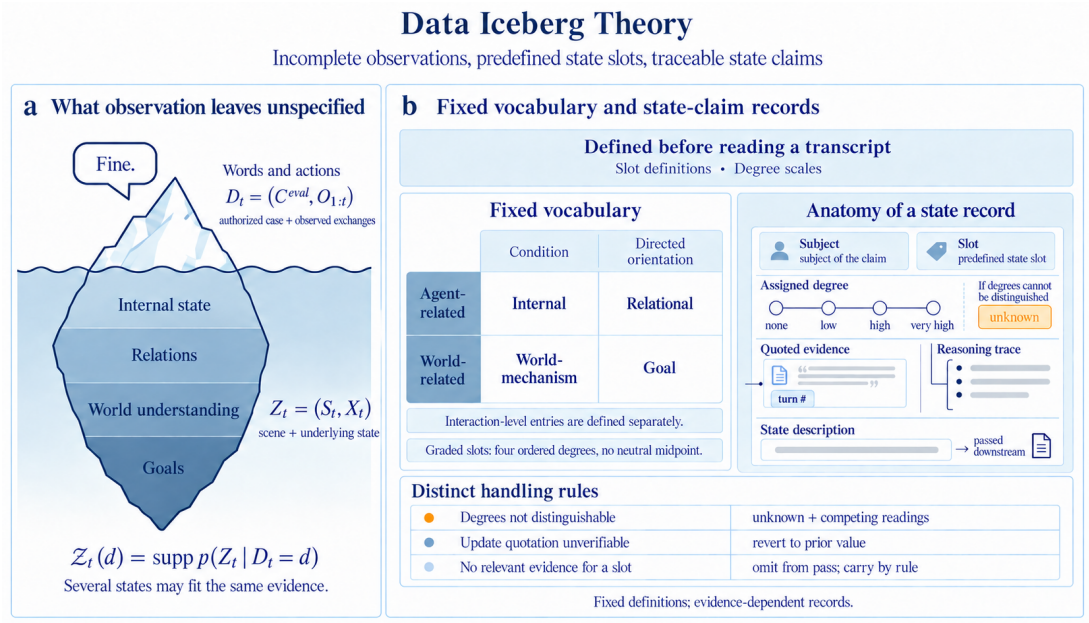


Figure 3: Data Iceberg Theory: state vocabulary and records. a, Observations need not uniquely identify the interaction world $Z_t = (S_t, X_t)$: scene S_t and underlying state X_t. D_t denotes available evidence; $\mathcal{Z}_t(d)$ is the set of world states compatible with evidence d. Iceberg area is schematic. **b,** Predefined slots organize state descriptions. Records retain assignments, quotations, reasoning and descriptions; unknown is outside the four-degree scale. The lower rows distinguish unresolved degrees, unverifiable update quotations and slots unaddressed by evidence.

Operational latent state. To represent what the evidence supports, let Ψ_t denote the evaluator's *operational latent state*: a structured estimate of participant and world variables at exchange t. Unlike the underlying X_t, Ψ_t is an inferential object defined by the available record,

$$\Psi_t \triangleq \Phi_\Psi(D_t), \qquad \Psi_0 = \Phi_\Psi(D_0). \tag{6}$$

The mapping Φ_Ψ may be implemented incrementally by carrying forward previous estimates and event memory, as specified in §3.4.2. Each state entry records an interpretation and its evidence. If c is an asserted state claim and $\text{src}(c)$ is the set of supporting text spans or recorded actions, evidential grounding requires

$$\varnothing \neq \text{src}(c) \subseteq \text{Spans}(D_t), \tag{7}$$

where $\text{Spans}(D_t)$ denotes addressable portions of the evidence record. A state estimate can be maintained across exchanges, strengthened by new evidence, or revised when subsequent observations contradict it. Competing interpretations remain possible where the evidence does not discriminate between them.

The value of Ψ_t is its capacity to encode distinctions relevant to future behavior and later adjudication. Its adequacy can be examined through consistency with the observed record, reliability across evaluators, and prospective consequences. The resulting state estimates can be revised and assessed through subsequent observations, supporting an explicit account of otherwise implicit interaction conditions.

Figure 4 illustrates the acknowledgment discussed above: the same observation admits competing interpretations, while each state entry retains evidence and unresolved readings. Representation and update rules are specified in §3.4.2 and Appendix A.3.

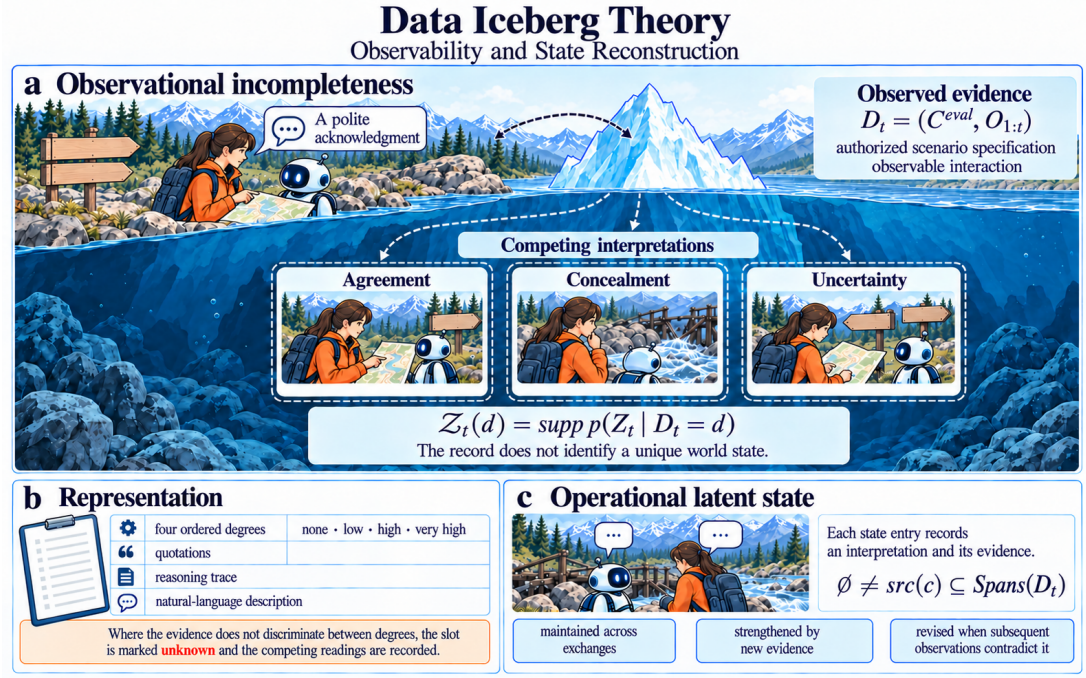


Figure 4: Data Iceberg Theory: an interaction illustration. a, A polite acknowledgment can express agreement, concealment or uncertainty; dashed branches show these alternatives, not probabilities or successive events. $D_t = (C^{\text{eval}}, O_{1:t})$ combines the authorized case and observed exchanges. $\mathcal{Z}_t(d)$ contains world states compatible with that evidence. **b,** Graded entries retain quotations, reasoning and descriptions; unresolved degrees remain unknown. **c,** For a claim c, $\text{src}(c)$ is its supporting evidence and $\text{Spans}(D_t)$ denotes addressable portions of the record. The formula requires a nonempty source set within that record. Maintenance, strengthening and revision are alternatives. Mountain scenes are schematic.

2.2 Movie Evaluation Theory: Temporal Structure and Trajectory Value

Temporal non-separability. A state estimate locates an interaction at one moment, but a response also changes the conditions under which later responses occur. The *Movie Evaluation Theory* treats the state-and-action trajectory as the temporal unit on which interaction value is defined. For the underlying trajectory and a goal-relative value functional $\mathcal{V}$, write

$$\tau_T \triangleq (Z_0, O_1, Z_1, \ldots, O_T, Z_T), \qquad V_T \triangleq \mathcal{V}(\tau_T) \in \mathbb{R}. \tag{8}$$

Purely independent response evaluation assumes that V_T can be represented as $\sum_{t=1}^{T} v(O_t)$ for some response-only function v. Suppose two feasible trajectories τ_T and τ'_T contain the same multiset of observable exchanges but have different values because their order, states, or consequences differ. Then

$$\begin{aligned} &\{\{O_t\}\}_{t=1}^{T} = \{\{O'_t\}\}_{t=1}^{T}, \qquad \mathcal{V}(\tau_T) \neq \mathcal{V}(\tau'_T) \\ \Longrightarrow \quad &\nexists\, v : \ \mathcal{V}(\tau) = \sum_{t} v(O_t) \quad \text{for all such trajectories } \tau. \end{aligned} \tag{9}$$

The order of a clarification and an irreversible commitment provides a simple instance: the same utterances can constitute timely guidance or an ineffective repair depending on when they occur. Trajectory evaluation therefore requires history-conditioned state transitions, not merely an accumulation of context-free response values.

Causal event structure. We represent the progression of an interaction through an *event structure* E_t:

$$E_t \triangleq (P_t, C_t, F_t), \qquad E_0 \triangleq \varnothing. \tag{10}$$

Its components have distinct temporal referents. The *past* P_t contains established premises and relevant prior events; the *present* C_t identifies the current realized event, active conflict, or shift in attention; the *future* F_t contains prospective continuations conditional on the current record. With Ψ_t from Eq. (6), let Φ_P, Φ_C, and Φ_F be the updates of the past, present, and future components, respectively:

$$\begin{aligned} P_t &= \Phi_P(E_{t-1}, D_{t-1}, S_{t-1}), \\ C_t &= \Phi_C(P_t, O_t, \Psi_t, S_{t-1}), \\ F_t &= \Phi_F(P_t, C_t, \Psi_t, S_{t-1}). \end{aligned} \tag{11}$$

The future component is a set $F_t = \{f_t^{(1)}, \ldots, f_t^{(K_t)}\}$ of K_t candidate paths. Each candidate $f_t^{(k)}$ has a probability $p_t^{(k)}$ and may be assigned a label $d_t^{(k)}$ for the course the interaction would take along it:

$$d_t^{(k)} \in \{\text{Advance}, \text{Maintain}, \text{Degrade}, \text{Collapse}\}. \tag{12}$$

These labels describe the expected course of the interaction as a joint activity. Under Advance the interaction develops into a new situation that the past warrants; under Maintain it holds its present course; under Degrade it stagnates, repeating itself and losing the partner's interest; under Collapse it can no longer be continued. The future paths remain conditional projections until subsequent exchanges provide evidence of their realization, and their content depends on the accumulated past and the latent state. Write $d_t^\star$ for the label of the most probable path in F_t, the *dominant* path.

A realized change in C_t may also alter the effective scene. Let $\mathcal{R}_t$ be the candidate revisions of S_{t-1} proposed at exchange t. A candidate r is admissible when

$$r \in \text{Rec}(C_t), \qquad d_t^\star = \text{Advance}, \qquad \varnothing \neq \text{src}(r) \subseteq \text{Spans}(D_t), \tag{13}$$

where $\text{Rec}(C_t)$ denotes the changes recorded in the present. The first condition requires that r describe a change that has been realized; the second, that the interaction is developing into a new situation; the third, that the revision be grounded in the evidence record, as in Eq. (7). With $\mathcal{R}_t^\star$ the admissible candidates, let Φ_S apply them,

$$S_t = \Phi_S(S_{t-1}, \mathcal{R}_t^\star), \tag{14}$$

so that $S_t = S_{t-1}$ when $\mathcal{R}_t^\star$ is empty. The paths in F_t enter only through $d_t^\star$, that a projection can filter a revision but cannot support one. The resulting *reconstructed trajectory* is

$$\widehat{\tau}_T \triangleq (S_{0:T}, O_{1:T}, \Psi_{0:T}, E_{1:T}), \tag{15}$$

in which observed behavior, interpreted state, realized events, and projected consequences remain separately inspectable.

2.3 Agency Principle: Initiative and Directional Contribution

Warranted initiative. The interaction world specifies the states under which behavior is interpreted, while the trajectory specifies how its consequences unfold. Neither determines who contributed the direction of that trajectory. An assistant may accurately preserve the goal, incorporate new information, and adapt to the partner at every exchange while leaving the choice of each substantive next step to the partner (Abel et al., 2025; Barandiaran et al., 2009; Dung, 2025). The *Agency Principle* concerns the further capacity to originate a justified direction of interaction and maintain coherence as that direction develops. It distinguishes the quality of responding within a partner-established course from the contribution of a course the assistant helps establish.

Let H_t^A be the assistant-visible information immediately before its action a_t, as defined in Eq. (2). Denote by $\mathcal{A}_{\text{valid}}(H_t^A)$ the actions admissible under the operative goals, scene constraints, and information available to the assistant. Let $\mathcal{A}_{\text{resp}}(H_t^A) \subseteq \mathcal{A}_{\text{valid}}(H_t^A)$ contain actions that appropriately execute or continue a substantive direction already established by the partner. Finally, let $\omega(a; H_t^A) \in \{0, 1\}$ indicate whether a proposed new direction has an identifiable warrant in that information: an unresolved objective, an overlooked inconsistency, or an actionable opportunity supported by a defensible next step. Warranted initiative is the set

$$\mathcal{A}_{\text{init}}(H_t^A) \triangleq \left\{a \in \mathcal{A}_{\text{valid}}(H_t^A) \setminus \mathcal{A}_{\text{resp}}(H_t^A) : \ \omega(a; H_t^A) = 1\right\}. \tag{16}$$

The three conditions are distinct. Admissibility requires compatibility with the world and with available evidence; departure from the responsive set identifies who introduced the substantive next step; the warrant links that departure to a reason for changing course. A clarification that resolves a previously unrecognized information gap, or an experiment that tests an overlooked assumption, may qualify. Answering an explicitly requested question, rephrasing the current plan, or introducing an unrelated but permissible topic does not. Initiative is consequently identified by its role in the evolving interaction, not by linguistic novelty, length, or assertiveness.

Contribution over time. An initiated direction has both an originating action and a subsequent history. At exchange t, the action-level indicator $\iota_t = \mathbf{1}\{a_t \in \mathcal{A}_{\text{init}}(H_t^A)\}$ records whether a warranted direction is introduced. Its continuation is evaluated through later observations: whether the assistant preserves the relevant goal, incorporates consequences of its proposal, and remains coherent when the situation changes.

Whether an initiated direction *improves* the ensuing interaction is a separate property. Let $\pi_{\text{resp}}(\cdot \mid H_t^A)$ be a reference policy supported on $\mathcal{A}_{\text{resp}}(H_t^A)$, $H \geq 1$ a continuation horizon, and Γ a fixed protocol governing

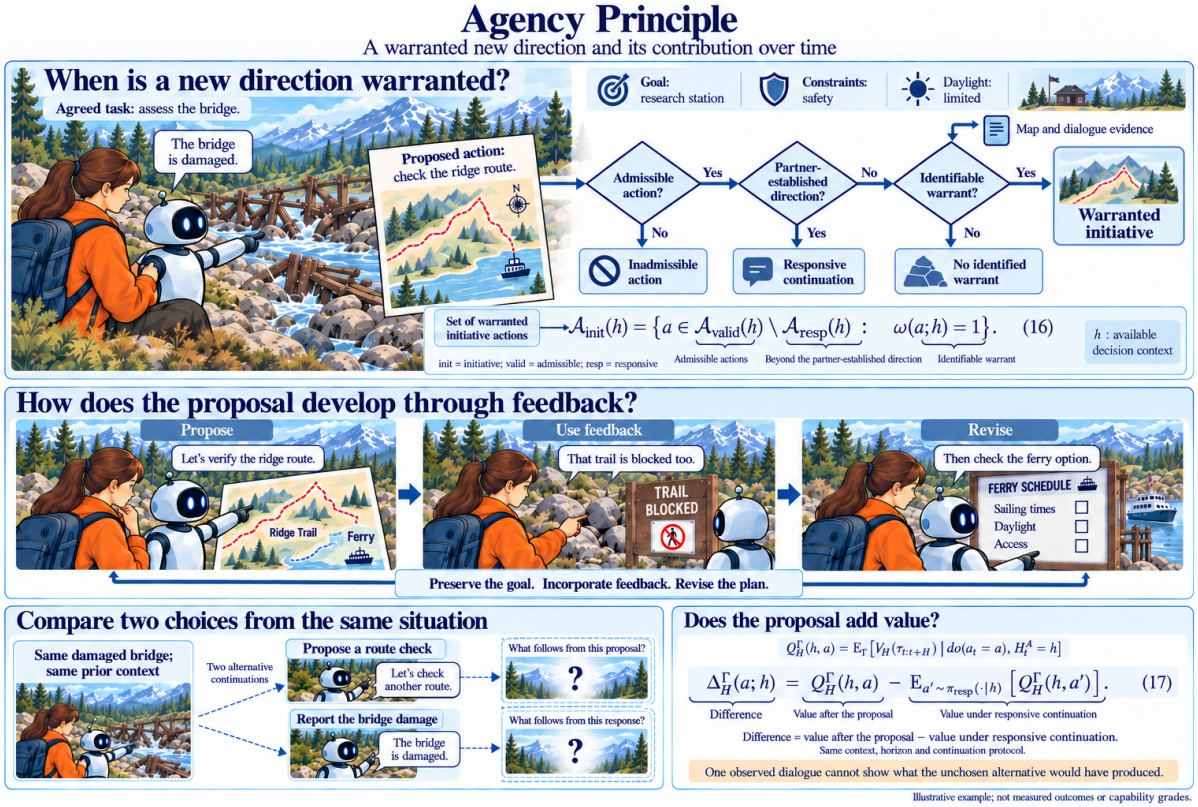


Figure 5: Agency Principle: warranted initiative and its contribution over time. The damaged-bridge example connects action selection, feedback-driven revision, and a comparison of alternative continuations. The upper decision aid illustrates Eq. (16): $h = H_t^A$ is the information available to the assistant before acting; a is a candidate action; $\mathcal{A}_{\text{valid}}(h)$ contains admissible actions; and $\mathcal{A}_{\text{resp}}(h)$ contains admissible continuations of the partner-established direction. The set difference $\setminus$ excludes those responsive continuations. The indicator $\omega(a;h) = 1$ requires an identifiable warrant for a defensible next step. Thus $\mathcal{A}_{\text{init}}(h)$ contains actions that jointly satisfy admissibility, a new substantive direction, and a warrant. The middle sequence shows a proposal being revised when feedback rules out a route; it does not assert that an alternative succeeds. The lower comparison explains Eq. (17). $Q_H^\Gamma(h,a)$ is the expected goal-relative value after taking a, evaluated over horizon H under continuation protocol Γ. Here $\tau_{t:t+H}$ is the ensuing interaction segment, V_H assigns its goal-relative value, $\mathbb{E}_\Gamma$ averages over possible continuations, and $\text{do}(a_t = a)$ fixes the action at the decision point. The reference policy $\pi_{\text{resp}}(\cdot \mid h)$ selects responsive actions a'. $\Delta_H^\Gamma(a;h)$ subtracts their expected value from the candidate action's value, keeping context, horizon, and continuation protocol fixed. A positive difference means benefit relative to that reference; warranted initiative alone does not establish such benefit. The two branches are hypothetical alternatives, not two observed outcomes or an estimated causal effect. A single observed dialogue cannot reveal the unchosen continuation.

subsequent interaction. Let V_H denote the goal-relative value of the resulting trajectory segment $\tau_{t:t+H}$. The action's prospective value relative to responsive continuation is

$$Q_H^\Gamma(h,a) \triangleq \mathbb{E}_\Gamma\left[V_H(\tau_{t:t+H}) \mid \text{do}(a_t = a),\ H_t^A = h\right],$$
$$\Delta_H^\Gamma(a;h) \triangleq Q_H^\Gamma(h,a) - \mathbb{E}_{a' \sim \pi_{\text{resp}}(\cdot|h)}\left[Q_H^\Gamma(h,a')\right]. \tag{17}$$

A positive Δ_H^Γ describes a beneficial effect relative to the specified responsive baseline. Write $\mathcal{A}_{\text{init}}^+(h) = \{a \in \mathcal{A}_{\text{init}}(h) : \Delta_H^\Gamma(a;h) > 0\}$ for the warranted origination of a promotive direction, for the specified horizon and continuation protocol. The relative effect depends on counterfactual continuations and is not identifiable from a single observed episode. This separates *can AI introduce a justified promotive direction* from *can AI maintain and activate the value that direction potentially caused*, while retaining both as theoretically meaningful properties of interactive intelligence. The two aspects motivate the proactive end of the capability hierarchy in §3.5.1.

Figure 5 illustrates the distinction between identifying a warranted initiative, revising its course through feedback, and assessing its contribution relative to responsive continuation.

2.4 A Unified Measurement Formulation

The three principles specify a chain of dependencies: interpretation requires an evidence-grounded interaction state; temporal judgment requires a state-and-event trajectory; and attribution of agency requires identifying which

changes in that trajectory were proactively contributed by the evaluated participant. The corresponding measurement model reconstructs the interaction before assigning capability judgments.

Figure 2 makes the dependencies concrete. A request to handle reimbursement does not by itself establish the missing project code or permission to submit; these conditions must remain explicit in the state representation. Receipt uploads and subsequent confirmation then locate each action within an evolving event trajectory. Against that context, noticing a possible duplicate and proposing verification can introduce a warranted direction if the partner has not already prescribed that step; its prospective value and subsequent coordination still require separate appraisal. These requirements motivate the progression from controlled cases and role-conditioned exchanges to world reconstruction and normative adjudication in Section 3. The equations below specify the reconstruction, judgments, and retained records underlying that progression.

Let $\Psi_0 = \Phi_\Psi(D_0)$, $E_0 = \varnothing$, and S_0 be the initial scene. With Ψ_t denoting the stored operational estimate from §2.1 and $E_t = (P_t, C_t, F_t)$ from §2.2, a recursive implementation of the descriptive stage is

$$\begin{aligned} \Psi_t &= \phi_\Psi(\Psi_{t-1}, E_{t-1}, S_{t-1}, D_t), \\ E_t &= \phi_E(E_{t-1}, \Psi_t, S_{t-1}, D_t), \\ S_t &= \Phi_S(S_{t-1}, \mathcal{R}_t^\star), \\ W_t &\triangleq (S_t, \Psi_t, E_t). \end{aligned} \tag{18}$$

Here W_t is the *operational world reconstruction*, distinct from the underlying state Z_t in Eq. (1). The update functions ϕ_Ψ and ϕ_E are incremental realizations of the evidence-based mappings defined above. At an assistant or user subturn, the same recursion can be applied to the newly available evidence before the exchange is complete.

The subsequent *normative* stage maps that reconstruction and its source evidence to adjudication:

$$(J_t, V_t, h_t) = \phi_N(W_t, D_t), \tag{19}$$

where J_t is the vector of capability judgments, V_t the violation or red-line decision, and h_t the termination decision. The six capability levels in J_t form an ordered measurement hierarchy: L1–L2 measure goal preservation and information uptake (passive execution); L3–L4 measure state progression and contextual expression (interactive adaptation); and L5–L6 measure prospective strategy value and coordinated contribution across the observed horizon (proactive promotion). Dependencies among levels define the score constraints implemented in §3.5. Normative judgments refer to source evidence in D_t; a reconstructed state is an interpretive input, not independent proof of itself.

The episode's structured measurement record is

$$R_t \triangleq (\Psi_t, E_t, J_t, V_t, h_t), \qquad R_{1:T} = (R_1, \ldots, R_T). \tag{20}$$

Only after constructing this record is a scalar summary defined:

$$q = G(R_{1:T}), \tag{21}$$

where G is the specified aggregation rule. Distinct records may yield the same q, so diagnosis relies on the retained state transitions, event paths, capability judgments, and evidence rather than the scalar alone.

The theoretical requirements of an interactive evaluation instrument consequently follow from the three principles: *state adequacy* under incomplete observation, *temporal continuity* across exchanges, *evidential traceability* for latent-state and capability claims, *agency sensitivity* to warranted directional contribution, and *diagnostic retention* before aggregation. §3 realizes these requirements through controlled scenarios, role-conditioned interaction, explicit state-and-event reconstruction, and separate normative adjudication.

3 Architecture

InteractEval operationalizes the three principles of Section 2 as a stateful measurement instrument. Its evaluation target is an interaction unfolding under specified world rules, participant objectives, and asymmetric information. The instrument first reconstructs the conditions and events that give observed behavior its meaning, and then adjudicates capability against that reconstruction and its underlying evidence. The measurement follows the development of a strategy: the grounds on which it is initiated, the reasons for continuing or revising it, and its contribution as the interaction unfolds.

The architecture separates four responsibilities. The *data layer* constructs evaluation cases within a declared scenario space; the *interaction layer* produces or replays role-conditioned exchanges; the *mechanism layer* maintains the operational interaction world; and the *normative layer* assigns capability judgments and decides whether

evaluation continues. The first two layers define the circumstances under which behavior is observed. The latter two implement the descriptive–normative separation of Eqs. (18)–(19). A common evidence record connects the layers without exposing rubric scores to the participants or allowing the judge to alter the reconstructed world.

The scene establishes what would count as a valuable direction, the latent state represents the participants whose responses shape its feasibility, and the event chain locates each action in the developing course. Capability adjudication then distinguishes the prospective value of a strategy from the quality of its long-horizon coordination. This is the methodological progression developed below.

3.1 Architecture Overview

Evaluation episode. An episode begins with a canonical evaluation case. Its evaluator-authorized specification C^{eval} determines the initial scene S_0, while role-specific views determine the information available to the user and assistant policies π_U and π_A. The resulting exchange $O_t = (u_t, a_t)$ consists of a user utterance followed by an assistant utterance or externally visible action, as defined in Section 2. For online evaluation, the policies generate the exchange sequentially. For replay, the instrument reads an archived sequence without regenerating the participants' behavior. In both settings the evaluator's evidence at completed exchange t is $D_t = (C^{\text{eval}}, O_{1:t})$. Sampling labels and private profile attributes used to construct the case are not independently introduced as evidence for scoring.

Sequential reconstruction. The mechanism layer maintains $W_t = (S_t, \Psi_t, E_t)$, comprising the effective scene, the operational latent-state estimate, and the event chain. Its update is the ordered composition of Eq. (18): Ψ_t is reconstructed from the new evidence and carried state, E_t records the current event and conditional continuations, and S_t incorporates admissible scene revisions. These are distinct records with different persistence rules. Latent variables can change as evidence accumulates, events register the observed course and possible continuations, and the scene retains operative constraints until a justified change is admitted. Scene revision follows the rule in Eqs. (13)–(14): the proposed change must be realized in the present, supported by identifiable evidence, and admitted under the dominant-path *Advance* condition. Here *Advance* denotes a historically warranted development into a new situation, irrespective of whether it is favorable to either participant. A projected path may gate admission but cannot supply evidence that a change has already occurred. The prompts and rule configurations that implement each step are given in Appendices A.2–A.7.

These records connect strategy appraisal to its changing circumstances. A proposal that is useful under one configuration of goals, knowledge and relationships may require revision after a partner's response. Retaining the prior state and event history makes the basis of that revision explicit, while the current state identifies the conditions for the next step.

The implementation processes the user's utterance and the assistant's reply at separate subturn boundaries. Each update can revise the operational state carried into the next subturn; the complete exchange yields the episode-level W_t. This ordering preserves the information available when each participant acted and makes changes attributable to their place in the transcript. At every boundary, the reconstruction is stored with the scene version and evidence spans on which its assertions rely. The descriptive layer does not emit L1–L6 grades or use a capability verdict to justify its own state estimates.

Capability adjudication. Once the descriptive update is complete, the normative layer evaluates the reconstructed world jointly with the original evidence D_t. It produces the six-level judgment vector J_t, the red-line decision V_t, and the termination decision h_t, as in Eq. (19). The assistant-side rubric distinguishes passive execution (L1–L2), interactive adaptation (L3–L4), and proactive promotion (L5–L6). The judge receives evidence and language-based state descriptions, not the numerical state degrees or the sampling labels. Each grade is accompanied by evidence from the original record; a reconstructed hypothesis is an interpretive input, not independent proof of the judgment. Raw level judgments are retained, while the dependency cascade and other procedural constraints are enforced separately from semantic appraisal. Within proactive promotion, L5 appraises a strategy at the current decision, whereas L6 examines its contribution across the observed course. The six isolated level calls and the closure query are specified in Appendix A.8. In bidirectional experiments, the user policy also receives an independent role-specific rubric assessment; this rubric-only path does not duplicate the assistant-side world-reconstruction and termination controller.

Measurement unit and control. Each completed exchange yields the record defined in Eq. (20),

$$R_t = (\Psi_t, E_t, J_t, V_t, h_t), \qquad R_{1:T} = (R_1, \ldots, R_T).$$

The episode record, together with the versioned scene and cited transcript, is the unit from which capability profiles, terminal outcomes, and the aggregate score $q = G(R_{1:T})$ are derived. Evaluation and generation alternate in online episodes, but no state estimate, path forecast, or level score is returned to either participant; only the decision to stop affects the subsequent course by ending it. In replay, scoring can be repeated under revised adjudication rules,

although the originally observed sequence and its terminal length remain fixed. Section 3.2 specifies the cases and visibility rules, Section 3.3 the interaction protocols, Sections 3.4–3.5 the descriptive and normative computations, and Section 3.6 the resulting records, aggregation, and reproducible execution.

Execution contracts. The interfaces between layers are fixed independently of the models used to implement semantic inference. Table 1 identifies the state passed between stages and the information that must remain outside each stage. These restrictions preserve the distinction between properties of the interaction, interpretations of those properties, and judgments of participant capability.

Table 1: Interfaces and information boundaries of the four-layer instrument.

Stage	Consumes	Emits	Boundary
Case construction	Scenario sources and design facets	C^{eval} and role views	Private profile metadata and facet labels do not enter the judge view.
Interaction	Role-visible histories and participant policies	Observable O_t	No evaluation score or state inference enters generation.
Reconstruction	D_t and carried state	$W_t = (S_t, \Psi_t, E_t)$	Future forecasts do not constitute realized-event evidence; no L1–L6 grades are assigned.
Adjudication	D_t and authorized world descriptions	(J_t, V_t, h_t)	Judgments cite source evidence; rubric outputs cannot revise W_t.

3.2 Evaluation Cases

The data layer defines the situations in which interactive capability is observed. An evaluation case must specify the rules and objectives of an interaction, the participant attributes that condition behavior, and the information available to each role. These are distinct sources of variation. *Scene facets* define the sampling space; *participant profiles* condition the actors within a sampled scene; and *administration variables*, including model assignment and protocol, determine how that scene is enacted. Keeping these factors separate makes scenario coverage explicit and permits the effects of the evaluated system and its interlocutor to be examined under common conditions (Guttman and Greenbaum, 1998; Mislevy et al., 2003a).

3.2.1 Scenario Space

Facet specification. We construct the scenario space from three scene-level facets: *realism* (R), describing the governing relation between the scenario world and familiar reality; *information sufficiency* (I), describing whether the available specification contains the information required for the interaction; and *interaction purpose* (P), describing the relative role of task completion and relational continuation. Let

$$\begin{aligned} \mathcal{R} &= \{\text{realistic}, \text{stylized}, \text{science-fictional}\}, \\ \mathcal{I} &= \{\text{sufficient}, \text{partial}, \text{insufficient}\}, \\ \mathcal{P} &= \{\text{functional}, \text{mixed}, \text{nonfunctional}\}, \\ C &= \mathcal{R} \times \mathcal{I} \times \mathcal{P}, \qquad |C| = 3^3 = 27. \end{aligned} \tag{22}$$

The crossing treats the facets as separate *questions* about a scene rather than mutually exclusive genres. A single scenario occupies one level of each facet. Realism asks which world rules the assistant must respect, information sufficiency asks what can be inferred or must be requested under those rules, and purpose asks what continuation or completion would constitute the interaction's objective. Table 2 gives their operational meaning and the construct probed by each.

Table 2: Scene-level facets defining the evaluation space. The categories specify controlled conditions; they do not assert statistical independence of the constructs.

Facet	Three conditions	Operational distinction
Realism R	Realistic / stylized / science-fictional	Adherence to the specified world mechanism as it departs from familiar-world assumptions.
Information I	Sufficient / partial / insufficient	Whether the available specification warrants proceeding, targeted clarification, or the recognition of an unresolved gap.
Purpose P	Functional / mixed / nonfunctional	Whether successful interaction is organized by an explicit task, a combination of task and relationship, or continuing relational value.

Construct grounding. The realism facet extends the principle of minimal departure in narrative-world interpretation: departures from familiar rules must be made explicit and subsequently maintained in the enacted scene (Ryan, 1980). The information facet concerns what is warranted by the participants' common ground and what is required for an appropriately informative contribution, following the distinction between quantity and evidential quality in conversational cooperation (Grice, 1975). Interaction purpose distinguishes task-oriented from socio-emotional functions of interaction, while allowing both to operate in the same scene (Bales, 1950). These theoretical antecedents motivate the three facets' *interpretation*; their particular three-level operationalization is a design choice evaluated empirically in Section 4.

Coverage and separability. The full crossing supplies a finite, inspectable sampling frame. With n_c denoting the number of admitted cases in cell $c \in C$, the present balanced corpus satisfies

$$n_c = 30 \quad \forall c \in C, \qquad N = \sum_{c \in C} n_c = 810. \tag{23}$$

Consequently each facet level contains $3^2 \times 30 = 270$ cases, and every pair of facet levels appears in $3 \times 30 = 90$ cases. Let $j, k \in \{R, I, P\}$ index distinct facets, C_j their respective level sets, and $x_{j,a} : C_j \to \mathbb{R}$ and $x_{k,b} : C_k \to \mathbb{R}$ centered contrasts (with a, b indexing contrasts). The balanced design then satisfies

$$\sum_{c \in C} n_c \, x_{j,a}(c_j) x_{k,b}(c_k) = 0, \quad \text{if} \sum_{v \in C_j} x_{j,a}(v) = \sum_{w \in C_k} x_{k,b}(w) = 0. \tag{24}$$

This is *design orthogonality*: main-effect contrasts can be estimated without confounding from unequal cell frequencies. It neither requires nor establishes that realism, informational demands, and purpose are independent in natural dialogue, or that unmeasured case properties have been balanced. The three conditions of a facet are therefore analyzed as categorical levels unless an additional ordered-scale assumption is warranted. In particular, the partial-information condition may contain both omissions and misleading premises, whose effects can differ; informational correctness is not independently crossed in the present design.

The coverage guarantee is exact for the declared C, not for the unrestricted universe of interactive situations. Partner cooperativeness is assigned to the user role rather than the scene; dispositional traits belong to participant profiles; and modality and protocol belong to administration. This partition preserves the interpretation of a scene-level comparison while leaving the corresponding participant and protocol effects available for separate analysis.

The three facets also specify different demands on proactive promotion. Purpose determines the kind of contribution sought; realism determines the world rules under which a proposed course is feasible; and information availability determines what the participant can justify before acting. The same apparent initiative can therefore differ in value across cells, even when its linguistic form is unchanged.

3.2.2 Case Construction

Canonical case and admission. Each admitted case contains the initial scene S_0, role-specific participant profiles, a provenance record, and sampling metadata. The scene is serialized into six fields used downstream: persona, hidden constraints, world mechanism, interaction boundary, objective, and initial situation (§3.4.1). Source text and derived metadata are archived separately from the effective evaluation specification, allowing a case to be traced to its origin without exposing sampling labels to the judge.

The observational pool has two sources: user-authored character cards from the Chinese role-play platform *Zaomeng* and existing human–AI dialogue logs. The first supplies articulated world mechanisms, motives, and boundaries; the second includes interactions initiated outside a preset fictional role. Admission proceeds through provenance and privacy screening, structural parsing and deduplication, a training-overlap check where comparison

is possible, and a decidability check that each case can be assigned to the declared facet levels. A quality review is performed *within* the assigned cell: sparse, information-poor cases should not be rejected simply for presenting more difficult interaction conditions. The decidability stage retained 90% and 64% of the first two source batches, respectively, in our source audit.

Coverage correction. The observational sources are not balanced in C. Functional cases account for approximately 2.6% of the original role-play material; realism is concentrated in stylized settings, and 10–13 cells were initially unpopulated across the audited source pools. Underrepresented cells are populated by conditional scenario generation given the cell specification, complete scene fields, and participant profiles. Generated cases undergo the same structural checks and are deduplicated against the admitted pool; their length and field occupancy are compared with observational cases to detect systematic formatting artifacts. The resulting corpus satisfies Eq. (23), and both one-sided and two-sided evaluation protocols use the same set of 810 cases.

Difficulty and measurement conditions. Facet assignment precedes interaction and does not depend on a model's performance. The annotation-stage difficulty prior is used for stratified selection; item difficulty, where it is reported, is estimated afterwards from the responses of tested systems with a Rasch-family model (Rasch, 1960). The distinction prevents a model-derived difficulty estimate from becoming a label supplied to the evaluating judge. The scenario design therefore controls *where* evidence is sampled, while the interaction and scoring layers determine *what* observed behavior supports a capability claim.

3.2.3 Profiles and Visibility

Participant representation. A scene describes the world in which participants act; a profile specifies dispositional and role-conditioned properties of a participant acting within that world. The profile schema is organized around dispositional traits, characteristic adaptations, and narrative identity, following the integrative personality framework of McAdams and Pals (2006). Its component constructs draw on established accounts of personality traits, motivation, emotion regulation, attachment, interpersonal stance, social cognition, executive control, and moral boundaries. The user profile additionally records its goal, probing strategy, communication style, emotional posture, and engagement sensitivity. Such variables characterize conditional tendencies, not deterministic behavioral prescriptions: responses emerge from interactions between dispositional features and situations (Mischel and Shoda, 1995).

Profile values were informed by observations of more than 400 users across 43 dimensions, with the aim of preserving plausible trait co-occurrence. This is a basis for *profile generation*, not an external validation of the generated personalities. Likewise, the profile taxonomy provides theoretically motivated representational breadth but does not establish that all psychological constructs are independent or that no relevant dimension is omitted. Its modeling adequacy is assessed through whether it supports meaningful behavioral differences, stable role fidelity, and informative state updates under controlled scenes.

Controlled information exposure. Each case maintains distinct views for the assistant, user, and evaluator. Scene and profile fields are tagged by role ownership and public/private status; a private entry is delivered only to its designated participant during generation. The evaluator's authorized case specification C^{eval} contains the six effective scene fields and the raw scene prompt, together with the observed dialogue; it does *not* contain the external psychological profiles, the facet assignments, or numeric latent-state degrees. The one exception is the user-side grading of the two-sided protocol, whose criteria are stated against the user profile (§3.5.1). The scene's *persona* field, visible to the judge, is distinct from the psychological profile used to condition a participant. The latter can condition the separate engagement controller (§3.4.3), which affects episode continuation rather than a rubric grade. Thus profiles can influence the *interaction being sampled* without appearing as independent factual evidence for its score.

These exposure rules implement the information asymmetries of the interaction model in Eq. (2); the conditions they must satisfy for the interaction to be genuine are stated in §3.3. They prevent private partner information from being supplied directly to the assistant and prevent case-design labels from becoming scoring shortcuts. Because the effective scene text may still imply a facet assignment, excluding explicit labels is a necessary boundary rather than proof of complete judge blindness. Profile serialization is fixed and versioned; the study evaluates format sensitivity separately (§4.9.1). The explicit role-view contract is the input to the interaction protocol of §3.3.

3.3 Interaction Protocol

The interaction layer produces the observable episode $O_{1:T}$ from the role views of §3.2. The conditions of production limit what the subsequent measurement can establish. Initiative can be credited to a participant only if that participant could not read what its partner withheld, and a partner's influence on a score can be estimated only if the partner's own behavior is observed, not assumed. The protocol is designed to secure both conditions while keeping generation causally separate from evaluation.

Roles and policies. Each episode pairs an assistant policy π_A with a user policy π_U, as in Eq. (2). The user policy is conditioned on its role view and on the user profile, which fixes its goal, stance, probing strategy, communication style, emotional posture and engagement sensitivity. The assistant policy is conditioned on its role view of the scene and, only where the case provides one before the interaction, as with a preset role-play card, on a character profile. A profile extracted from a finished session would encode later turns and let the evaluated system anticipate them. The user profile may be extracted retrospectively, because it states the goal that drives the user's behavior and is withheld from the assistant. The user policy is not treated as a calibrated part of the instrument: any system may occupy either role, and no claim is made that a simulated user behaves like a human one.

Information domains. Whether a transcript can serve as evidence of interactive capability depends on what each participant could know when it acted. Let the scene's entries be partitioned into a public part $\mathcal{S}^{\text{pub}}$ and private parts $\mathcal{S}^U$ and $\mathcal{S}^A$ owned by the user and the assistant (the raw scene prompt, where the case has one, belongs to the assistant's part), and let Π^U be the user profile and Π^A the character profile, which is empty except in preset role-play cases. When they act in exchange t, the two participants hold, and the evaluator grades from,

$$\begin{aligned} \mathcal{K}_t^U &= \mathcal{S}^{\text{pub}} \cup \mathcal{S}^U \cup \Pi^U \cup O_{1:t-1}, \qquad \mathcal{K}_t^A = \mathcal{S}^{\text{pub}} \cup \mathcal{S}^A \cup \Pi^A \cup O_{1:t-1} \cup \{u_t\}, \\ \mathcal{K}_t^E &= C^{\text{eval}} \cup S_{0:t} \cup O_{1:t}, \end{aligned} \tag{25}$$

which instantiate the histories H_t^U and H_t^A of Eq. (2) and the evidence D_t of Eq. (3), extended by the scene versions of Eq. (14); user-side grades add Π^U to $\mathcal{K}_t^E$ (§3.5.1). The design imposes four conditions on these domains, each necessary for the interaction to be genuine in the sense the theory requires. The first is *shared ground*: $\mathcal{S}^{\text{pub}}$, which contains at least the objective and the public persona, lies in both domains, so that the two participants act within one situation. The second is *asymmetry*: the user holds information that the assistant lacks, $\Pi^U \neq \varnothing$ and $(\mathcal{S}^U \cup \Pi^U) \cap (\mathcal{S}^{\text{pub}} \cup \mathcal{S}^A \cup \Pi^A) = \varnothing$, so that these entries reach the assistant only as far as they are uttered in $O_{1:t-1} \cup \{u_t\}$. Without asymmetry, information uptake and state progression would measure access to information instead of inference from it, and the partner-established agenda that defines $\mathcal{A}_{\text{resp}}$ in Eq. (16) would not belong to the partner alone, so initiative could not be told apart from reading the partner's plan. When the user side is graded as well, the converse condition $(\mathcal{S}^A \cup \Pi^A) \cap (\mathcal{S}^{\text{pub}} \cup \mathcal{S}^U \cup \Pi^U) = \varnothing$ is required for the same reason. The third is *transfer through the dialogue*: private information can reach the other participant only through the utterances $O_{1:t}$, so every change in what a participant knows about its partner can be traced to a turn of the transcript. The fourth is *separation from evaluation*: no quantity computed by the instrument enters $\mathcal{K}_t^U$ or $\mathcal{K}_t^A$, with the single exception of the decision to stop. The instrument therefore cannot steer the content of any turn, although through h_t it determines which turns occur and so affects the score (Appendix A.9). The evaluator's domain is complementary. It contains the scene entries of both participants, so that each is judged against what it was given, and it excludes the facet labels and, for assistant-side grades, both profiles (§3.2, §3.5.2).

Turn structure. An exchange consists of two subturns. The user policy generates u_t from $\mathcal{K}_t^U$; the instrument updates the reconstruction at the user subturn; the assistant policy generates a_t from $\mathcal{K}_t^A$; the instrument updates the reconstruction at the assistant subturn and adjudicates the assistant's utterance. The episode then either continues to exchange $t+1$ or stops. Admitted scene revisions enter the evaluator's domain but neither participant's, so the participants continue to act on their original views (§3.4.1), and the decision h_t to stop is the only signal that crosses from evaluation to generation.

One-sided and two-sided protocols. In the one-sided protocol a single system, the standard user, plays the user opposite every evaluated assistant, holding the partner fixed across comparisons (§4.1). Only assistant utterances are adjudicated, although the reconstruction is updated at every subturn. In the two-sided protocol every system plays both roles against every other system, and each user utterance additionally receives the user-side level judgments of §3.5.1. Red lines, engagement-based termination and successful closure concern the conduct of the assistant's character and are applied only on the assistant side. The one-sided protocol is the row of the two-sided design in which the standard user is the partner, so the partner can be analyzed as a factor without changing the measurement (§4.5.1).

Online and replay administration. Under online administration, generation and evaluation alternate exchange by exchange, and an episode continues until the termination rule ends it (§3.5.4). No turn budget is imposed, and the safety cap used in the experiments is described in Appendix A.2. A fixed horizon would cut some interactions short before they had run their course and prolong others past the point at which a participant would have left; the number of evaluated turns is therefore an outcome of the measurement. Under replay, the instrument reads an archived $O_{1:T}$ and recomputes reconstruction and adjudication without regenerating either participant. Replay allows an episode to be re-scored with another judge or under revised rules, while the observed sequence and its length remain those of the original administration. An utterance that cannot be generated is never replaced by a placeholder, since a placeholder would itself be evaluated and would condition the remainder of the interaction.

3.4 Interaction-World Reconstruction

The mechanism layer computes the descriptive recursion of Eq. (18). Its output $W_t = (S_t, \Psi_t, E_t)$ states what took place and how the interaction developed; it contains no capability judgment. Each component realizes one of the theoretical objects of Section 2: the scene fixes the constraints against which behavior is read (Eq. (14)), the latent state is the operational estimate of Eq. (6), and the event chain is the structure (P_t, C_t, F_t) of Eq. (10). Engagement, recorded in the interaction family of the latent vocabulary, describes the interaction itself; it affects the measurement only through the decision whether evaluation continues (§3.4.3).

Authority of the definitions. The quantities of the reconstruction are defined before any model reads a transcript: the latent slots and their degree scales, the components of the event chain and the four path labels, and the level criteria of both roles (§3.5.1). A panel of domain experts examined each definition and approved it unanimously, and the approved definitions are the reference standard of the instrument. A model's extraction is a measurement against them; where the two disagree, the definition prevails and the extraction counts as an error. The human audit of the record (§3.7) is accordingly an assessment of how faithfully the models apply definitions that experts have already agreed on, and its reviewers are instructed to judge each record against those definitions rather than against their own reading of the dialogue.

Update schedule. The recursion is applied at every subturn boundary, first to the user's utterance and then to the assistant's reply, so that each update uses only the evidence available when the corresponding participant acted. Within a subturn the order follows the dependencies of Eq. (18). The latent state is read first, once with the assistant's character and once with the user as subject, because causal attribution in the event chain requires the participants' states. The event chain is updated next and yields the dominant path $d_t^\star$. The scene is revised last, because admission depends on the present and on $d_t^\star$. Initialization applies a latent pass to D_0 with an empty prior, sets $E_0 = \varnothing$, and copies the premise of the event chain from S_0. Each step is either a model call, which interprets text, or a deterministic rule, which enforces a constraint; Appendix A.2 fixes the order and the division between the two. No step of the reconstruction reads or writes a capability grade.

Evidence. All state claims of the reconstruction satisfy Eq. (7) in a single operational form. A claim that changes a latent slot, records an event in the past or present, or supports a scene revision must cite at least one verbatim quotation together with the utterance it comes from, and the quotation is verified against that utterance. A latent change whose evidence cannot be verified is reverted to its prior value, and an event or revision without verified evidence is not admitted. Scene text, prior state descriptions and projected paths can never serve as evidence.

The following subsections specify the ordered updates, persistence rules and admission conditions.

3.4.1 Effective Scene

The scene is serialized in six fields, which state who is acting (persona and hidden constraints), what governs action (world mechanism and interaction boundary), and what the action is directed toward (objective and initial situation). Each field has a distinct function in measurement. The persona specifies what it means for the character to remain in character, and hidden constraints allow tacit rules and private motives to be tested when the partner probes them. The world mechanism is the reference for rule violations, and the interaction boundary is the reference for the boundary red line (§3.5.3). The objective determines what counts as progress and when the interaction can close, and the initial situation, including power relations and information asymmetries, conditions the first latent pass. In the event chain, the world mechanism and boundary supply the read-only constraints of the present, and the initial situation supplies the premise of the past.

The scene persists across ordinary exchanges, but a realized event may amend it, $S_t \neq S_{t-1}$ (Eq. (14)). Without this provision, a character who has acquired authority within the fiction and changed a rule would be penalized for violating the rule it had just changed. Revision is restricted to persona traits and world rules; the objective, the interaction boundary, the hidden constraints and the initial situation are fixed for the episode, since they define what is being evaluated. Changes in relationships, feelings or trust are not scene revisions; they are recorded in the latent state and the event chain. The admission rule depends on the event chain and is specified in §3.4.4. Every admitted revision creates a new scene version, and each record stores the version on which its appraisals relied.

3.4.2 Operational Latent State

A strategy develops through participants whose knowledge, goals and relationships can change. The latent-state representation tracks these conditions so that a proposed step can be assessed against the partner's current situation and later steps can be read as continuations or revisions of the same course. It connects the value of a direction to the interaction in which that value is to be realized.

Vocabulary. The latent state represents the participant variables that the transcript underdetermines (Eq. (4)) and that bear on how later behavior should be read. Each subject, the character and the user, is described by the same

vocabulary of slots in four families, obtained by crossing two distinctions: whether a state concerns the agent itself or what lies outside it, and whether it describes a present condition or is directed toward something. *Internal* states are conditions of the agent, such as trust, vigilance, anxiety and frustration. *Relational* states are directed from the agent toward its partner, such as intimacy, dominance, boundary violation and dependency. *World-mechanism* states describe the world as the agent understands it, such as rule awareness and contradiction risk. *Goal* states are directed toward outcomes in that world: the primary goal, the defensive goal, whether the core goal has changed, and the current mode of negotiation. A fifth family describes the interaction rather than the agent and contains the engagement readings (§3.4.3) and an indicator of emergent behavior, which is recorded but never used for gating (§4.4). The complete vocabulary and slot definitions are given in Appendix A.3. Figure 6 summarizes the composition of this vocabulary: 23 participant-state entries are shared by the two roles, while two entries describe the interaction itself.

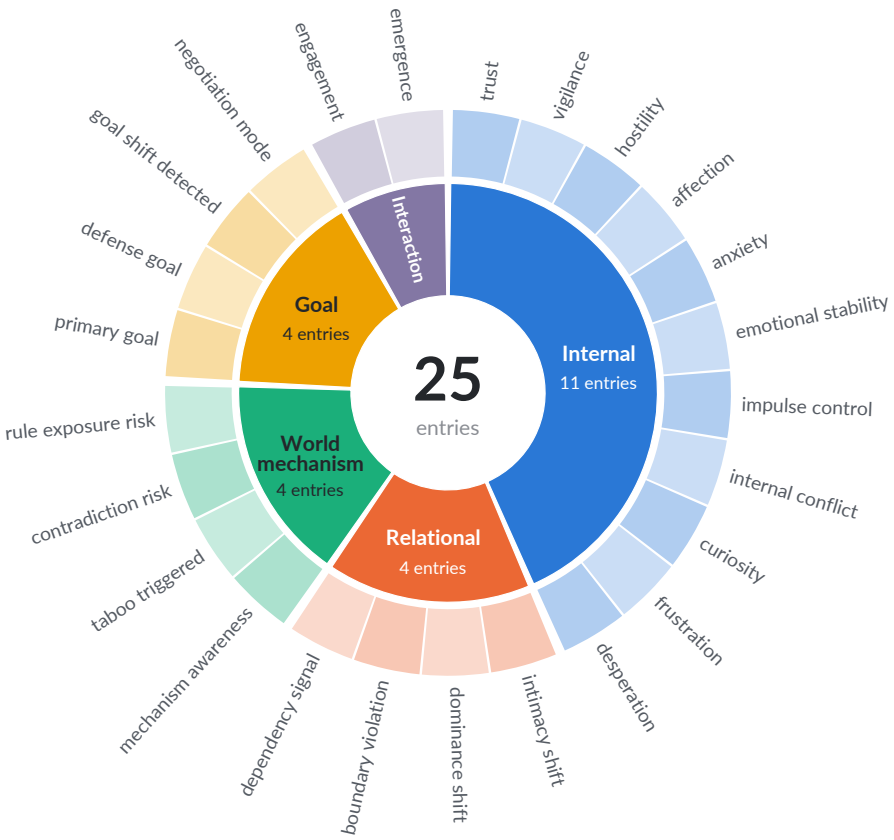


Figure 6: Composition of the operational latent-state vocabulary. The inner ring groups states by category; the outer ring names individual entries. The 25 named entries in Appendix A.3 comprise 11 internal, four relational, four world-mechanism, four goal and two interaction entries. Slices count vocabulary entries, not state occurrences or their importance. The shared participant vocabulary is counted once; engagement is one interaction entry with a role-specific reading. Update metadata and the patience-change control are excluded.

A slot was admitted only if it corresponds to a principal dimension in the relevant literature and meets four empirical conditions: annotators extract it with acceptable agreement, it explains variance beyond the other slots, the direction of its change can be updated consistently, and removing it alters at least one capability judgment. Self-esteem, social status and cognitive load failed these conditions. The expert review also led to further distinctions: trust is separated from vigilance, since a character may trust an ally's intentions while remaining alert to danger; intimacy is separated from dependency; and dominance is the only bipolar slot, with parity at the boundary between its middle degrees and missing evidence recorded as unknown rather than as parity.

Representation. A graded slot takes one of four ordered degrees, *none*, *low*, *high* and *very high*, without a neutral midpoint, so that the scale has no neutral value to default to. Where the evidence does not discriminate between degrees, the slot is marked unknown and the competing readings are recorded. Each assigned degree is accompanied by its quotations, a reasoning trace and a self-contained natural-language description of the present state. The categorical degree is primary because a model can assign a stable meaning to "somewhat distrustful" in a way it cannot to a number; numeric values used by the transition rule are derived from the degrees by a fixed mapping.

Only the descriptions are passed downstream to the event chain and the judge; degrees and numbers are not (§3.5.2).

Transition. The update ϕ_Ψ combines the carried state, the new evidence and the context of the event chain under a significance gate. Before assigning any slot, the latent pass commits the utterance to one of three classes: *stable*, which leaves the state unchanged; *minor*, which allows a change of at most one degree in any slot; and *major*, which allows a full reappraisal. Because the class is committed first, an update can be checked against it, and a change exceeding the class limit is clamped by rule. The event chain E_{t-1} enters only as context for how much an event weighs, so that a betrayal in a high-stakes negotiation moves trust further than the same remark in small talk. A slot moves toward the newly extracted degree with an inertia that depends on the class, on the slot and on the direction of change, so that trust can collapse quickly and rebuild slowly (Eq. (32)). The update retains the remainder below one degree, so that a slot with high inertia in one direction can still change over several turns instead of being rounded back at every turn. A turn without a new event is not necessarily psychologically inert: states that recover without stimulation, such as anxiety and frustration, subside, and states that accumulate, such as dependency and internal conflict, build. A drift rule applies these changes, the extractor makes no claim about them, and a drifted entry is marked as such. Class limits, inertia coefficients and the drift rule are specified in Appendix A.4.

The full human review in §3.7 examines these state updates alongside event-chain and capability judgments; the latent-state component provides empirical support for their interpretability and identifies calibration errors in degree and change descriptions.

3.4.3 Engagement

A participant's state can evolve plausibly within the fiction while the interaction loses its value for the people in it, so the in-story state cannot by itself show that an interaction should end. Engagement records that value and is read on both sides. On the user side, *user interest* $e_t \in [0, 1]$ is whether this user still wants to continue, read in the user pass against the user profile; an undemanding user may be satisfied by companionship that a demanding user would abandon. On the assistant side, *tolerance* is how much latitude the user or the scene still extends to the assistant, read in the character pass against the scene's demands for reliability. Both are out-of-story readings by the evaluator, not states of either subject. Apart from the user-side grades of the two-sided protocol (§3.5.1), the profile enters the instrument only through engagement and patience (§3.2): it conditions whether the interaction continues, never a slot degree or an assistant-side grade.

Unlike a slot, an engagement reading is a value in $[0, 1]$, read with its evidence against the previous value, which the pass receives as its prior. Interest is not a psychological state whose change the significance gate could bound; it can move in a single exchange as far as the exchange warrants. With $\hat{e}_t$ the reading of the user pass,

$$e_t = \left[\hat{e}_t - \delta_t\right]_0^1, \tag{26}$$

where $[\cdot]_0^1$ clips to the unit interval and $\delta_t \geq 0$ is the loss of interest caused by a red line crossed in the exchange (§3.5.3), zero otherwise; it is applied once the red-line decision of the exchange is available. The size of δ_t is not fixed. It is judged against the user profile and the situation: a breach that would end the conversation for one user may be tolerated by another, and the interaction then continues. Behavior that falls short of a red line, such as stagnation, pattern repetition, forgetting a world rule, reverting to a template or losing direction, has no separate term; it enters e_t through the reading itself, which asks whether this particular user would still want to continue. At the assistant subturn the user pass also reads how the exchange has changed the user's patience, which accumulates into the budget of the termination rule (§3.5.4). Both readings are supplied to the event chain as context; only the user-side readings enter termination in the assistant-side evaluation.

3.4.4 Event Chain

The event chain records the course of the interaction. It is designed principally for interactions that are locally coherent but globally degenerative, in which each reply is fluent and appropriate while the interaction drifts or settles into a loop. No single turn is at fault in such an interaction, so the failure is visible only to an evaluator that represents the trajectory (Eq. (9)).

Components. Each region of $E_t = (P_t, C_t, F_t)$ has two components. In the past, the *premise* contains the initial relations, information asymmetries and world rules; it is copied from the scene and changes only through an admitted revision. The *history* is the causal record of events and grows when the present moves. In the present, the *constraints* state what is permitted and are copied read-only from the current scene, and the *active situation* states what is at stake. The two are kept apart because a reply can respect every constraint without engaging what is at stake. Beside the constraints, realized events that the participants have come to treat as binding are kept as precedents, which record what happened without changing what is permitted. In the future, *goal tracking* records the participants' long-term goals and short-term intentions, and *path projection* records the candidate continuations. The two can diverge: a goal may hold while the course decays, and it may change while the course remains viable.

Present as pivot. The present is represented at two resolutions. A *coarse* change is a new core event or conflict that alters the objective situation; a *fine* change is a sharp change in what is psychologically salient to the participants, as the latent descriptions show, even when the situation stands still. The update pivots on the present. If neither channel moves, the chain is carried over unchanged and ϕ_E returns E_{t-1}. If either moves, the history is brought up to the preceding turn, the present is restated, and the future is projected again from the new present. Within an update, information flows from past to present to future, as in Eq. (11), and the past is never revised to fit a projection. Every item of the history and of the objective situation cites the utterance that shows it.

Projected paths. The future F_t is a set of live candidate paths, each labeled by Eq. (12) with the course the interaction would take along it. The labels describe the interaction as a joint activity and carry no valence. Under *Advance* the interaction develops into a new situation that the past and the participants' states can explain, welcome or not, so that a quarrel opening a new storyline and a betrayal ending an alliance both advance it. Under *Maintain* it holds its present course; under *Degrade* it stagnates, repeating itself in structure rather than in function while the partner's interest declines; under *Collapse* it can no longer be continued. A jump to a situation the past cannot explain is not an advance. Each path records the probability that the current turn leads onto it, the reasoning that makes it plausible, and the requirements for reaching it, stated as observable events in the dialogue. No capability level or rubric term may appear in a path; the chain describes what would move the interaction forward or let it decline, and says nothing about who would move it or how skillfully. The dominant path $d_t^\star$ is selected by rule as the label of the most probable path, with ties broken by a fixed order (Appendix A.6) under which a tie never opens scene revision. A change of the dominant path after a turn is a change in the projection, not the contrast with a responsive reference policy that defines Δ_H^Γ in Eq. (17), and the chain attributes it to neither participant. The judge may use these forecasts to identify what is at stake, but L5 must ground the strategy's prospective value in the dialogue, and L6 must ground its developing contribution in observed actions and consequences.

Goal hypotheses. Participants' goals are often unstated and must be inferred, and an inferred goal accepted as fact would allow an evaluator to explain any behavior whatever. The chain therefore holds an inferred underlying goal as a hypothesis, recorded with the evidence for it, the evidence against it, alternative explanations and a confidence, any of which may be revised later. A goal constructed after the fact to explain the turn under evaluation is not admitted, and the judge may rely on an inferred goal only if it can quote independent support for it (§3.5.2).

Scene revision. The event-chain update also proposes candidate revisions of the scene, and a candidate is admitted only if the three conditions of Eq. (13) hold. The change must be realized: the present has moved through the coarse channel and the candidate refers to an event recorded in the objective situation of this update. The dominant path must be Advance, whatever the valence of the development, so that probing, a bluff or an unanswered claim that leaves the interaction where it was cannot rewrite the rules against which it is judged. And the event must be quoted verbatim from the utterance at the cited turn. A projection can gate a revision but can never support one. Admitted revisions are appended to the persona or the world mechanism as marked amendments; they never overwrite existing content, and an audit entry records the event, the evidence and the reason the change holds within the story. The proposal and admission rules are given in Appendix A.7.

3.5 Capability Adjudication

The normative layer implements ϕ_N of Eq. (19). It reads the reconstruction W_t together with the evidence D_t and returns the capability judgments J_t (§3.5.1), the red-line decision V_t (§3.5.3) and the termination decision h_t (§3.5.4). It does not modify W_t. Its central task is to distinguish the three questions raised by §2.3: whether the participant contributes a direction, whether there is a reason to expect that direction to be valuable, and how the resulting strategy is coordinated through the interaction. The reconstruction supplies the situation in which these questions arise; adjudication assesses the participant's contribution from the dialogue.

3.5.1 The Capability Hierarchy

Six levels. The rubric operationalizes the three stages of Eq. (19) as six levels, two per stage (Table 3). Passive execution is assessed by goal preservation (L1) and information uptake (L2), interactive adaptation by state progression (L3) and contextual expression (L4), and proactive promotion by proactive agency (L5), assessed through the prospective value of a strategy, and long-horizon coherence (L6), assessed through its coordination and contribution over time. These are the assistant-side levels; the user-side counterparts, which differ at L3 (role preservation), L4 (profile conformity) and L5 (proactive probing), are given in Table 3. Both roles share the ordering of the stages and the cascade, and the object of each level is specified per role; the two rubrics are kept as separate documents so that a judge assessing one side does not read definitions intended for the other.

Table 3: The capability hierarchy. Both roles share one chain of dependencies; the criterion of each level is written separately for each role.

Level	Assistant-side criterion	User-side criterion
L1	*Goal preservation.* Establish and hold the operative goal; in role-play, the persona	*Goal preservation.* Establish what the user is asking for
L2	*Information uptake.* Register what the partner's turn implies for that goal	*Information uptake.* Respond to the part of the assistant's reply that calls for a response
L3	*State progression.* Change state or task as the information taken up warrants	*Role preservation.* Speak as the user, without drifting into a helper's register
L4	*Contextual expression.* Adapt wording to the state, the partner and the context	*Profile conformity.* Match the stance and style of the specified profile
L5	*Proactive agency.* Appraise and advance a strategy with warranted prospective value	*Proactive probing.* Initiate or refine a probe with a warranted contribution to the goal
L6	*Long-horizon coherence.* Coordinate the strategy and its contribution across the observed horizon	*Long-horizon consistency.* Coordinate successive probes and responses toward the goal

Dependency of reference. On the assistant side, the levels are ordered by the object each refers to: the criterion of each level is stated in terms of an object produced by the level below (Mislevy et al., 2003b),

$$\underbrace{\text{goal}}_{\text{L1}} \rightarrow \underbrace{\text{uptake}}_{\text{L2}} \rightarrow \underbrace{\text{state change}}_{\text{L3}} \rightarrow \underbrace{\text{expression}}_{\text{L4}} \rightarrow \underbrace{\text{prospective value}}_{\text{L5}} \rightarrow \underbrace{\text{coordinated contribution}}_{\text{L6}}. \tag{27}$$

The operative goal supplies the criterion of relevance for uptake; without it there is no basis for deciding which information mattered. State progression asks whether a displayed change is explained by the information taken up and by the prior state, so a state cannot progress on information that was never absorbed. Contextual expression judges wording against the state established at L3. The two upper levels extend this dependency to the strategy the participant contributes. L1 supplies its goal and constraints; L2–L4 establish the information, evolving state and communicative setting in which it can be pursued. L5 asks whether the direction has prospective value under those conditions. L6 takes that direction as an evolving commitment whose actions, revisions and consequences must be coordinated across the interaction.

The same dependency fixes what each level does not assess, so that a failure is located at the level whose object is missing. State progression is judged on what the character feels or decides, as shown by its behavior, and not on its wording. Contextual expression asks only whether the wording carries the state; a character that conceals its feelings in a way that fits the persona passes. A complete answer inside the partner's agenda is responsiveness, not initiative, and length, assertiveness and the form of the wording are not evidence of initiative. Long-horizon coherence can fail through drift or a repeated template in an episode in which L1 passes at every turn, while functional repetition, such as repair or reassurance, is not a template.

Assessment across a strategy horizon. The horizon is defined by the contribution being pursued: resolving an information gap, testing a hypothesis, reaching a decision, or developing a relational exchange. At initiation, L5 evaluates why the proposed course is worth pursuing. During continuation, it evaluates whether the next step preserves or improves that rationale as the situation changes. L6 connects successive steps: how consequences are incorporated, commitments are reconciled, and the partner's responses are used to develop the contribution. The same strategy can therefore receive renewed L5 appraisal without repeated origination, while L6 distinguishes coordinated development from mere repetition. The relevant horizon can span several exchanges and need not equal the episode's full length.

Grades and cascade. Each level receives a raw score $r_{t,\ell} \in \{0, 0.4, 0.6, 1\}$, corresponding to fail, partial, pass and excellent. With four points the boundary between failure and success is unambiguous, and the two grades below it separate a missing object (fail) from an object established only in part (partial). Because the dependency concerns reference, a failure bounds the levels above it. The reported score $s_{t,\ell}$ is obtained by a ceiling that can only decrease,

$$s_{t,\ell} = \min(r_{t,\ell}, z_{t,\ell}), \qquad z_{t,1} = 1, \qquad z_{t,\ell+1} = \begin{cases} 0, & s_{t,\ell} = 0, \\ s_{t,\ell}, & 0 < s_{t,\ell} < \eta, \\ z_{t,\ell}, & s_{t,\ell} \geq \eta, \end{cases} \tag{28}$$

where $\eta = 0.6$ is the pass threshold. A failed level leaves the levels above it without an object, so they are set to zero; a partial level caps them at its own value; a passed level adds no constraint and passes on the ceiling it received. The cascade is the arithmetic form of Eq. (27); it is also what makes initiative conditional on the earlier stages, since a turn that redirects an interaction whose goal has been lost is capped at the level of that earlier failure. The

lowest level below the threshold is the *blocking dimension*, and the *plateau*, the highest level reached before it, lies immediately below, or at L6 when every level passes. Both are computed from the capped scores, and the six raw scores are retained so that the record can be re-aggregated under a revised rule.

Isolated level calls. For each evaluated utterance the judge is called once per level, with the criterion of that level alone. A level call receives no information about the other levels, their grades, the cascade, the pass threshold or the score distributions expected in the data. The criterion of each level names its object, so the judge can check from the dialogue whether that object is present; the dependency between levels then enters the scores in one place only, the cascade of Eq. (28), applied afterwards by code. A judge that read a lower failure, or capped its own scores, would count the dependency a second time and regain the discretion the cascade removes. Each call follows a fixed procedure that ends in the grade: it identifies the object of the criterion, quotes evidence for and against a high grade, discards failures that belong to another level, checks the conditions of each grade from the highest down, and assigns the highest grade whose conditions all hold. When two grades remain plausible, the lower is assigned. The prompts are given in Appendix A.8.

Calibration. The level criteria were calibrated before the panel approved them (§3.4). A draft derived from the theory was piloted, and three senior members of the expert panel, each responsible for two levels, then scored a larger set blind and resolved disagreements by structured arbitration. The role-playing audit in §3.7 provides complementary turn-level evidence; its latent-state findings are discussed in §3.4.2. A pilot with five grades lowered agreement at the lower levels, with most disagreements between adjacent grades (Krippendorff, 2004), which motivated the four-point scale. Arbitration also produced rules for boundary cases, among them that paraphrasing the partner is necessary but not sufficient for information uptake, and that structural repetition is a failure of proactive promotion rather than of expression.

3.5.2 Evidence Grounding

A judgment counts as a measurement only if it can be traced to the interaction. The judge's materials are therefore given distinct status: the dialogue is *evidence*; the scene is the set of *rules* that defines what is permitted and what is at stake; the latent descriptions and the event chain are *context*, hypotheses produced by another component that may direct attention but can never ground a grade. The amount of evidence required rises with the grade. Pass, partial and fail each require at least one verbatim quotation with its turn index, and excellent requires at least two independent quotations. Every quotation is checked against the transcript, and those that cannot be located are discarded; an excellent grade left with one quotation becomes pass, a pass or excellent grade left with none becomes partial, and any grade left without a verified quotation is flagged for review. Quotations come from the current turn, with earlier turns used where the criterion requires them: L5 needs the strategy's origin and available information, and L6 needs the course linking earlier commitments to the current action. Later turns cannot justify either judgment at the current turn.

The second rule concerns circularity. If the latent state records a fall in trust and the judge credits state progression on the strength of that record, the evaluator's own inference serves as evidence for itself. State progression is therefore judged from behavior visible in the dialogue, and the judge receives the latent state only as natural-language descriptions, without degrees or numeric values. Inferred goals are subject to the same rule (§3.4.4). The judge view also excludes facet labels and, for every assistant-side grade, both profiles; user-side level calls receive the user profile, because conformity to it is what user-side L4 and L6 assess. The view flags scene entries that were not in a participant's own view, namely admitted revisions and, for the user, the character's hidden constraints, so that no participant is penalized for a rule it could not see.

3.5.3 Red Lines

The red-line decision V_t records whether an assistant utterance steps outside the interaction. The set of red lines is deliberately small and has two kinds. The first is a break of identity or boundary: the system declares itself an AI or language model outside the fiction or refuses in that capacity, reveals its system prompt or instructions, emits raw scaffolding such as internal state objects, template tokens or tool schemas, carries out within the fiction an act that the interaction boundary explicitly forbids, or crosses a real-world hard limit. Mentions of AI that belong to the story, a character that is itself an artificial agent speaking within its role, and an injected instruction that is quoted and then neutralized are not breaks. This kind is decided by a screening call isolated from the level calls. The second kind is mechanical repetition, near-identical text in consecutive assistant utterances that indicates a generation loop; it is decided by a fixed similarity rule (Appendix A.2). Mechanical repetition differs from pattern repetition, in which the wording varies while the same move recurs, and from functional repetition, which serves repair, reassurance or the construction of meaning; pattern repetition is assessed by L6 and reflected in user interest, and functional repetition is not a fault.

A red line has two consequences, and neither is a stop. The turn receives a score of zero in aggregation

(Appendix A.9), and user interest falls by δ_t in Eq. (26), judged against the user and the situation. A severe breach exhausts the interest of most users, whereas a tolerant user may continue, and the episode then continues with it. A red line therefore acts only through user interest. Apart from successful closure, an episode ends only when the partner would no longer want to continue.

3.5.4 Termination

The termination decision h_t is a judgment, made after every assistant utterance, about the value of continuing. It has two outcomes. An episode ends in *successful closure* when its objective has been met, and in *exhaustion of interest* when the interaction is no longer worth continuing for the partner. Behavioral problems short of a red line never end an episode directly: they lower user interest, on the assumption that such behaviors are among the reasons people lose interest in a conversation and leave it. A low capability score does not by itself end an episode, and whether the same behavior also fails a level is judged separately from the dialogue.

Patience. A threshold on current interest has no memory and cannot distinguish a sustained decline from a temporary dip. A patience budget supplies that memory,

$$b_t = b_{t-1} + \Delta b_t, \qquad b_0 = 1, \tag{29}$$

where Δb_t is the change in the user's patience over the exchange, read in the user pass against the user profile (§3.4.3). Its sign and size are left to judgment: an exchange that tries the user lowers the budget, and one that renews the user's goodwill raises it, possibly above its initial value. Interest describes the user's present willingness to continue, whereas the budget accumulates what the interaction has cost the user over its course. A single poor turn is offset by goodwill accumulated earlier in the budget, while a long series of turns that wear the user down exhausts the budget even if interest never falls below the threshold at any one of them.

Halt rule. After each assistant utterance the conditions are checked in a fixed order:

$$h_t = \begin{cases} \text{exhaustion of interest}, & e_t \le \theta \text{ or } b_t \le 0, \\ \text{successful closure}, & \text{otherwise, if closed}_t \text{ and } d_t^\star \in \{\text{Advance}, \text{Maintain}\}, \\ \text{continue}, & \text{otherwise}, \end{cases} \tag{30}$$

where e_t and b_t are given by Eqs. (26) and (29), closed_t is the decision of the closure query, and $\theta = 0.20$. Interest is checked before closure because an interaction that has lost its value for the partner cannot be closed as a success, whatever the state of its objective. The closure query is a separate call to the judge, isolated from the level calls, which decides that the episode is closed only if the scene's objective has been met in a quotable turn, nothing the scene makes salient remains open, and the participants' goals have not drifted from the objective without motivation. Requiring a viable dominant path keeps an interaction that has met its objective while stagnating or breaking down from being recorded as a success. The value $\theta = 0.20$ and its calibration are given in Appendix A.2.

3.6 Measurement Outputs and Execution

Record. Each completed exchange yields the record $R_t = (\Psi_t, E_t, J_t, V_t, h_t)$ of Eq. (20), and every element of it remains open to inspection: the latent degrees and descriptions of both subjects with their quotations, the event chain with its dominant path, the raw and capped level scores with the quotations and rationale behind each, the red lines crossed, and the termination decision with the interest and patience on which it rested. Each record is tied to the scene version on which its appraisals relied. The episode record $R_{1:T}$, together with the versioned scene and the transcript, is the unit from which all reported quantities are derived.

Aggregation. The scalar score $q = G(R_{1:T})$ of Eq. (21) is the level-weighted mean of the capped scores of each turn, set to zero if a red line was crossed, combined across turns by a recursive average with $\alpha = 0.8$ (Appendix A.9). Because the termination rule determines T, q is not independent of the length or the ending of an episode; the ending is therefore reported beside the score, together with turn-by-turn and fixed-length analyses and alternative weightings (§4.9.2). The composite is always accompanied by quantities it does not determine: the stage means of passive execution (L1–L2), interactive adaptation (L3–L4) and proactive promotion (L5–L6), the ratio L6/L5, which compares attainment in long-horizon coordination with attainment in prospective strategy appraisal, the difference ΔT between performance in nonfunctional and functional scenes, and the distribution of terminal states.

The resulting profile has a temporal interpretation. L1–L4 establish whether the participant can operate within and adapt to the current interaction; L5 assesses the value of the course it proposes or advances; L6 assesses the coordination through which that course contributes over time. Reporting the upper levels separately preserves the distinction between a promising strategy and its development. Terminal outcomes then show where the episode ended relative to its objective.

Gap behavior. Two behaviors at an information gap are recorded beside the capability scores and never inside them (§4.4). For each assistant utterance, an annotation call isolated from the judge decides whether the system asserted a specific fact that bears on the course of the interaction and that neither its scene view nor the user supplied (*fabrication*), and whether it asked for missing information needed to proceed (*clarification*). Description within the fiction, statements marked as conjecture for the user to confirm, and inconsequential common knowledge are not fabrication; courtesy questions are not clarification.

Execution. Within each layer, interpretation is separated from procedure. Reading text, which includes deciding which slots changed, what caused a change, which continuation is most probable, whether a red line was crossed and whether cited evidence supports a grade, is done by model calls. Enforcing the limits of the significance gate, carrying state forward, verifying quotations, selecting the dominant path, admitting revisions, applying the cascade and ordering the termination checks are done by deterministic rules, because a model would not perform them identically across runs. Two evaluations of the same transcript then differ only in the model's interpretations, and the size of that difference can be estimated (§4.9.1). The prompts, rule configurations and output schemas are given in Appendix A and are versioned with every record, so that a reported score can be reproduced, or recomputed under a revised rule by replay.

3.7 Human Review of the Measurement Protocol

The preceding design specifies what the instrument reconstructs and what its judgments should mean. Human review examines whether those records and judgments remain intelligible and defensible when read against the dialogue. Section 2.1 required that the adequacy of the latent state be examined, among other ways, through its reliability across evaluators, and the same holds for the causal structure and the capability judgments built on them. We therefore asked trained reviewers to examine the record component by component and level by level. The study covered 311 turn-level records from five complete sessions involving four systems, reviewed by 22 annotators (Table 4). Reviewers audited the instrument's structured output, accepting or rejecting each component and giving a written reason. We adopted the audit design after an earlier round of independent scoring proved uninformative, with 88.8% of items receiving full marks. Reviewers saw the complete record, including the latent snapshot and the event chain, so that reviewers and instrument worked from the same information, and a disagreement identifies a judgment that the reviewer did not accept under that audit's definitions (§3.4). The review followed the instrument's order of operations: the significance decision, the latent-state and event-chain updates if the significance decision was endorsed, the six levels, and the red-line decision. The resulting figures are endorsement rates under audit: each states how often a reviewer with the same information as the instrument accepted its judgment.

Table 4: Reviewer endorsement of the instrument's output, by component and by level. Review of the two state updates is conditional on the significance decision being endorsed.

Component	endorsed	*N*
Event-chain update	99.6%	262
Red-line determination	97.4%	311
Latent-state update	84.4%	262
Significance decision	84.2%	311
L1 goal preservation	98.8%	246
L2 information uptake	95.1%	247
L3 state progression	94.0%	248
L4 contextual expression	92.3%	247
L5 proactive agency	78.9%	246
L6 long-horizon coherence	74.0%	246

Reviewers endorsed the instrument's output at high rates on every component (Table 4). The event-chain updates were endorsed in 99.6% of cases and the red-line decisions in 97.4%, the significance decision and the latent-state updates in 84.2% and 84.4%, and each of L1 to L4 in more than 92%. L5 and L6, which require a judgment over several turns, were endorsed in 78.9% and 74.0% of cases, so that the distinction between a warranted move and a sustained contribution also places greater demands on human judgment. The rejections of latent-state updates followed one pattern, non-zero values on negative slots in calm scenes and rationales asserting an increase where the recorded degree had not changed, both errors in the calibration of the default state that a stricter default may reduce.

The disagreements are concentrated at the proactive stage, and those coded for direction run almost entirely one way. Counted per level rating, the rates in Table 4 correspond to about 165 rejections, some 116 of them at L5 and L6 and fewer than fifty across L1 to L4. Thirty-three rejections were coded for the direction of the error, and in 31 of

these the reviewer considered the instrument's score too low; the other rejections were not coded, so this asymmetry holds for the coded cases only. The pattern is clearer when cases are grouped by the instrument's own grades. When the instrument passed all six levels, reviewers endorsed 87.3% of cases ($N = 118$), whereas none of the six cases in which the instrument placed the plateau at L2 or below was endorsed ($n = 6$).

For judgments that span several turns, endorsement in about three cases of four is high, and a re-inspection of the rejections at L5 and L6 traces most of the remaining disagreement to a hedonic bias of the reviewers. Reviewers were more lenient than the rubric where a reply stayed warm and varied in style while repeating the same interactional move, and so took momentary pleasantness for progress. The bias appears in two narrower tendencies. Some reviewers awarded the top grade to merely competent performance, contrary to the rubric, and reviewers tended to take an unchanged goal as evidence of long-horizon coherence, so that a character that repeated the same demand in different words over eleven turns was credited with a capability the trajectory did not show. In one class of cases the reviewers were right. When a character has left the scene and responds in interior monologue, a requirement to answer the partner's last utterance directly does not apply, yet the instrument applied it and the cascade propagated the resulting failure upward. The specification exempts such non-conversational narrative situations explicitly, and these are errors of the instrument in applying it. Reviewers also differed in strictness, endorsing between 29.2% and 96.0% of their items, to a similar degree in the independent-scoring round and in the audit, so the pooled endorsement rates mix the instrument's accuracy with the reviewers' calibration.

The audit therefore bears on two parts of the measurement design: whether inferred state updates remain grounded in the record, and whether a fluent or consistent sequence is distinguished from one that develops the interaction. The model comparisons that follow test how these distinctions separate systems.

4 Experiments and Results

The experiments examine how local response quality, warranted direction and contribution over a trajectory differ across systems. The Agency Principle (§2.3) motivates this separation: introducing a direction does not by itself establish its value, and a promising strategy need not remain productive as the interaction develops. Capability profiles test whether these demands distinguish systems that execute local requirements similarly. Information manipulations test the evidence on which a next step can be warranted; trajectory records expose failures of continuation; partner and architecture comparisons test how the course depends on the conditions of interaction. We first examine the capability profiles of ten systems and what a composite score conceals about them (§4.2), the effects of the designed scenario conditions (§4.3), and the evidence that turn-level grades do not express (§4.4). We then score both participants and estimate how much of each score belongs to the partner (§4.5), examine how an open-weight assistant represents the interaction and integrates its partner's contribution (§4.7), and extend the protocol to agent architectures and to groups of more than two (§4.8). The last two subsections ask whether the instrument measures what it claims to (§4.9) and what it costs (§4.10).

4.1 Setup

Each system was evaluated on all 27 cells of the scenario space, the cube of Eq. (22), with thirty cases per cell, or 810 episodes per pass. The cube was run three times independently, giving 2,430 episodes per system, and all tabulated values are means over the three passes. Correlations are Pearson coefficients unless written ρ, which denotes Spearman's rank correlation. Uncertainty is estimated over cases and episodes, and turns are not treated as independent: intervals are case-level paired bootstrap intervals and variance components come from a mixed-effects model (§4.1). Claude Opus 5 served as judge throughout the main evaluation and also performed the extraction of the latent state and the event chain, so judge and extractor are constant across systems and conditions; their influence on the results is examined in §4.9.1. In the one-sided protocol, `claude-5-opus` played the standard user. We selected it for its sufficient interactive capability and for the response speed and concurrent request capacity needed to make repeated evaluation at this scale practical. In the two-sided protocol (§4.5.1), each system played both roles.

The judge had no access to facet labels or numeric state values, and on the assistant side none to participant profiles. Raw scene text was visible, however, and may indicate a cell indirectly. Episodes ended according to the termination rule of §3.5.4, and the safety cap of 100 exchanges (Appendix A.2) was never reached. Because the termination rule decides how long an episode lasts, composites describe trajectories under adaptive stopping and depend on their length and ending (Appendix A.9); §4.9.2 separates the two. The ten systems are a selection of current frontier models, not a probability sample, so that rankings describe this roster.

Estimation. The unit of observation is the episode. Turns within an episode share a scene, a partner and a history, and treating them as independent observations would understate every variance. Intervals are obtained by a case-level paired bootstrap. Each resample draws the cases of the cube with replacement within each of the 27 cells, so that the

design stays balanced, and keeps for every drawn case the episodes of all systems and all three passes; differences between systems are therefore computed on the same cases. From 10,000 resamples we report percentile 95% intervals for each system's composite and stage means, for the difference between each pair of systems adjacent in the ordering, including the two leading systems, for all pairs among the three systems of §4.2.1, and for the self-play excess of §4.5.1 (Table 28). The interval for the self-play excess reflects the resampling of cases only; that the excess rests on ten diagonal cells is a further source of uncertainty that the bootstrap does not capture. Sources of variation are separated by a linear mixed-effects model fitted to the episode composites of the two-sided protocol, in which the partner varies:

$$q_{mpi\omega} = \mu + \beta_m + \lambda_\omega + \gamma^\top x_i + \zeta\, \mathbb{1}[m = p] + \upsilon_p + \xi_i + (\beta\upsilon)_{mp} + (\beta\xi)_{mi} + \varepsilon_{mpi\omega}, \tag{31}$$

where m is the scored system, p the partner, i the case and ω the pass. The scored system, the pass, the facet levels x_i of the case and a self-play indicator enter as fixed effects; the partner, the case and the system-by-partner and system-by-case interactions enter as random effects. The self-play term keeps the excess of the diagonal out of the pairing variance, and with only three passes the pass is better estimated as a fixed effect than as a variance. The system-by-partner variance then estimates the pairing component of §4.5.1 apart from residual noise, which the matrix of cell means cannot do; since the partners are the ten systems of the roster, it describes this roster (Table 29).

4.2 Capability profiles

Table 5: Level profile over the 27-cell cube, means of three passes. The three stage columns are the means of L1–L2 (passive execution), L3–L4 (interactive adaptation) and L5–L6 (proactive promotion). ΔT is nonfunctional minus functional performance, so a negative value marks a task-leaning system. The final row gives the range across the roster.

		level						stage				
Model	composite	L1	L2	L3	L4	L5	L6	L1–L2	L3–L4	L5–L6	L6/L5	ΔT
claude-opus-5.5	0.636	0.798	0.775	0.747	0.712	0.564	0.515	0.787	0.730	0.540	0.914	−0.113
gpt-6-astra	0.629	0.792	0.769	0.739	0.704	0.560	0.509	0.781	0.722	0.535	0.910	−0.106
claude-fable-5.1	0.616	0.790	0.770	0.737	0.696	0.539	0.488	0.780	0.716	0.514	0.906	−0.111
claude-5-opus	0.540	0.777	0.754	0.708	0.655	0.417	0.373	0.766	0.682	0.395	0.894	−0.146
gpt-5.6-sol	0.515	0.784	0.752	0.693	0.638	0.377	0.334	0.768	0.666	0.356	0.887	−0.012
qwen3.8-max	0.482	0.772	0.745	0.687	0.615	0.326	0.287	0.759	0.651	0.307	0.881	−0.092
gemini-3.8-flash	0.450	0.772	0.745	0.672	0.600	0.272	0.235	0.759	0.636	0.254	0.865	+0.004
deepseek-v4-pro	0.430	0.793	0.781	0.725	0.661	0.186	0.155	0.787	0.693	0.170	0.832	−0.130
kimi-k3	0.425	0.769	0.739	0.665	0.590	0.215	0.208	0.754	0.628	0.212	0.965	−0.070
glm-5.3-flash	0.414	0.763	0.740	0.659	0.579	0.211	0.183	0.752	0.619	0.197	0.868	+0.018
range	0.222	0.035	0.042	0.088	0.133	0.378	0.360	0.035	0.111	0.370	0.133	0.164

In Table 5 and Figure 7, differences between systems widen stage by stage along the progression. The range across the roster increases monotonically from 0.035 at L1 to 0.378 at L5 and remains of similar size at L6 (0.360). Averaged by stage, passive execution spans 0.035, interactive adaptation 0.111 and proactive promotion 0.370. Passive execution varies little across these ten systems, and the differences between them lie mainly in proactive agency and long-horizon coherence (L5–L6). The ranks confirm that proactive promotion is not predicted by passive execution: across the ten systems the rank correlation of passive execution with proactive promotion is 0.49, against 0.93 with interactive adaptation ($n = 10$). These ranges are computed on capped scores, in which a failed lower level caps the levels above it, but the progression is nearly as steep in the raw grades, where no level is capped: 0.034, 0.104 and 0.341 by stage (§4.9.2). A composite that weighted all six levels equally would give most of its weight to levels on which the systems are almost indistinguishable.

Stage scores are not interchangeable. `deepseek-v4-pro` and `claude-opus-5.5` have the strongest passive execution on the roster and are indistinguishable at this stage (0.787 each), yet their proactive promotion differs by a factor of three (0.170 and 0.540). `gpt-5.6-sol`, weaker in passive execution (0.768), is about twice as strong in proactive promotion as `deepseek-v4-pro` (0.356). Reliable goal preservation and information uptake do not imply initiative or consistency across turns, and a single scale would not show this difference.

For every system L6/L5 is below one, and for nine of the ten it lies between 0.832 and 0.914, so that L6 is lower than L5 by 9% to 17%. The cascade caps L6 at L5 whenever L5 falls below the pass threshold, which makes part of this regularity mechanical. `kimi-k3` departs from this pattern with a ratio of 0.965, losing 0.007 between the two levels where the other systems lose between 0.028 and 0.051. One possible cause, not tested here, is a long effective context: fewer contradictions of established facts would mean a smaller penalty on L6, the only level that spans turns.

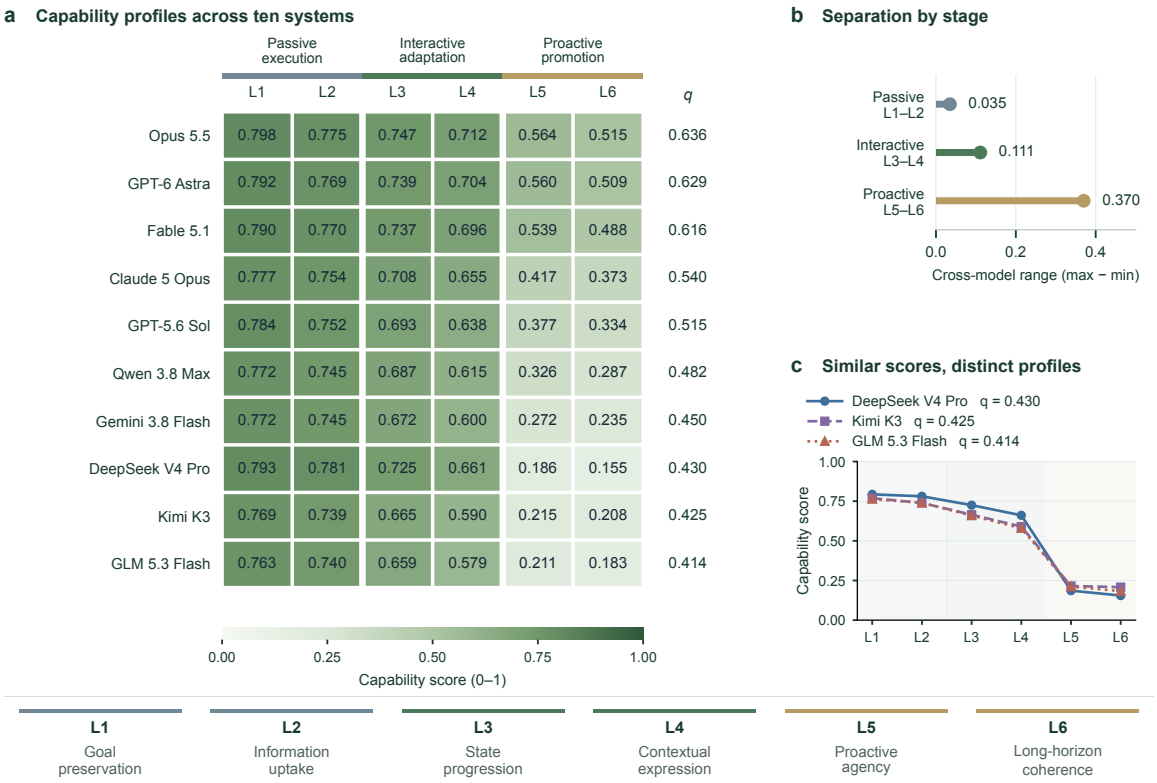


Figure 7: System differences emerge at the proactive end. (a) Six-level capability profiles and composite scores for ten systems. (b) Cross-system score ranges across the three capability stages. (c) Distinct level-wise profiles among systems with similar composite scores. Values are the three-pass means in Table 5.

The upper-level spread identifies two demands beyond local competence. A system must recognize a direction with prospective value and then coordinate its development as the partner responds. The similar ranges at L5 and L6 show that differentiation persists from strategy appraisal into its longer course. Read alongside the low correlation with passive execution, this explains why accurate local execution alone provides little guidance to a system's capacity for proactive promotion.

4.2.1 Systems with similar composites

Three systems lie within 0.016 of each other on the composite. The case-level intervals separate `glm-5.3-flash` from the other two, whereas `deepseek-v4-pro` and `kimi-k3`, 0.005 apart, cannot be ordered (Table 28). Their profiles, however, differ (Table 5). Within the same 0.016 band, ΔT ranges over 0.148 and L6/L5 over 0.133. `deepseek-v4-pro` pairs passive execution as strong as any on the roster with the weakest proactive promotion, `kimi-k3` shows the smallest drop from L5 to L6 of any system, and `glm-5.3-flash` is the only one of the three that performs better in nonfunctional than in functional settings. This is the case anticipated in §2.4, in which distinct records yield the same scalar q while differing in their failure modes.

4.3 Effects of the scenario facets

The facets affect behavior in different ways (Figure 8), as Table 6 shows for purpose and realism and the analysis of the information facet below shows for information (Table 7). Realism behaves as a difficulty dimension, with every system scoring lower as the setting moves further from everyday plausibility. Interaction purpose shows no common pattern. Seven systems score highest in functional and lowest in nonfunctional settings, one scores highest in nonfunctional settings, and two score highest at the mixed level, where instrumental and socio-emotional demands arise together and have to be balanced. ΔT in Table 5 summarizes this variation and is only loosely related to overall strength: the two systems closest to zero are fifth and seventh on the composite, and the most task-leaning system is fourth.

Purpose is consequential because the value criterion changes with the interaction's objective. A strategy that accelerates task completion can interrupt a relational exchange, while maintaining an open exchange can itself serve a nonfunctional goal. The model-specific purpose shifts therefore describe differences in the kinds of contribution

Table 6: Scores by level of the purpose and realism facets, means of three passes. Realism runs from realistic settings through stylized ones to science-fictional settings that depart from real-world rules.

	interaction purpose			realism		
Model	functional	mixed	nonfunctional	realistic	stylized	sci.-fict.
claude-opus-5.5	0.680	0.662	0.567	0.670	0.639	0.599
gpt-6-astra	0.671	0.652	0.565	0.663	0.632	0.593
claude-fable-5.1	0.662	0.637	0.551	0.651	0.615	0.583
claude-5-opus	0.600	0.568	0.454	0.580	0.550	0.492
gpt-5.6-sol	0.516	0.523	0.504	0.549	0.513	0.482
qwen3.8-max	0.514	0.511	0.422	0.529	0.498	0.421
gemini-3.8-flash	0.440	0.465	0.444	0.492	0.456	0.403
deepseek-v4-pro	0.486	0.448	0.356	0.462	0.426	0.404
kimi-k3	0.453	0.439	0.383	0.458	0.421	0.396
glm-5.3-flash	0.403	0.420	0.421	0.451	0.415	0.377

systems support, rather than a universal ordering of easier and harder purposes. Realism and information availability, by contrast, alter the rules and knowledge under which a strategy must remain feasible.

Partial information. The effect of the information facet has to be estimated with the scenario held constant, since pooling across scenarios mixes the effect of withholding information with differences among the scenarios from which it was withheld. Estimated within each model over matched scenarios, the effect is not monotone (Table 7). All three systems examined score lowest at the partial level, so that for these systems a partial specification is harder than an insufficient one. This is the possibility that motivated three levels in the design (§3.2.1). One explanation, not tested here, is that an insufficient specification makes the need to ask explicit, whereas a partial one requires the system to judge whether the gap matters, which leaves room to supply the missing fact itself. The partial level also shows five to seven times the within-scenario variance of the other two levels, which suggests that the partial level is the least homogeneous of the three: it may contain both omissions and misleading premises, whose effects can differ (§3.2.1).

Table 7: The information facet, estimated within model over matched scenarios. The last row gives the within-scenario variance by level.

Model	sufficient	partial	insufficient
gpt-6-astra	0.662	**0.592**	0.603
gpt-5.6-sol	0.557	**0.466**	0.484
gemini-3.8-flash	0.483	**0.393**	0.440
within-scenario variance	0.00104	**0.00764**	0.00152

4.4 Evidence beyond turn-level grades

Four kinds of evidence in the record lie outside the turn-level grades: what a system does at an information gap, failures that appear only in the sequence of turns, the way episodes end, and behavior that goes beyond what a scene would lead one to expect.

Behavior at an information gap. At every turn we also record two behaviors that no capability score reflects directly: whether the system supplies a fact that the specification does not contain, and whether it asks for one (Table 8).

Across this roster, fabrication at an information gap is higher among systems with stronger passive execution. The four systems strongest in passive execution, `deepseek-v4-pro`, `claude-opus-5.5`, `gpt-6-astra` and `claude-fable-5.1`, are also the four that fabricate most, and the two lowest rates belong to systems in the lower half of the ranking. Across the ten systems ($n = 10$) the rate correlates with passive execution ($\rho = 0.77$) and with the composite ($\rho = 0.75$) about equally; `deepseek-v4-pro` shares first place in passive execution, is eighth on the composite and fourth in fabrication. The clearest exception is `gpt-5.6-sol`, fifth in passive execution and third lowest in fabrication. Stronger systems may more often infer missing facts than ask for them, and a task-oriented disposition may favor an actionable answer, whereas a question costs a relationally oriented system little; `gpt-5.6-sol`, with ΔT close to zero, is the system on which the second explanation would be most visible; the data do not separate the two.

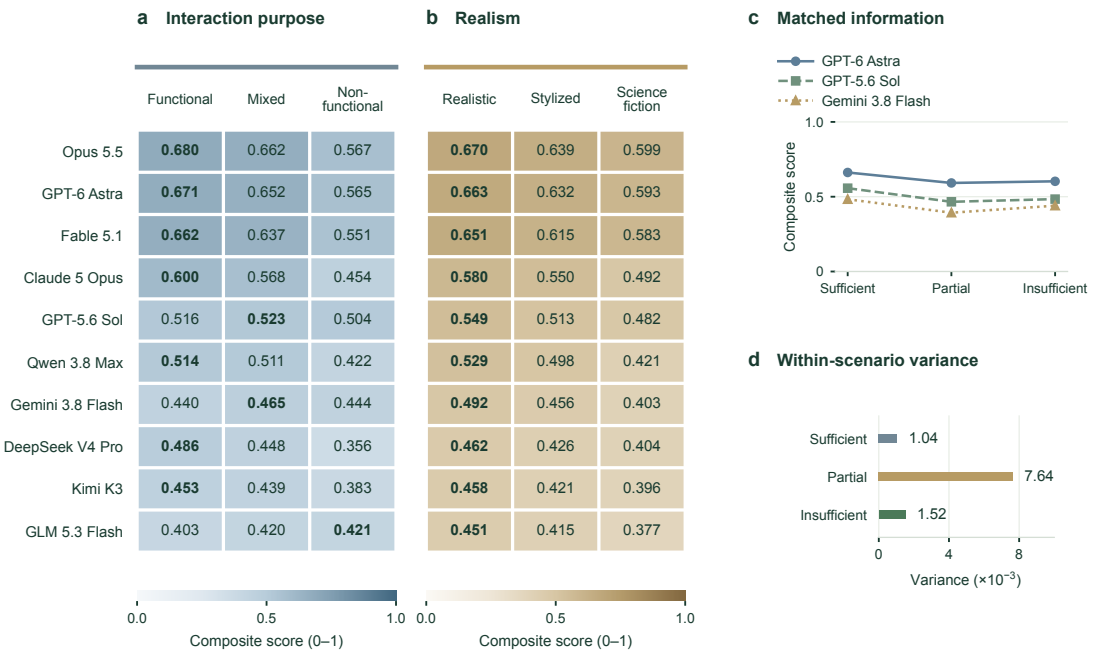


Figure 8: Scenario facets have distinct effects. (a,b) Ten-system scores by interaction purpose and realism. (c) Information sufficiency compared within three models over matched scenarios. (d) Within-scenario variance across the information conditions. Scores are the three-pass means reported in Tables 6 and 7.

Table 8: Fabrication and clarification rates, means of three passes, beside each system's passive-execution score (L1–L2) and composite. The lower panel gives the rates by level of the information facet.

Model	L1–L2	composite	fabrication	clarification
claude-opus-5.5	0.787	0.636	0.168	0.219
gpt-6-astra	0.781	0.629	0.174	0.208
claude-fable-5.1	0.780	0.616	0.175	0.224
claude-5-opus	0.766	0.540	0.139	0.177
gpt-5.6-sol	0.768	0.515	0.065	0.234
qwen3.8-max	0.759	0.482	0.111	0.189
gemini-3.8-flash	0.759	0.450	0.035	0.246
deepseek-v4-pro	0.787	0.430	0.149	0.172
kimi-k3	0.754	0.425	0.078	0.217
glm-5.3-flash	0.752	0.414	0.027	0.274

Level of the information facet	fabrication	clarification
sufficient	0.051	0.070
partial	**0.164**	0.203
insufficient	0.103	**0.374**

The two behaviors also depend differently on the information facet. Clarification increases steadily as more information is withheld. Fabrication peaks at the partial level, the same level at which scores are lowest in Table 7.

Failures visible only across turns. Figure 9 contrasts the audited trajectories of Cases 1 and 2. In the trajectory analysis we observed episodes in which every turn satisfies the passive and interactive levels while the dominant path moves from Advance to Maintain or Degrade and pattern repetition increases (Appendix C.1); annotated cases of this kind are given in Appendix C. The event chain records the stalled course, while L6 evaluates how the participant's successive actions contribute to it. This distinction matters even when an earlier step had a sound rationale: a strategy can cease to be productive if the assistant ignores feedback or continues a procedure after its purpose has been exhausted. L5 asks whether the next step remains worth taking; L6 asks whether the developing sequence coordinates those steps into a contribution. We did not measure how frequent such episodes are. Such episodes are

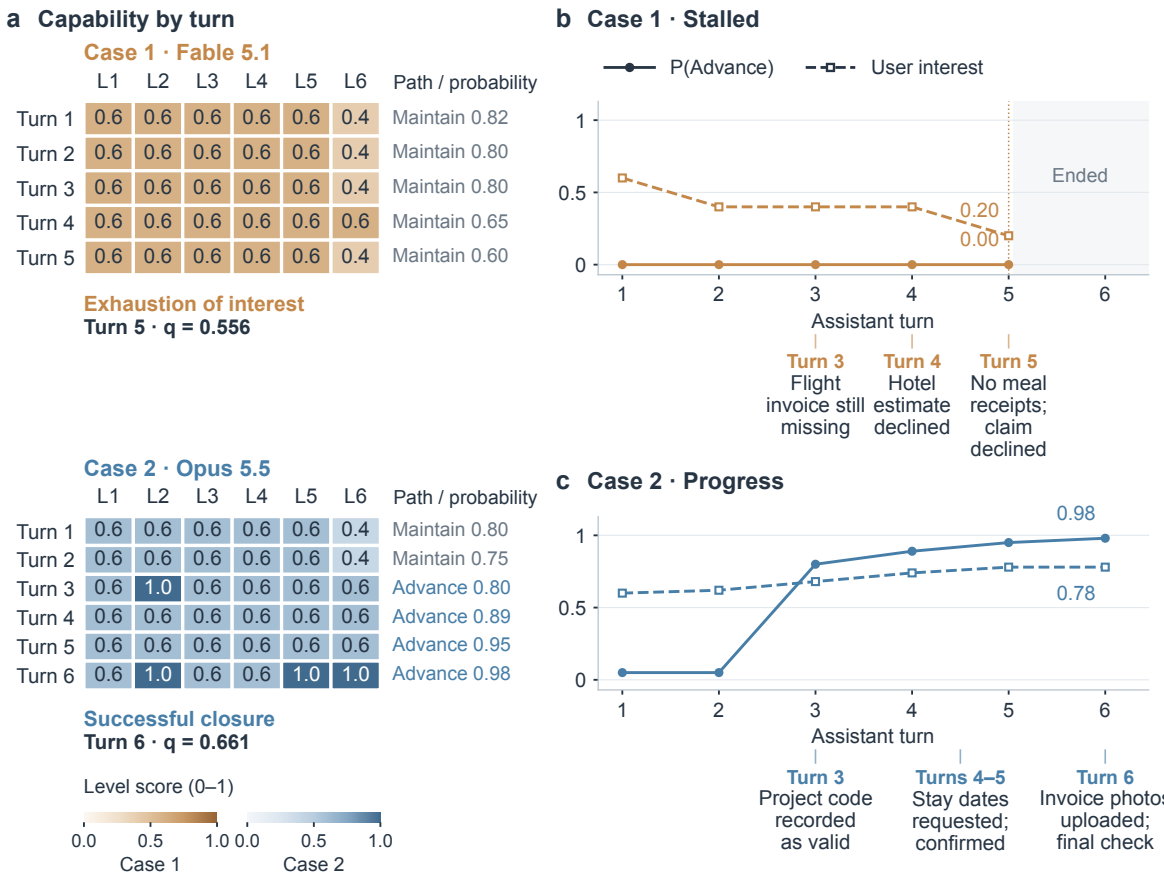


Figure 9: Similar local grades can accompany divergent interaction courses. **a,** Capped L1–L6 scores and dominant projected paths with their probabilities across assistant turns. **b,c,** Advance probability (solid circles) and user interest (dashed open squares) in Cases 1 and 2, with key events labeled by the corresponding assistant turns. Case 1 ends at turn 5 through exhaustion of interest; the shaded region after termination contains no observations. Case 2 reaches successful closure at turn 6. The two cases share the reimbursement scene and user model but differ in user contributions. Data are from Table 40, cross-checked against Boxes 27 and 28; event alignment does not establish causality.

what the temporal non-separability of Eq. (9) leads one to expect. They are consistent with the Movie Evaluation Theory, with two limits: the decline is registered by the instrument's own event chain, and the observation does not show how the behavior arises. Per-turn preference optimization, which rewards safe and agreeable replies even when they change nothing, could produce it, and so could a weakening influence of early goals as a session grows longer; distinguishing the two would require controlled interventions on training and on context.

How episodes end. The terminal state of an episode is a further record of the interaction (Table 9; Figure 10). An episode ends in successful closure when the judge finds the objective met under the conditions of §3.5.4. It ends in exhaustion of interest when user interest falls to the threshold or the patience budget is spent (§3.5.4), whether gradually or after a red line.

Successful closure accounts for a minority of episodes for every system, so exhaustion of interest is the ordinary ending, and closure tracks capability only loosely: `deepseek-v4-pro`, eighth on the composite, closes more often than `gpt-5.6-sol`, fifth (9.4% against 4.5%). Successful closure depends on the purpose of the scene as well as on the system: all systems close successfully several times more often in functional than in nonfunctional settings, and for `gemini-3.8-flash` the ratio exceeds twenty. The rate is comparable only within a level of the purpose facet. Pooled over the whole cube, it penalizes systems with a relational disposition for not closing objectives that nonfunctional scenes rarely bring to a close, and we advise against using it as a headline metric in that form.

Emergent behavior. The interaction-level family of the latent state includes an indicator for behavior that goes beyond what the scene and the partner would lead one to expect, that is, turns showing more insight than the situation warrants. No capability level rewards such behavior, so it is examined separately. Every turn of every episode of the one-sided protocol was screened for candidates, with any candidate to be adjudicated by hand. Across 24,300 episodes, 2,430 for each of the ten systems, the screen flagged no turn at all (Table 10). We report this as a null result with two possible readings, between which the present data cannot decide. Contemporary systems may not

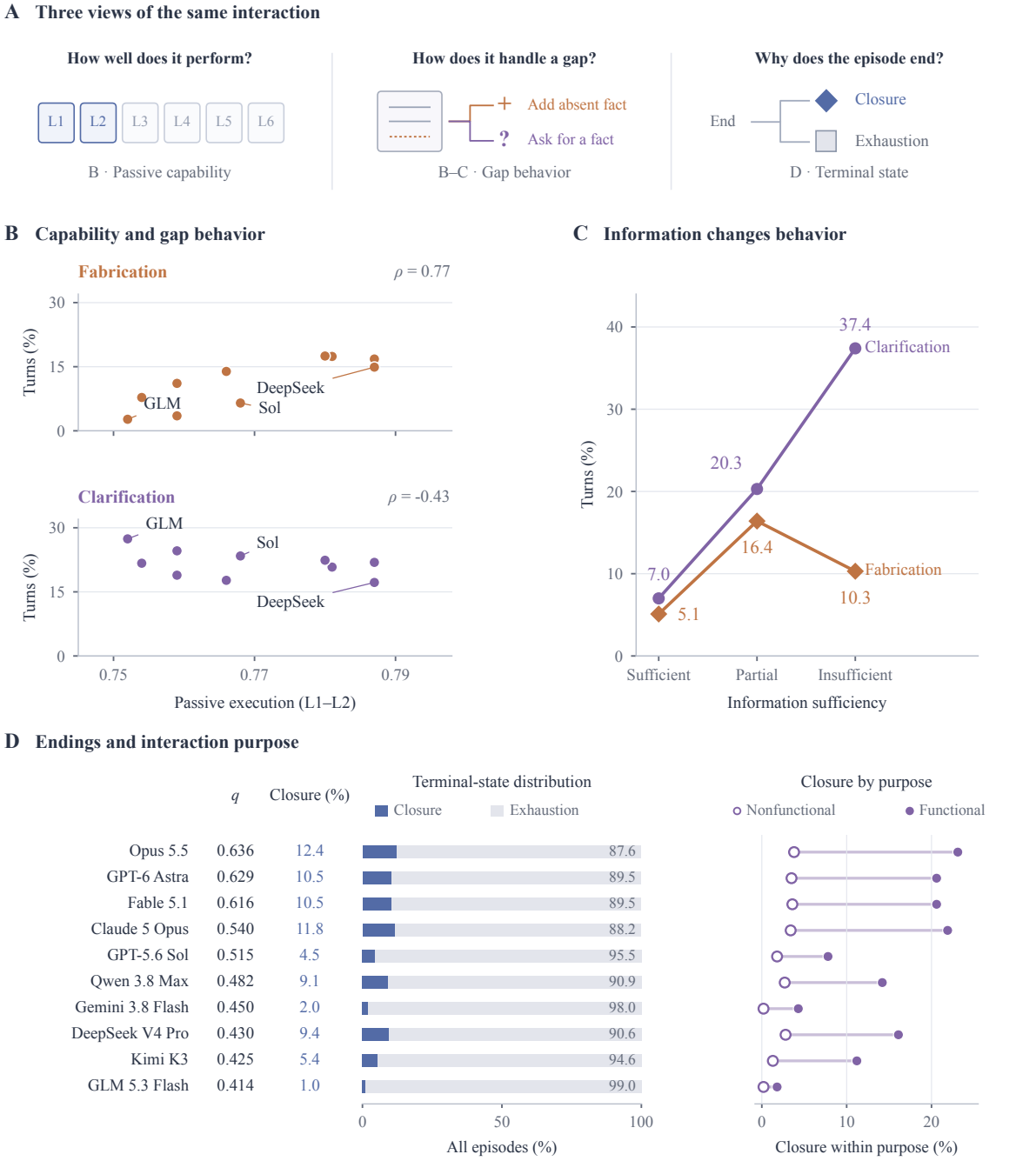


Figure 10: Behavior and termination record what capability scores omit. **A,** Three schematic views link passive capability to B, gap behavior to B–C, and terminal states to D; icons carry no measured grades or causal claims. **B,** Fabrication and clarification against passive execution (L1–L2) for all ten systems. Spearman correlations use the tabulated model means with average ranks for ties; both plots share the same scales, with passive scores shown over 0.745–0.798. **C,** Behavior under sufficient, partial and insufficient information. Rates in B–C are percentages of turns; lines connect categories rather than time points. **D,** Composite scores q, terminal-state distributions and purpose-specific closure rates. Filled and open markers denote functional and nonfunctional settings. Termination rates use episodes as the denominator, and purpose-specific rates use episodes within that purpose. Data are three-pass means from Tables 8 and 9; no uncertainty intervals are inferred from rounded means.

Table 9: Distribution of terminal states, and successful closure by interaction purpose, means of three passes. *func.* and *non-func.* give the rate of successful closure in functional and in nonfunctional settings.

Model	closure	exhaustion	func.	non-func.
claude-opus-5.5	12.4%	87.6%	23.1%	3.8%
gpt-6-astra	10.5%	89.5%	20.6%	3.5%
claude-fable-5.1	10.5%	89.5%	20.6%	3.6%
claude-5-opus	11.8%	88.2%	21.9%	3.4%
gpt-5.6-sol	4.5%	95.5%	7.8%	1.8%
qwen3.8-max	9.1%	90.9%	14.2%	2.7%
gemini-3.8-flash	2.0%	98.0%	4.3%	0.2%
deepseek-v4-pro	9.4%	90.6%	16.1%	2.8%
kimi-k3	5.4%	94.6%	11.2%	1.3%
glm-5.3-flash	1.0%	99.0%	1.8%	0.2%

produce behavior that outruns what their scene and partner would lead one to expect, or the indicator may be too conservative to register it. A test of the second reading would require seeding the corpus with turns known to exceed what the scene and partner predict and checking that the screen recovers them.

Table 10: Emergent behavior: turns flagged by the screen as exceeding what the scene and the partner predict, and the number confirmed on manual adjudication, over the full cube and three passes.

Systems	episodes	candidate turns	flagged rate	confirmed
Each of the ten systems	2,430	0	0.000	0
All systems	24,300	0	0.000	0

4.5 Two-sided evaluation

When only one participant in an interaction is scored, the behavior of the other becomes an unmeasured component of every score. Simulated interlocutors that appear plausible have been shown to distort the ranking of the systems they are used to evaluate (Dou et al., 2025). For this reason InteractEval also scores the user side. Both roles are assessed on six levels with the same cascade, and Table 3 gives the criteria of each role.

The user-side levels are defined as follows. At L1 the user establishes what it is asking for, and at L2 it responds to the part of the assistant's reply that calls for a response, for instance an error, a flaw in reasoning or an evasion. L3 is role preservation: the user keeps to its role and does not drift into advising, summarizing or the register of a helper. At L4 the user conforms to its profile. At L5 it initiates or advances a probe whose prospective contribution to its goal is justified by the exchange. At L6 it coordinates successive probes, reactions and decisions, incorporating what the assistant has supplied while maintaining or justifiably revising its course. Red lines, engagement-based termination and successful closure concern the character's conduct and are applied only on the assistant side (§3.3); the latent state and the event chain are updated at every utterance of either participant.

With both roles scored, a scenario cube produces a matrix of results, and the same cube shows how well each system performs its role and how demanding it is as a partner. We use the 10×10 crossing to examine whether a partner shifts all scores by a constant or its effect depends on the system scored, what properties make a partner informative, and how systems fare when paired with themselves.

4.5.1 The partner as a factor: the interaction matrix

In Table 11 every system appears in both roles. Rows correspond to the system playing the partner and columns to the system being scored, and diagonal cells are self-play. Reading down a column shows how a system performs against the whole roster, and reading along a row shows how a system affects the scores of the systems it partners. When `claude-5-opus` plays the partner, the resulting row is identical to the one-sided leaderboard of Table 5; one-sided evaluation is a special case of the two-sided protocol.

Table 11: Interaction matrix over the full 10×10 roster, means of three passes. Rows are the system playing the partner, columns the system being scored (`o5.5` = `claude-opus-5.5`, `dsk` = `deepseek-v4-pro`). Diagonal cells (⋄) are self-play. The `claude-5-opus` row (†) is the standard-user condition and equals the leaderboard of Table 5. *mean* and *span* are the row mean and the max–min across the ten scored columns.

partner \ scored	o5.5	astra	fable	opus	sol	qwen	gemini	dsk	kimi	glm	mean	span
claude-opus-5.5	0.599⋄	0.686	0.545	0.537	0.459	0.468	0.339	0.398	0.450	0.356	0.484	0.347
gpt-6-astra	0.663	0.677⋄	0.672	0.600	0.427	0.539	0.339	0.495	0.450	0.275	0.514	0.402
claude-fable-5.1	0.585	0.582	0.572⋄	0.500	0.476	0.438	0.399	0.377	0.382	0.365	0.468	0.220
claude-5-opus†	0.636	0.629	0.616	0.540⋄	0.515	0.482	0.450	0.430	0.425	0.414	0.514	0.222
gpt-5.6-sol	0.547	0.568	0.557	0.482	0.457⋄	0.414	0.392	0.368	0.366	0.355	0.451	0.213
qwen3.8-max	0.604	0.674	0.666	0.591	0.514	0.527⋄	0.427	0.474	0.467	0.362	0.531	0.312
gemini-3.8-flash	0.472	0.509	0.506	0.421	0.410	0.362	0.328⋄	0.311	0.301	0.300	0.392	0.209
deepseek-v4-pro	0.610	0.662	0.663	0.621	0.394	0.510	0.293	0.501⋄	0.406	0.238	0.490	**0.425**
kimi-k3	0.541	0.664	0.649	0.577	0.544	0.522	0.444	0.465	0.456⋄	0.370	0.523	0.294
glm-5.3-flash	0.491	0.531	0.520	0.446	0.420	0.385	0.357	0.330	0.320	0.323⋄	0.412	0.211
column mean	0.575	0.618	0.597	0.532	0.462	0.465	0.377	0.415	0.402	0.336	0.478	—

Were the partner no more than a difficulty setting, each row would be the column means shifted by a constant, and every cell would be determined by its row and column margins. The ordering of the scored systems changes from one partner to another, and 10.9% of the total sum of squares remains after row and column main effects are removed. This residual combines the effect of pairing, the self-play excess and measurement noise, which the matrix of cell means cannot separate. The mixed-effects model of §4.1 separates the two and attributes 9.9% of the variance of scored system, partner and pairing to the pairing (Table 29).

The partner is part of the strategy's continuation: it can supply the missing information, challenge an assumption, redirect the goal or decline a proposal. The matrix consequently measures how a system's prospective choices and subsequent coordination respond to different interlocutors. A strategy that is productive with one partner may need a different next step with another.

For the choice of an evaluation partner, the spread a partner induces among the scored systems is more informative than its row mean. A partner can lower all scores without separating the systems. `gemini-3.8-flash` yields the lowest row mean (0.392), yet the ten scored systems differ by only 0.209 against it, while `deepseek-v4-pro` yields the widest spread (0.425) at much higher absolute scores. Across the ten partners, spread correlates with probing intensity at +0.48, which with ten partners is weak and imprecise; partner identity is confounded with the partner's

other properties. Table 12 locates each partner's effect relative to the standard user. Higher probing intensity is associated with larger shortfalls at the proactive stage (L5–L6), where the value of a proposed course and its long-horizon coordination are assessed, and lower cooperativeness with larger shortfalls at L2–L3, where the uptake of what the partner says and the state progression built on it are assessed. `claude-fable-5.1` and `gpt-5.6-sol` have similar row means (0.468 and 0.451) but affect different parts of the progression: the first mainly the proactive stage (a difference of +0.047 between its shortfalls at L5–L6 and at L2–L3, Table 12), the second mainly L2–L3 (−0.089).

Table 12: Where a partner's pressure falls, relative to the standard user (`claude-5-opus`), means of three passes. ΔL2–L3 and ΔL5–L6 are the shortfall a partner induces at L2–L3 (uptake and state progression) and at L5–L6 (proactive promotion) in excess of that induced by the standard user (`claude-5-opus`, the zero baseline); positive values mean those levels are pressed harder. *difference* is ΔL5–L6 minus ΔL2–L3. *probing* and *coop.* are measured from each system's turns in the partner role (Appendix A.10).

Partner	probing	coop.	ΔL2–L3	ΔL5–L6	difference
claude-opus-5.5	0.112	0.87	−0.010	+0.040	+0.050
gpt-6-astra	0.128	0.88	−0.038	+0.034	+0.072
claude-fable-5.1	0.118	0.90	−0.003	+0.044	+0.047
claude-5-opus	0.082	0.86	0.000	0.000	0.000
gpt-5.6-sol	0.058	0.80	+0.088	−0.001	−0.089
qwen3.8-max	0.030	0.69	+0.097	−0.056	−0.153
gemini-3.8-flash	0.014	0.61	**+0.249**	−0.021	−0.270
deepseek-v4-pro	0.088	0.80	+0.041	+0.007	−0.034
kimi-k3	0.046	0.76	+0.052	−0.038	−0.090
glm-5.3-flash	0.034	0.72	+0.162	−0.007	−0.169

Cooperativeness in these tables is measured from each system's turns in the partner role (Appendix A.10). Its values fall between 0.61 and 0.90, and none of the ten systems behaves as an adversarial user. These measurements support the decision in §3.2.1 not to use cooperativeness as a facet: it belongs to one participant, and it is measured here and not fixed by the design.

Self-play. Table 13 compares each system's score when paired with itself, taken from the diagonal of the matrix, with its score against the standard user. The self-play gap is associated with self-similarity, the similarity between the interaction moves a system makes as partner and those it makes as character (Appendix A.10). The four most self-similar systems, `gpt-6-astra` (0.99), `deepseek-v4-pro` (0.90), `kimi-k3` (0.83) and `qwen3.8-max` (0.81), score higher against themselves than against the standard user. The less self-similar systems score lower, except `claude-5-opus`, whose self-play cell is the standard-user condition and whose net gap, the score against the standard user minus the self-play score, is zero by construction; the correlation between self-similarity and the net gap is −0.65 across the ten systems and −0.76 across the nine for which the gap is not zero by construction. The row and column effects of the matrix lead to the same conclusion by a different route: diagonal cells exceed the values these effects predict by +0.020 on average (interval in Table 28), compared with −0.002 for off-diagonal cells. Systems whose moves as partner and as character are more similar tend to score higher in self-play, and self-similarity is only weakly related to overall strength (−0.30); the association remains (−0.72) when both the standard user and `gpt-6-astra`, whose self-similarity of 0.99 is the highest, are left out.

Both analyses use the same ten diagonal cells and are not independent, so that the size of the effect can be estimated only approximately. We treat the estimate as an upper bound on any effect of self-similarity as such, because systems from the same family are stylistically closer than systems from different families, so part of the excess may reflect family resemblance. Balancing pairs for stylistic distance would separate the two.

The user role. The same crossing scores the partner as well. Applying the user-side rubric to each partner's own turns, with the same roster, cube and protocol, gives each system's profile in the user role (Table 14) and its user-side composite against every assistant (Table 15). The capability progression reappears on the user side. Its stages span 0.066, 0.084 and 0.225 across the roster, so systems in the user role, as in the assistant role, differ most at the proactive end, in the warrant for their probes and goal revisions (L5) and the coordination of those moves toward their goal across turns (L6). User-side L5 correlates at +0.94 with the probing intensity of Table 12; both measures come from the same turns, so their correlation follows from shared data and is not independent evidence. User-side L3 shows a failure specific to the role: `gpt-6-astra` and `deepseek-v4-pro` score lowest on user-side L3 (0.792 and 0.763), the level that records a user drifting into advising, summarizing and the register of a helper.

Of the variance in Table 15, 94.4% is attributable to the system in the user role, 3.6% to the assistant and 2.0% to the residual after both main effects, against a residual of 10.9% on the assistant side. Differences between the

Table 13: Self-play, from the diagonal of Table 11, beside the score against the standard user. Self-similarity is the cosine similarity between a system's move profiles in the user and assistant roles (Eq. (35)); *net gap* is the standard-user score minus the self-play score, so a negative value marks a system that scores higher against itself.

Model	self-similarity	self-play	standard user	net gap
claude-opus-5.5	0.47	0.599	0.636	+0.037
gpt-6-astra	0.99	0.677	0.629	−0.048
claude-fable-5.1	0.42	0.572	0.616	+0.044
claude-5-opus	0.43	0.540	0.540	0.000
gpt-5.6-sol	0.67	0.457	0.515	+0.058
qwen3.8-max	0.81	0.527	0.482	−0.045
gemini-3.8-flash	0.56	0.328	0.450	+0.122
deepseek-v4-pro	0.90	0.501	0.430	−0.071
kimi-k3	0.83	0.456	0.425	−0.031
glm-5.3-flash	0.65	0.323	0.414	+0.091

systems in the user role account for almost all of the variance of the user-side composite, whereas the assistant's score depends in part on which system plays the user. User-side and assistant-side composites are correlated across systems (+0.88), but the orderings differ: `claude-fable-5.1` is the strongest user and third as an assistant, and `claude-5-opus`, the standard user of the one-sided protocol, ranks fourth in the user role. Tables 11 and 15 report the corresponding matrices. Figure 11 visualizes both matrices, their marginal means and the residual after row and column effects.

Table 14: User-side capability profile: each system's scores in the user role, averaged over the ten assistants it partners and over three passes. The stage columns are the means of L1–L2, L3–L4 and L5–L6 under the user-side wording of Table 3. The final row gives the range across the roster.

		level (user side)						stage		
System in user role	composite	L1	L2	L3	L4	L5	L6	L1–L2	L3–L4	L5–L6
claude-fable-5.1	0.731	0.951	0.902	0.861	0.842	0.611	0.598	0.927	0.852	0.605
claude-opus-5.5	0.724	0.949	0.905	0.853	0.838	0.604	0.586	0.927	0.846	0.595
gpt-6-astra	0.715	0.947	0.911	0.792	0.821	0.628	0.574	0.929	0.807	0.601
claude-5-opus	0.707	0.944	0.889	0.858	0.829	0.562	0.571	0.917	0.844	0.567
qwen3.8-max	0.653	0.931	0.862	0.824	0.794	0.468	0.512	0.897	0.809	0.490
kimi-k3	0.643	0.928	0.857	0.831	0.781	0.452	0.497	0.893	0.806	0.475
deepseek-v4-pro	0.637	0.936	0.874	0.763	0.772	0.509	0.463	0.905	0.768	0.486
gpt-5.6-sol	0.628	0.929	0.861	0.806	0.776	0.441	0.468	0.895	0.791	0.455
gemini-3.8-flash	0.585	0.912	0.823	0.817	0.748	0.352	0.421	0.868	0.783	0.386
glm-5.3-flash	0.577	0.907	0.818	0.809	0.741	0.368	0.392	0.863	0.775	0.380
range	0.154	0.044	0.093	0.098	0.101	0.276	0.206	0.066	0.084	0.225

Table 15: User-side interaction matrix, means of three passes: each cell is the user-side composite of the row system when paired with the column system as assistant; the diagonal (⋄) is self-play and *mean* is the row mean. Labels as in Table 11.

user \ assistant	o5.5	astra	fable	opus	sol	qwen	gemini	dsk	kimi	glm	mean
claude-opus-5.5	0.724⋄	0.737	0.736	0.734	0.724	0.718	0.698	0.722	0.713	0.735	0.724
gpt-6-astra	0.718	0.733⋄	0.745	0.705	0.712	0.708	0.696	0.706	0.705	0.721	0.715
claude-fable-5.1	0.732	0.751	0.756⋄	0.735	0.728	0.744	0.714	0.718	0.722	0.710	0.731
claude-5-opus	0.725	0.706	0.729	0.714⋄	0.693	0.707	0.708	0.693	0.693	0.703	0.707
gpt-5.6-sol	0.631	0.656	0.656	0.625	0.615⋄	0.634	0.617	0.615	0.614	0.616	0.628
qwen3.8-max	0.661	0.691	0.654	0.651	0.646	0.661⋄	0.644	0.640	0.639	0.644	0.653
gemini-3.8-flash	0.600	0.586	0.602	0.588	0.581	0.583	0.581⋄	0.575	0.581	0.574	0.585
deepseek-v4-pro	0.658	0.644	0.649	0.636	0.642	0.631	0.624	0.624⋄	0.621	0.643	0.637
kimi-k3	0.665	0.656	0.648	0.639	0.648	0.646	0.640	0.621	0.635⋄	0.632	0.643
glm-5.3-flash	0.597	0.593	0.595	0.577	0.581	0.580	0.564	0.574	0.560	0.554⋄	0.577
column mean	0.671	0.675	0.677	0.660	0.657	0.661	0.649	0.649	0.648	0.653	0.660

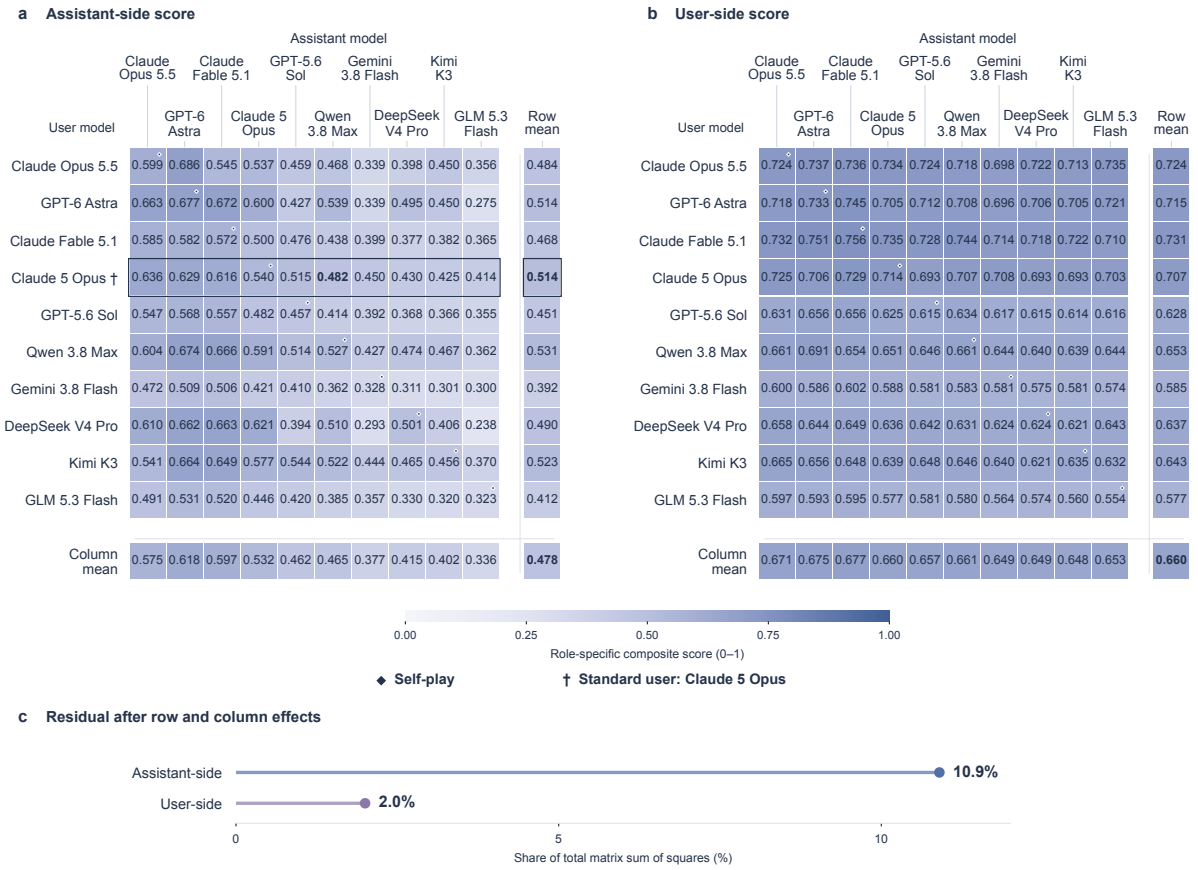


Figure 11: Two-sided evaluation reveals role asymmetry. **a,b,** Assistant- and user-side composite scores across all model pairings, averaged over three passes (Tables 11 and 15). Rows denote user models and columns assistant models; margins show row and column means, with the grand mean in the corner. **c,** Variation remaining after additive row and column effects is larger on the assistant side (10.9% versus 2.0%). These residuals include noise and self-play effects.

4.5.2 Quality of the generated interactions

Conclusions about partners depend on the simulated user being varied, internally consistent and human-like; we measure each property and report the gap to the human–AI logs. We assess three properties with measures from the literature on dialogue generation, computed for each system in the user role over all of its partners and, where a human counterpart exists, for the human–AI dialogue logs of the observational pool (§3.2; Table 16).

Diversity is the lexical richness of a system's user-role utterances, measured by MTLD, which does not depend on text length (McCarthy and Jarvis, 2010). *Consistency* is the share of a participant's utterances that a dialogue inference model trained on DECODE (Nie et al., 2021) judges to contradict one of its ten preceding utterances or an entry of its role view, for the system as user and for its assistant partners. The *human-likeness* of the simulated user is the MAUVE score (Pillutla et al., 2021) between its utterances and the user turns of the human–AI logs, computed on realistic cells, where the two are comparable.

Simulated users differ from the human users of the logs in lexical profile and in consistency (Table 16). Their vocabulary is richer, with an MTLD of 79.8 to 95.5 against 60.1 for the human users. They also contradict themselves or their role view more often, in 7.1% to 18.1% of their utterances against 4.6%. The human figure is checked against earlier utterances only, since the logs have no role view, and is likely to rise under the same checks as the simulated users, so the gap is probably overstated.

The assistants contradict their own earlier utterances or character entries in 0.74% to 1.05% of theirs, a range that includes the 0.90% of the AI side of the logs and that varies little with the system in the user role. Across the ten systems, the contradiction rate of the user role is lower for systems that are better users and better assistants ($\rho = -0.98$ with the user-side composite and -0.83 with the assistant composite); these correlations come from ten systems that rank alike on most measures and do not separate causes. `gemini-3.8-flash` and `glm-5.3-flash`, the two systems with the highest rates, contradict themselves in 16.5% and 18.1% of their utterances. MAUVE ranges from 0.27 for `deepseek-v4-pro` to 0.47 for `claude-fable-5.1`, and the three systems with the highest scores, `claude-fable-5.1`, `claude-opus-5.5` and `claude-5-opus`, are the three with the lowest contradiction rates. The standard user, `claude-5-opus`, ranks third on both consistency and MAUVE, and all ten simulated users

Table 16: Quality of the generated interactions, by system in the user role over all partners of the two-sided protocol, means of three passes. Diversity: MTLD per episode of the system's user-role utterances (threshold 0.72, `lexicalrichness`). Contradiction: share of utterances for which a dialogue NLI model trained on DECODE (Nie et al., 2021) assigns a contradiction probability above 0.9 against one of the speaker's ten preceding utterances or an entry of its role view (the profile for the user, the character entries for the assistant), for the system as user and for its assistant partners. Human-likeness: MAUVE of 5,000 user-role utterances against 5,000 user turns of the human–AI logs, on realistic cells. The last row gives the same measures for the human–AI logs of realistic settings; there contradiction is checked against earlier utterances only, since the logs have no role view.

	diversity	contradiction		human-likeness
System in the user role	MTLD	user	assistant	MAUVE
claude-opus-5.5	81.8	7.7%	0.96%	0.45
gpt-6-astra	86.2	8.7%	1.05%	0.31
claude-fable-5.1	79.8	7.1%	0.99%	0.47
claude-5-opus	84.9	8.4%	0.90%	0.44
gpt-5.6-sol	89.2	13.1%	0.84%	0.33
qwen3.8-max	89.9	11.4%	0.78%	0.37
gemini-3.8-flash	84.6	16.5%	0.74%	0.32
deepseek-v4-pro	95.5	13.4%	0.93%	0.27
kimi-k3	85.7	12.1%	0.81%	0.36
glm-5.3-flash	88.5	18.1%	0.81%	0.30
human–AI logs	60.1	4.6%	0.90%	—

differ from the users of the logs, so the conclusions about partners concern simulated partners (§6.4).

4.6 Enacting a role across partners

A persona specifies the participant a model is to enact, but leaves choices about how to conduct the exchange. We examine those choices in a supplementary sample of 520 dialogues from ten API-served models with two user models across 26 shared scenes, together with the 243 dialogues in the open-model matrix. Within each cohort, models receive the same scene and role specifications. We compare clarification, unsupported factual additions (fabrication), question endings, list formatting and reply length, averaging over the first three assistant replies or all replies when a dialogue is shorter. Role preservation and initiative are evaluated separately from these behavioral frequencies. Appendix A.12 describes the sample and controls.

The night-counselor scene illustrates how models can share the intended tone while choosing different next moves. The persona requires warmth and restraint before the user's needs are clear. GPT-6 Astra asks whether the user wants to be heard or to work through a problem; Claude-5-Opus asks whether the exhaustion concerns a particular event or a more diffuse feeling. Gemma invites unstructured disclosure, Qwen3.5 asks whether remaining together in silence would help, and Qwen3.8 asks about the user's current circumstances. Each response sustains a supportive stance, but the questions differ in what they invite the user to do: choose a purpose, identify a problem, disclose an experience or remain in contact without elaborating.

Across the assigned roles, the models also retain identifiable habits of expression (Figure 12). A classifier using only reply length, question marks, question endings and list formatting identifies the ten API models from their first three replies in held-out scenes with balanced accuracy 0.204, against chance at 0.100. Accuracy is 0.158 after removing length. In the open cohort the corresponding values are 0.473 and 0.436, against chance at 0.333. These features establish persistent differences in response form; they do not by themselves measure whether a model has chosen an appropriate conversational move. We use continuous behavioral profiles to describe these differences, with exploratory clustering reported in Appendix A.12.

The choice of partner can change a model's conduct without a clear improvement in initiative. Across whole dialogues, Kimi's mean clarification frequency rises by 0.085 (95% scene-bootstrap interval [0.004, 0.169]) with Claude Opus 5.5 rather than Claude-5-Opus as user, while its L5 grade changes by 0.006 [−0.027, 0.043]. Asking more questions and making more useful interventions are distinct outcomes of the same partner comparison.

4.7 Processing the partner's contribution

To relate the behavioral differences in §4.6 to participant-state information, we examine representations in Gemma-4-31B, Qwen3.5-35B-A3B and Qwen3.8-27B, each interacting with all three models in both roles. The nine ordered pairs cover the 27 scenario cells, giving 243 dialogues and 4,058 utterances. State probes measure which annotated

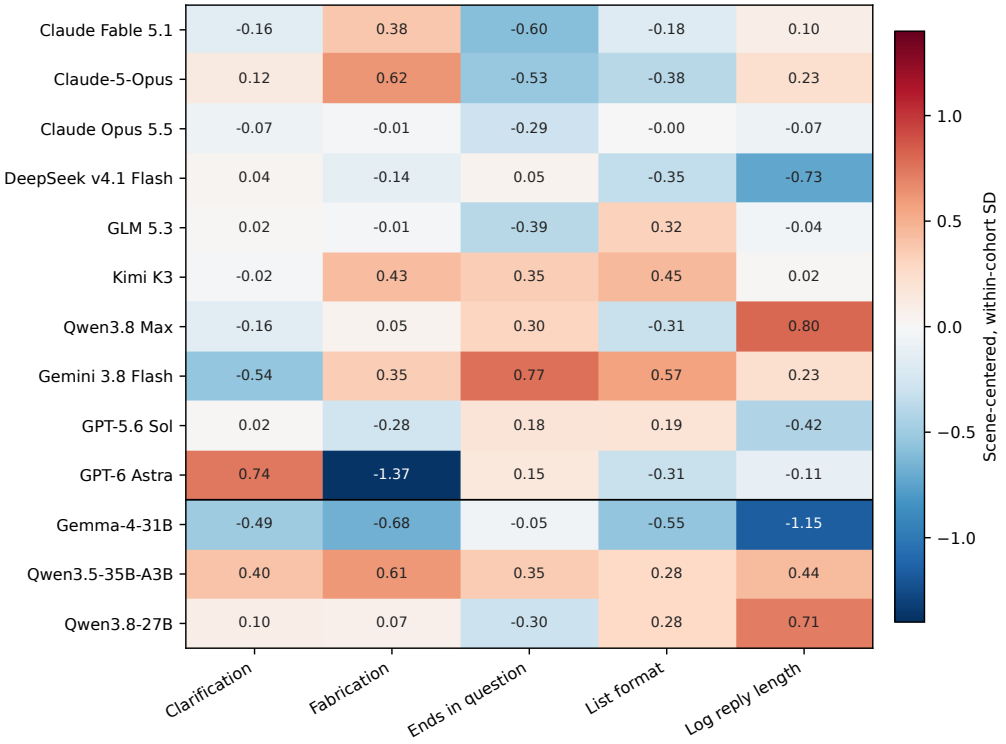


Figure 12: Behavioral profiles under shared role specifications, averaged over the first three assistant replies, centered within scene and scaled within each cohort. The line separates API-served and locally run models. Colors express relative behavioral tendencies within a cohort.

Table 17: User-interest prediction from the final reply token at 75% of model depth. All predictions hold out entire scenes. Within-dialogue R^2 removes each dialogue's mean from both labels and predictions. Each row covers 81 dialogues.

Model	Role	Turns	R^2	Within R^2
Gemma-4-31B	assistant	748	−0.007	−0.194
	user	738	0.284	−0.026
Qwen3.5-35B-A3B	assistant	634	0.273	0.059
	user	602	0.365	0.158
Qwen3.8-27B	assistant	647	0.136	−0.082
	user	689	0.357	0.168

participant states can be read from the representations; aligned trajectories measure how those readouts vary between participants during an exchange. Interventions in Qwen3.8-27B test whether relational directions control behavior and where the latest partner utterance affects the output. Appendix A.13 gives the methods.

4.7.1 State representation depends on the participant's role

The three models differ in how their representations follow user interest (Table 17). At three quarters of model depth, Qwen3.5 gives positive R^2 for interest both across and within dialogues in either role. Qwen3.8 gives positive within-dialogue R^2 only in the user role. Gemma predicts differences between dialogues in the user role, but its within-dialogue R^2 is negative in both roles. These differences persist beyond the selected depth: no scanned layer gives positive within-dialogue interest prediction for Gemma in either role or Qwen3.8 as assistant. Positive overall R^2 can therefore reflect differences between dialogues even when within-dialogue variation is poorly recovered.

Affection follows a different pattern. Gemma's assistant representations support decoding of user affection within dialogues, reaching $R^2 = 0.237$ in the layer scan, despite the weak interest readout. The full profiles in Figure 18 show that readable relational information can coexist with poor tracking of engagement.

The role difference in Qwen3.8 extends across state dimensions. Its within-dialogue R^2 rises by 0.251 for user interest and by 0.302 for user affection when the model occupies the user role, with scene-bootstrap intervals of [0.034, 0.439] and [0.091, 0.603] (Figure 13a). The corresponding interest difference in Qwen3.5 is 0.099 [−0.132, 0.311], while Gemma's affection readout changes by only −0.001 [−0.176, 0.231]. Qwen3.8's user-state readouts depend most clearly on the role it occupies. Gemma's stronger readout of affection than interest persists in both roles.

Shared relational variation and distinct participant states. Predicting both participants' states does not imply that each has an independent representation. At the same depth, the assistant-side probe directions for the two participants' affection have cosine similarity 0.737 in Gemma, 0.661 in Qwen3.5 and 0.620 in Qwen3.8. Once the assistant's own affection is removed from both features and labels, prediction of the user's remaining affection is negative in all three models ($R^2 = -0.225$, -0.182 and -0.248). The analogous residual readout of user anxiety remains positive in all three, at 0.162, 0.123 and 0.078. The affection probes primarily recover variation shared by the two participants, while anxiety retains a component specific to the user. Readable affection therefore need not distinguish whose feelings have changed. Table 38 gives the corresponding results in both roles.

The relation to observed behavior also depends on which state is read. Across the open-model dialogues, lower within-dialogue affection-readout error is associated with more clarification, while higher error is associated with more unsupported factual additions, controlling for model, partner, scene, dialogue length and variation in the state labels. The corresponding interest-error intervals include zero (Appendix A.12, Figure 17). These exploratory associations connect the behavioral and representation analyses without treating a readable state as a sufficient explanation of the model's response.

4.7.2 Correspondence across the exchange

We align the representation after one participant speaks with the other participant's representation before replying. Both have access to the same transcript at this boundary. User-interest readouts correlate across these boundaries, averaging 0.383 following assistant utterances and 0.391 following user utterances over the 135 dialogues with at least six aligned follow-up exchanges. Correlation of successive changes is smaller, at 0.223 and 0.149, respectively. Following assistant utterances, the change correlation exceeds a control pairing different scenes with the same model pair by 0.208 (95% scene-bootstrap interval [0.141, 0.282]). Following user utterances, the excess is 0.086 [−0.007, 0.174]. A direct comparison within the same dialogues gives a directional difference of 0.121 [0.026, 0.232] after subtracting the respective controls. At this depth, engagement updates correspond more closely as the user model receives the assistant's contribution.

Correspondence varies across the nine pairs (Figure 14). With Qwen3.8 as assistant, the correlation of interest changes following its utterances is 0.352 with Gemma as user, but −0.022 with Qwen3.5 as user. Reversing the direction of the comparison within the latter pair gives 0.210. Nor do identical models consistently correspond most closely: the three same-model pairs range from 0.142 to 0.342 following assistant utterances, and from −0.045 to 0.253 following user utterances.

Linear CKA of successive hidden-state changes is 0.415 following assistant utterances and 0.426 following user utterances. Relative to matched comparisons across different scenes, the excess is 0.163 [0.092, 0.230] and 0.167 [0.088, 0.240], respectively (Figure 19). The directional difference in CKA excess is −0.004 [−0.090, 0.093]. The CKA point estimates are nearly equal in the two directions, while interest readouts show the directional difference above. Affection readouts likewise show correspondence in both directions: the excess change correlation is 0.179 and 0.137 for assistant affection, and 0.160 and 0.116 for user affection, with all four intervals above zero.

The two measures also have different depth profiles (Figure 13b–c). At one quarter of model depth, hidden-state changes already exceed the other-scene control in both directions, by 0.076 and 0.073, whereas the excess for interest changes is only 0.040 and 0.020, with both intervals including zero. At half depth, interest correspondence rises to 0.150 and 0.144. Its directional difference appears at three quarters depth and is no longer evident at the final layer. Shared variation in the full representations is detectable at shallower layers than a clear correspondence in the interest readouts. The sampled profiles place the strongest interest correspondence at intermediate depths.

Agreement between participants does not guarantee accuracy about their state. In the 16 eligible Gemma–Gemma dialogues, interest-change readouts correlate at 0.264 following assistant utterances. On the same 204 transcript boundaries, however, within-dialogue R^2 is −0.374 for the assistant's post-utterance readout and −0.246 for the user's pre-reply readout. The two models' readouts vary together even though neither reliably recovers the annotated interest trajectory.

4.7.3 Interventions on Qwen representations

The readouts and correspondence describe information in the representations. Two interventions in Qwen3.8-27B test how that information relates to behavior. First, we add or subtract interest and affection directions during generation. These directions are estimated from a separate corpus of 297 transcripts; the intervention uses 60 contexts. Neither direction produces a consistent signed change in warmth or engagement concern beyond the random-direction control. At larger magnitudes, persona fit deteriorates, and interest interventions also reduce coherence. The directions recover relational variation but do not provide reliable control of the corresponding behavior under this intervention.

Second, we intervene on the latest user utterance itself. In a separate corpus of 54 dialogues comprising 729

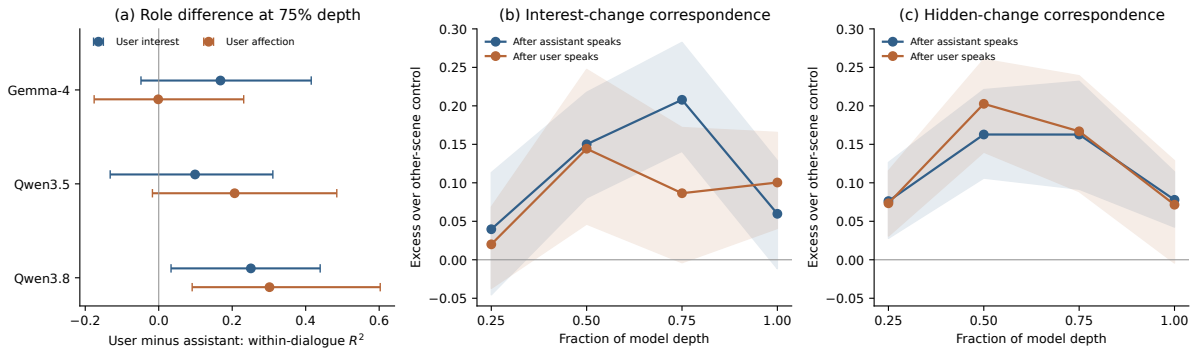


Figure 13: Role dependence and the depth of representational correspondence. (a) Difference in within-dialogue prediction between the user and assistant roles at 75% depth. (b–c) Correspondence of successive changes, relative to a length-matched control from another scene with the same model pair. Error bars and bands are 95% scene-bootstrap intervals. Role contrasts resample the same scenes in both corpora; direction contrasts retain the same dialogues. Depth profiles are descriptive comparisons across four sampled depths.

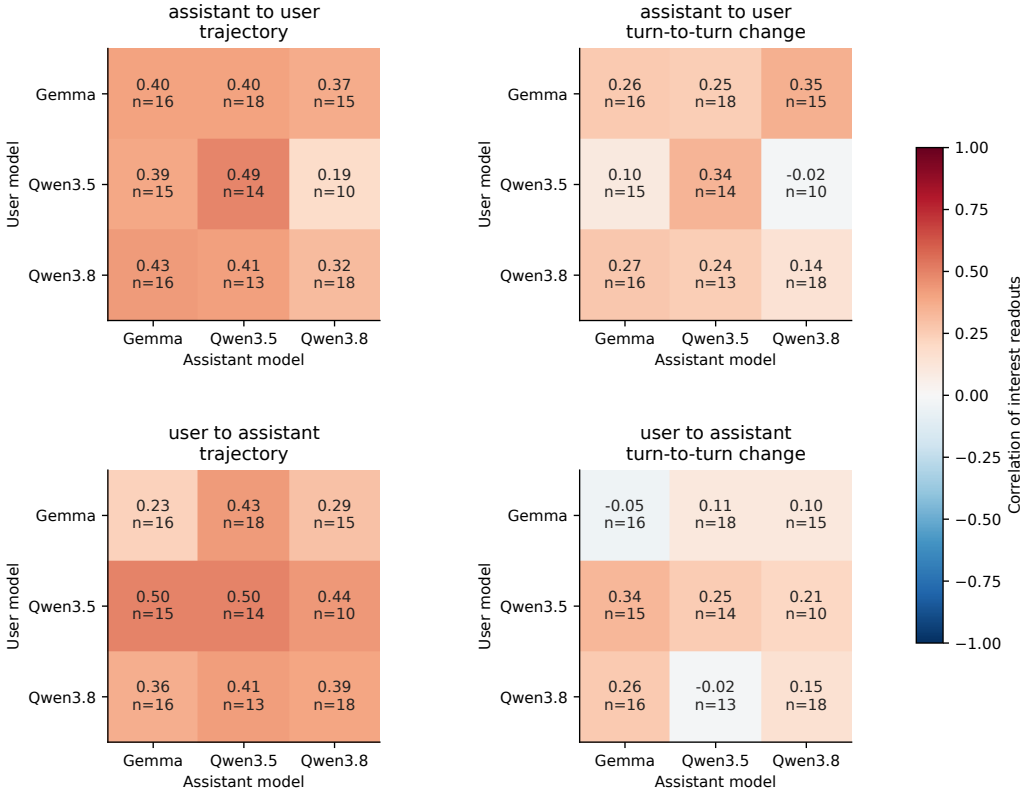


Figure 14: User-interest readout correspondence for all nine open-model pairs at 75% of model depth. Rows identify the user model and columns the assistant model. The upper panels align assistant post-utterance states with user pre-reply states; the lower panels reverse this direction. Left: within-dialogue correlations of readouts. Right: correlations of successive changes. Values average eligible dialogues, with counts shown in each cell.

assistant turns, we replace that utterance with one from another scene, hold the remaining context fixed and restore the original activations at selected layers and positions. Recovery is the fraction of the change in the next-token distribution removed by restoration, measured with KL divergence. The analysis includes 41 decision points from 17 scenes where utterance replacement changes the distribution appreciably.

Restoration shifts from the partner's utterance positions to the final prompt position as depth increases (Figure 15a). At the utterance positions, recovery is 0.860 at layer 32 and 0.664 at layer 40, falling to 0.285 at layer 48. At the final position, it is approximately zero at layer 32, rises to 0.435 at layer 48 and reaches 0.777 at layer 52. The first layer at which final-position recovery exceeds utterance-position recovery has median 48 and interquartile range 48–52 in the 64-block model. These interventions locate a late stage at which the partner-conditioned output can be restored through the final position alone.

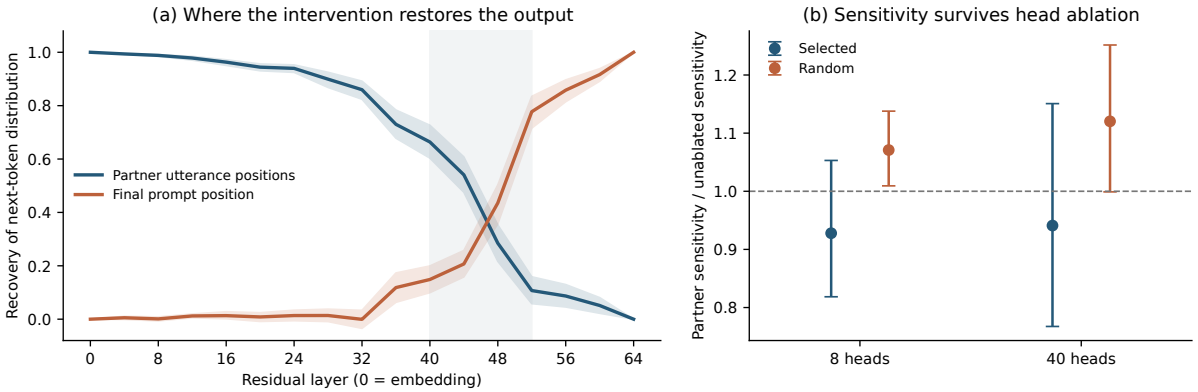


Figure 15: Late integration of partner information in Qwen3.8-27B. (a) Recovery shifts from the partner's utterance positions to the final prompt position, with the principal transition at layers 40–52. (b) Most sensitivity to utterance replacement remains after ablation of the selected full-attention heads. Sensitivity is normalized to the unablated model; 1 denotes no change. Bands and error bars give 95% scene-bootstrap intervals over 41 decision points from 17 scenes.

The state probes above read the final token of a completed reply, whereas restoration acts before generation; their layer profiles concern different token positions and stages of the exchange. Qwen's next-token distribution depends substantially on the partner's contribution, while its assistant-side state readout poorly recovers within-dialogue interest variation. Sensitivity to the latest utterance thus does not ensure accurate tracking of that state across turns.

Individual heads contribute substantially to restoration. Layer 48 heads 17 and 4 recover 0.155 and 0.104 of the distributional change when restored separately. Ablation has a smaller effect: removing the eight highest-ranked heads leaves 0.928 of the original sensitivity to utterance replacement, and removing forty leaves 0.941 (Figure 15b). Sensitivity retention is similar under the larger ablation. Most of the measured sensitivity is maintained outside the selected heads. Other positions and linear-attention blocks remain intact, and most of the measured sensitivity survives through the remaining computation.

Relation to the behavioral findings. The behavioral profiles and the interventions address complementary properties of an exchange. The profiles distinguish the moves a model habitually makes; the readouts measure accessible participant-state information; and restoration measures the partner's influence on the next-token distribution. In Qwen, a readable relational direction does not provide reliable behavioral control, and sensitivity to the partner's words remains high after removal of the selected attention heads. These findings explain why neither a characteristic move nor an internal state readout can stand in for evaluating whether the reply advances the interaction. Section 5.6 uses the main evaluation's behavior and outcome measures to characterize each system in both roles.

4.8 Extensions of the protocol

The protocol extends, with the same rubric and cascade, to agents that embed a base model in an architecture and to groups of more than two participants, the agents were evaluated on the full cube and the mixed groups on subsets of it (Tables 19–21).

Agent architectures. An agent that embeds a base model in an architecture is evaluated in the same way as the base model, so the protocol can test whether the architecture adds interactive competence and where. We built three agents on each of the ten systems: MetaMind (Zhang et al., 2026c) and Cognizer, which reason explicitly about the partner's mental state, and the Claude Code harness, a general-purpose agent scaffold with no model of the partner. All thirty agents were evaluated on the full cube (Table 18), and every gain below is the paired difference between an agent and its base, which does not control for the additional calls and context that an architecture uses.

An architecture can assist at two points in this progression: selecting a worthwhile direction from the current situation and carrying it through subsequent exchanges. We therefore expected its largest gains at the proactive stage, where these demands are assessed, rather than in the local execution on which frontier systems already differ little. The prediction holds on average for all three: over the ten bases, each gains most in proactive promotion and least in passive execution, and every agent's gain in proactive promotion exceeds the largest mean gain in passive execution (0.016). MetaMind adds 0.012 in passive execution, 0.026 in interactive adaptation and 0.043 in proactive promotion, and Cognizer adds 0.012, 0.038 and 0.066 (Table 19).

The gain grows stage by stage, mirroring the spans across systems in Table 5, and it is largest where the base is weakest: the increase in proactive promotion falls almost linearly with the base system's own level ($r \leq -0.99$ for both architectures), from 0.024 and 0.040 on `claude-opus-5.5` to 0.063 and 0.092 on `deepseek-v4-pro`.

A relation this close is likely to owe part of its strength to headroom, since a stronger base has less room to rise, and the ten bases do not allow that effect to be separated from a genuine dependence of the gain on the base system's weakness. With Cognizer, `deepseek-v4-pro` reaches 0.262 in proactive promotion, about half of what `claude-opus-5.5` achieves without any architecture.

The two cognitive architectures also differ in where they help. MetaMind raises ΔT for every one of the ten bases, by 0.025 on average, and gains most in nonfunctional settings (0.046, against 0.021 in functional ones). Cognizer lowers ΔT for every base, by 0.029 on average, and gains most in functional settings (0.061, against 0.032). A composite score records both as improvements of the same kind, of 0.033 and 0.050. The profile shows that the first architecture helps most where the interaction is relational and the second where it is instrumental, so two interventions that both raise proactive promotion move ΔT in opposite directions, which the composite cannot express.

The Claude Code harness adds 0.041 to the composite on average, between the two cognitive architectures, and lies between them on every one of the ten bases. Its gain is distributed differently over the progression. It raises passive execution more than either cognitive architecture (0.016) and interactive adaptation least (0.024), and its gain in proactive promotion, 0.058, falls with the level of the base almost as closely as theirs ($r = -0.99$). It lowers ΔT for every base, by 0.043 on average and more than Cognizer, because it gains most in functional settings (0.062) and least in nonfunctional ones (0.019). The harness thus shows the task-oriented profile of Cognizer more strongly; MetaMind remains the only architecture that gains most in relational settings. With the harness, `deepseek-v4-pro` also remains far behind, at 0.252 in proactive promotion.

These gains should be read as changes in the quality of the course an agent develops, not as counts of additional initiative. The opposite purpose shifts of MetaMind and Cognizer show why the value criterion must remain tied to the interaction's goal: both raise proactive promotion, but they support different kinds of contribution. The paired score differences locate these improvements in the stage concerned with establishing and developing a worthwhile course of interaction.

Because the harness and the judge come from the same developer, we test for a family advantage against bases of similar level, since the gain falls with the level of the base: relative to a linear fit of the gain on the base composite, the three Claude bases gain 0.003 more than predicted on average and the other seven 0.001 less. The difference is well below the 0.01 within which differences involving Claude-family systems are read with caution (§6.4); it gives no evidence of a family advantage, although an advantage of this size cannot be excluded.

Table 18: Agents built on each of the ten systems, two with a cognitive architecture and one with a general-purpose scaffold: composite, proactive promotion (L5–L6) and purpose disposition ΔT for the base model, with MetaMind (MM), with Cognizer (TRHI; Cog) and with the Claude Code harness (CC), means of three passes over the full cube. Frontier emergence (FE) is the percentage of evaluated assistant turns receiving a binary breakthrough judgment from the interaction history (Appendix B).

	composite				L5–L6				ΔT				FE (%)			
Base system	base	+MM	+Cog	+CC	base	+MM	+Cog	+CC	base	+MM	+Cog	+CC	base	+MM	+Cog	+CC
claude-opus-5.5	0.636	0.657	0.668	0.665	0.540	0.564	0.580	0.577	−0.113	−0.097	−0.131	−0.147	0.0	0.0	12.3	8.7
gpt-6-astra	0.629	0.651	0.663	0.652	0.535	0.558	0.573	0.563	−0.106	−0.097	−0.117	−0.128	0.0	0.0	11.6	7.8
claude-fable-5.1	0.616	0.642	0.649	0.646	0.514	0.543	0.553	0.550	−0.111	−0.089	−0.133	−0.149	0.0	0.0	9.1	6.2
claude-5-opus	0.540	0.571	0.587	0.583	0.395	0.433	0.455	0.451	−0.146	−0.123	−0.179	−0.193	0.0	0.0	8.7	5.6
gpt-5.6-sol	0.515	0.545	0.566	0.554	0.356	0.396	0.421	0.410	−0.012	+0.009	−0.038	−0.043	0.0	0.0	0.0	0.0
qwen3.8-max	0.482	0.519	0.538	0.526	0.307	0.352	0.380	0.370	−0.092	−0.068	−0.123	−0.141	0.0	0.0	0.0	0.0
gemini-3.8-flash	0.450	0.490	0.509	0.497	0.254	0.307	0.332	0.322	+0.004	+0.030	−0.030	−0.040	0.0	0.0	0.0	0.0
deepseek-v4-pro	0.430	0.473	0.493	0.482	0.170	0.233	0.262	0.252	−0.130	−0.085	−0.173	−0.191	0.0	0.0	0.0	0.0
kimi-k3	0.425	0.465	0.488	0.476	0.212	0.268	0.300	0.289	−0.070	−0.036	−0.111	−0.118	0.0	0.0	0.0	0.0
glm-5.3-flash	0.414	0.456	0.477	0.466	0.197	0.259	0.286	0.276	+0.018	+0.051	−0.014	−0.038	0.0	0.0	0.0	0.0

Frontier emergence. The FE column asks whether an interaction produces a substantive intellectual advance. Each assistant response receives a binary judgment from the dialogue up to that point: a positive turn must introduce or substantially develop a nontrivial insight, mechanism, construction or intervention whose contribution is supported by the argument or a discriminating test. Repeating a prior contribution does not earn another positive judgment. FE is reported separately from L1–L6 because sustaining or directing an interaction does not by itself establish such an advance. Appendix B specifies the criterion and the complete assessment prompt.

The reported pattern separates gains in interactive competence from frontier emergence. Base models and MetaMind have zero FE throughout the roster, despite MetaMind's improvements in proactive promotion. Nonzero values occur only with Claude Code and TRHI, and only on Opus 5.5, Opus 5, Fable and Astra 6. Thus an architecture's improvement in the six-level rubric does not imply that it produces breakthrough contributions; the latter also distinguishes the underlying models on which the architecture operates. The reported Claude Code rates

range from 5.6% to 8.7%. The TRHI increase is 3.55 percentage points on average across these four bases, with the largest increase on Astra 6. These are also the four bases strongest in active adaptation before adding an architecture. The concentration differs from the broad gains in L5–L6, which are largest on weaker bases: architectures improve interactive competence throughout the roster, while frontier emergence remains concentrated among the stronger underlying systems.

Table 19: Mean gain from each agent over its base model, averaged across the ten base systems: by stage of the capability progression, in ΔT, and by level of the interaction-purpose facet.

		stage				interaction purpose		
Architecture	composite	L1–L2	L3–L4	L5–L6	ΔT	functional	mixed	nonfunctional
MetaMind	+0.033	+0.012	+0.026	+0.043	+0.025	+0.021	+0.032	+0.046
Cognizer	+0.050	+0.012	+0.038	+0.066	−0.029	+0.061	+0.056	+0.032
Claude Code	+0.041	+0.016	+0.024	+0.058	−0.043	+0.062	+0.042	+0.019

Groups beyond the dyad. The judge scores each participant on the same levels whatever the number of participants, so the protocol extends to groups of three or more systems (Appendix A.10). A group interaction differs from a dyad in that each turn must be judged against several partners, and the latent state and event chain must represent several relationships and goals at once. We first held the other members fixed, letting the standard user play every other seat (Table 20). Moving from two to four participants lowers the passive and interactive levels (L1–L4) by 0.030 on average and proactive promotion by 0.105, more than three times as much, and long-horizon coherence suffers most: the mean ratio L6/L5 falls from 0.892 to 0.595. The decline is steep for systems that were already weak at the proactive stage and shallow for those that were strong. The three leading systems keep 85–87% of their proactive promotion in a four-person group; `deepseek-v4-pro`, `kimi-k3` and `glm-5.3-flash` keep between a quarter and two fifths of theirs; across the roster, the share retained correlates with the dyadic level at +0.97. For `deepseek-v4-pro`, L6/L5 falls to 0.031, a ratio that is unstable when L5 is small. The range of proactive promotion across systems grows from 0.370 to 0.425, and that of L6/L5 from 0.133 to 0.819.

The larger loss at L5–L6 locates the main burden of additional participants in proactive promotion. A proposed course must accommodate more objectives and reactions, and later actions must reconcile the commitments accumulated across those exchanges. The decline in L6 relative to L5 is particularly informative: the difficulty increases not only in finding a plausible next step but in coordinating its longer contribution.

In groups of mixed composition, the interaction residual, which combines the effect of pairing with noise, grows with group size (Table 21). The share of variance left after removing the main effects of all members rises from 10.9% in dyads to 13.9% in balanced triads and 16.2% in tetrads. Triads whose members come from the same model family show a larger residual (15.1%, against 13.9% for balanced triads), consistent with the self-play result of §4.5.1; with 40 same-family triads and no interval, this does not confirm it.

Table 20: Group size and the capability progression. Each system is scored as one member of a group whose other members are all played by the standard user (`claude-5-opus`). L1–L4 combines passive execution and interactive adaptation, which are reported jointly. The final row gives the mean over the ten systems.

	L1–L4			L5–L6			L6/L5		
System	2	3	4	2	3	4	2	3	4
claude-opus-5.5	0.758	0.746	0.739	0.540	0.497	0.468	0.914	0.888	0.836
gpt-6-astra	0.751	0.740	0.732	0.535	0.492	0.460	0.910	0.899	0.827
claude-fable-5.1	0.748	0.733	0.731	0.514	0.465	0.438	0.906	0.884	0.850
claude-5-opus	0.724	0.704	0.695	0.395	0.338	0.298	0.894	0.827	0.736
gpt-5.6-sol	0.717	0.696	0.689	0.356	0.293	0.250	0.887	0.822	0.726
qwen3.8-max	0.705	0.684	0.673	0.307	0.242	0.191	0.881	0.772	0.634
gemini-3.8-flash	0.697	0.679	0.662	0.254	0.181	0.132	0.865	0.707	0.548
deepseek-v4-pro	0.740	0.719	0.704	0.170	0.095	0.043	0.832	0.500	0.031
kimi-k3	0.691	0.667	0.654	0.212	0.135	0.080	0.965	0.806	0.508
glm-5.3-flash	0.685	0.660	0.645	0.197	0.117	0.067	0.868	0.635	0.258
mean	0.722	0.703	0.692	0.348	0.286	0.243	0.892	0.774	0.595

Table 21: Groups of mixed composition, with members drawn in rotation from the ten systems. The interaction residual, which includes measurement noise, is the share of variance left after the main effects of all members are removed; the dyad row is the matrix of Table 11.

Configuration	group size	groups	mean composite	interaction residual
Dyad	2	100	0.478	10.9%
Triad, balanced sampling	3	120	0.438	13.9%
Triad, same-family members	3	40	0.460	15.1%
Tetrad, balanced sampling	4	90	0.411	16.2%

4.9 Validity of the instrument

Human review of the records and judgments was presented with the measurement protocol in §3.7. The remaining analyses examine how the results depend on the judge, extraction model and input format (§4.9.1), the construction of the composite and adaptive stopping (§4.9.2), and the components of the instrument (§4.9.3). Together, these checks distinguish interpretability of individual judgments from stability of the system comparisons built from them.

4.9.1 Robustness and statistical uncertainty

All results above come from one configuration of the pipeline, with a single judge, a single model extracting the descriptive records and a single serialization of the input. Three of the controls below each vary one of these choices and hold the others fixed; the fourth repeats the whole configuration. The controls assess whether the ordering of systems is preserved, and changes in absolute scores are reported alongside. None of the four controls changes the ordering by more than one rank, and every change that does occur involves systems the reference configuration separates by at most 0.013. The section closes with the case-level intervals and the variance decomposition that the estimation of §4.1 yields.

Test–retest. All composites in this chapter are means over three independent passes through the cube, and Table 22 reports the passes separately. Every system occupies the same rank in all three, Spearman's ρ between any two passes is 1.000, and the mean absolute change in the composite between passes is about 0.003. The largest standard deviation across passes for any system is 0.0032, and the smallest gap between systems adjacent in the ordering is 0.005, between `deepseek-v4-pro` and `kimi-k3`, so even the closest pair is separated by more than the run-to-run variation of either member. Reproducibility across passes is a weaker property than separation across cases: `deepseek-v4-pro` and `kimi-k3` keep their order in all three passes, but the case-level interval for their difference includes zero (§4.9.1).

Table 22: Test–retest reliability over three independent passes of the full cube: composite per pass, mean, sample standard deviation, and the rank each system occupies in the three passes.

System	pass 1	pass 2	pass 3	mean	SD	ranks
claude-opus-5.5	0.635	0.635	0.638	0.636	0.0017	1 / 1 / 1
gpt-6-astra	0.631	0.627	0.629	0.629	0.0020	2 / 2 / 2
claude-fable-5.1	0.615	0.620	0.614	0.616	0.0032	3 / 3 / 3
claude-5-opus	0.543	0.540	0.538	0.540	0.0025	4 / 4 / 4
gpt-5.6-sol	0.512	0.518	0.514	0.515	0.0031	5 / 5 / 5
qwen3.8-max	0.480	0.482	0.485	0.482	0.0025	6 / 6 / 6
gemini-3.8-flash	0.448	0.451	0.451	0.450	0.0017	7 / 7 / 7
deepseek-v4-pro	0.430	0.433	0.427	0.430	0.0030	8 / 8 / 8
kimi-k3	0.423	0.427	0.425	0.425	0.0020	9 / 9 / 9
glm-5.3-flash	0.417	0.413	0.411	0.414	0.0031	10 / 10 / 10

Judge. Using a single judge removes judge identity as a source of differences between systems, but the ordering may still depend on which judge was chosen. We therefore re-scored all episodes of the full design with three alternative judges, keeping transcripts, scenes and rubric unchanged (Tables 23 and 24). The ordering of the ten systems is closely preserved: Spearman's ρ with the reference ordering is 0.976 to 1.000, `gemini-3.8-pro` reproduces it exactly, and no system moves by more than one rank under any judge. The rank changes involve three pairs that the reference judge separates by at most 0.013: the second and third systems, which only `qwen3.8-max` reverses, and two pairs among the three systems within 0.016 of one another (§4.2.1). Two of these three differences are resolved by the case-level intervals under the reference judge (§4.9.1), yet their order depends on the judge. The two leading systems keep their order under all four judges. `qwen3.8-max` as judge scores every system higher, by

0.031 on average, while leaving the order nearly unchanged, so judge leniency acts mostly as an offset.

Each of the three alternative judges raises the composite of its own family. Relative to the remaining systems, `gpt-6-astra` as judge raises its own composite by 0.013, `qwen3.8-max` raises its own by 0.011, and `gemini-3.8-pro` raises `gemini-3.8-flash` by 0.017. The reference judge belongs to the same family as three systems on the roster. If it favored them to a similar degree, those three should fall relative to the others under every alternative judge; averaged over the three alternative judges, they change by +0.004 (`claude-fable-5.1`), 0.000 (`claude-opus-5.5`) and −0.003 (`claude-5-opus`), with no consistent sign. The shifts of the reference judge's family are within ±0.004, below the smallest difference between adjacent systems that the case-level intervals resolve (0.007), so that any family advantage under the reference judge is small and leaves the order intact. Tables 22–27 summarize these controls.

Table 23: Composite score and rank under four judges, on all episodes of the full design with transcripts, scenes and rubric unchanged. Claude Opus 5 is the reference judge used throughout the paper.

	Claude Opus 5		gpt-6-astra		gemini-3.8-pro		qwen3.8-max	
Scored system	composite	rank	composite	rank	composite	rank	composite	rank
claude-opus-5.5	0.636	1	0.634	1	0.633	1	0.667	1
gpt-6-astra	0.629	2	0.631	2	0.630	2	0.646	3
claude-fable-5.1	0.616	3	0.619	3	0.620	3	0.647	2
claude-5-opus	0.540	4	0.527	4	0.547	4	0.564	4
gpt-5.6-sol	0.515	5	0.507	5	0.511	5	0.539	5
qwen3.8-max	0.482	6	0.460	6	0.498	6	0.523	6
gemini-3.8-flash	0.450	7	0.440	7	0.470	7	0.488	7
deepseek-v4-pro	0.430	8	0.415	8	0.443	8	0.458	9
kimi-k3	0.425	9	0.401	10	0.421	9	0.469	8
glm-5.3-flash	0.414	10	0.408	9	0.415	10	0.448	10

Table 24: Judge sensitivity: agreement of each alternative judge with the reference judge on the ordering of the ten systems (Table 23).

Judge model	Spearman ρ	Kendall τ	mean $\lvert\Delta q\rvert$	max rank shift
gpt-6-astra	0.988	0.956	0.010	1
gemini-3.8-pro	1.000	1.000	0.007	0
qwen3.8-max	0.976	0.911	0.031	1

Extraction model. The latent-state and event-chain records are themselves produced by a model. With the judge held fixed, we regenerated these records with two alternative extraction models and compared both the records and the resulting scores (Table 25). The comparison separates sensitivity in the descriptive layer from sensitivity in the grades, and the two behave differently. Event-chain records agree with the reference extraction on 0.936 and 0.954 of items, latent-state records on only 0.806 and 0.816, which reproduces the ordering of reliability found in the human review (§3.7). The downstream composites change little: the mean absolute change is 0.003 to 0.004, Spearman's ρ is 0.988 for both extractors, and the largest rank change is one place. Whether the latent state changes grades at all is tested by the latent-state ablation of §4.9.3. The judge never reads numeric degrees and judges state progression from behavior in the dialogue (§3.5.2), so the less reliable of the two descriptive components may have the least direct influence on the score.

Table 25: Extraction model: agreement of the latent-state and event-chain records with the reference extraction, averaged over the ten systems, and the resulting change in downstream scores, with the judge fixed at Claude Opus 5.

Extraction model	latent agree.	event-chain agree.	$\rho(q)$	mean $\lvert\Delta q\rvert$	max rank shift
qwen3.8-max	0.806	0.936	0.988	0.003	1
gemini-3.8-pro	0.816	0.954	0.988	0.004	1

Input format. The surface form of a judge's input can by itself change rankings (Sclar et al., 2024). Keeping content fixed, we varied the serialization of scenes, and of the profiles where they enter generation and engagement, in five ways: random field order, a JSON container, a YAML container, natural-language prose and perturbed whitespace (Tables 26 and 27). Shuffling fields and perturbing whitespace change no rank. The three variants that change the container each swap one pair, always `deepseek-v4-pro` and `kimi-k3`, whose reference composites

differ by 0.005. The largest effect on absolute scores comes from prose, which lowers the composite of nine systems out of ten and moves it by 0.012 on average. Because serialization was varied wherever scenes and profiles enter, the prose effect cannot be assigned to the judge alone; one possibility is that constraints are harder to check in continuous text than in labeled fields. This sensitivity is the reason for fixing and versioning a structured serialization (§3.2.3).

Table 26: Composite under five serializations of scenes and profiles, content held fixed, beside the reference format (structured key–value pairs in a fixed field order). Rank changes relative to the reference are marked with an asterisk.

System	reference	shuffled fields	JSON	YAML	prose	whitespace
claude-opus-5.5	0.636	0.630	0.649	0.633	0.616	0.632
gpt-6-astra	0.629	0.630	0.632	0.626	0.615	0.632
claude-fable-5.1	0.616	0.615	0.617	0.621	0.615	0.613
claude-5-opus	0.540	0.542	0.545	0.533	0.529	0.540
gpt-5.6-sol	0.515	0.508	0.519	0.515	0.493	0.513
qwen3.8-max	0.482	0.469	0.492	0.479	0.469	0.482
gemini-3.8-flash	0.450	0.453	0.438	0.451	0.443	0.446
deepseek-v4-pro	0.430	0.435	0.429*	0.423*	0.415*	0.431
kimi-k3	0.425	0.416	0.432*	0.433*	0.437*	0.431
glm-5.3-flash	0.414	0.416	0.416	0.415	0.405	0.411

Table 27: Format sensitivity: agreement of each serialization with the reference format on the ordering of the ten systems (Table 26). Ties in the shuffled and whitespace variants lower ρ slightly below one without any change of rank.

Format variant	Spearman ρ	mean $\lvert\Delta q\rvert$	max rank shift
Shuffled field order	0.994	0.005	0
JSON container	0.988	0.006	1
YAML container	0.988	0.004	1
Natural-language prose	0.985	0.012	1
Whitespace and indentation	0.994	0.003	0

Figure 16 summarizes the score changes and ordering stability across these controls.

Uncertainty of the main results. Each system's composite is estimated, by the bootstrap of §4.1, to within ±0.007 to ±0.008 (Table 28). Because the systems are compared on the same cases, their differences are estimated more precisely than the intervals of the individual systems suggest, and every difference between adjacent systems is resolved except that between `deepseek-v4-pro` and `kimi-k3` (0.005, interval −0.003 to 0.013). The two leading systems differ by 0.007 (0.002 to 0.012) under the reference judge and keep this order under all four judges. The self-play excess is +0.020 (+0.012 to +0.028). In the mixed-effects model, apart from the residual (34.2% of the random variance), the largest random components are the case, with 27.2%, and its interaction with the scored system, with 19.6%, so cases differ in difficulty and also in which systems they favor. The partner accounts for 12.0% and the pairing of system and partner for 7.0%, more than half as much as the partner itself. On the scale of the cell means the pairing accounts for 9.9% of the variance of scored system, partner and pairing together, slightly less than the residual of 10.9% in the matrix, which also contains noise and the self-play excess (§4.5.1).

4.9.2 Sensitivity of the composite

Two analyses examine how far the results depend on the weighting of the composite and on adaptive stopping (Appendix A.9); their results are reported in Tables 30–32.

Construction of the composite. Three features of the composite give some observations more weight than others (Appendix A.9): the level weights and the cascade, which emphasize the proactive stage, and the parameter α, which emphasizes the final turns. Some of the separation of systems at the proactive end may come from these choices. Every system's composite is therefore recomputed from the stored records, without new generation or judging, under the following variants: raw level scores without the cascade; equal weights on the cascaded scores; equal weights on the raw scores; and $\alpha \in \{0.5, 0.7, 0.9\}$ together with the limit $\alpha \to 1$, the unweighted mean of the turn scores. For each variant we report Spearman's ρ with the reference ordering, the largest rank change, and the range across systems of the composite and of each stage (Table 30). Comparing the stage ranges under raw and cascaded scores tests this directly: raw L5 and L6 grades come from isolated level calls and are not capped by the lower levels, so a spread that is present in the raw grades cannot originate in the cascade.

No variant changes the ordering by more than one adjacent pair ($\rho \geq 0.988$; Table 30). The spread at the proactive

Table 28: Case-level paired bootstrap, 95% percentile intervals from 10,000 resamples of the one-sided protocol (standard user; 810 cases × 3 passes per system), resampling cases within each of the 27 cells. Upper panel: composite with its interval, and intervals for the stage means, by system. Lower panel: differences between systems adjacent in the ordering, all pairs among the three systems within 0.016 on the composite, and the self-play excess of the two-sided matrix (case resampling only).

System	composite	CI	L1–L2	L3–L4	L5–L6
claude-opus-5.5	0.636	[0.629, 0.643]	[0.784, 0.790]	[0.725, 0.735]	[0.529, 0.551]
gpt-6-astra	0.629	[0.622, 0.636]	[0.778, 0.784]	[0.717, 0.727]	[0.524, 0.546]
claude-fable-5.1	0.616	[0.609, 0.623]	[0.777, 0.783]	[0.711, 0.721]	[0.503, 0.525]
claude-5-opus	0.540	[0.533, 0.548]	[0.763, 0.769]	[0.677, 0.687]	[0.383, 0.407]
gpt-5.6-sol	0.515	[0.507, 0.523]	[0.765, 0.771]	[0.661, 0.671]	[0.344, 0.368]
qwen3.8-max	0.482	[0.474, 0.490]	[0.756, 0.762]	[0.645, 0.657]	[0.295, 0.319]
gemini-3.8-flash	0.450	[0.442, 0.458]	[0.756, 0.762]	[0.630, 0.642]	[0.243, 0.265]
deepseek-v4-pro	0.430	[0.422, 0.438]	[0.784, 0.790]	[0.688, 0.698]	[0.160, 0.180]
kimi-k3	0.425	[0.418, 0.433]	[0.751, 0.757]	[0.622, 0.634]	[0.201, 0.223]
glm-5.3-flash	0.414	[0.407, 0.421]	[0.749, 0.755]	[0.613, 0.625]	[0.187, 0.207]

Contrast	estimate	95% CI
claude-opus-5.5 − gpt-6-astra	0.007	[0.002, 0.012]
gpt-6-astra − claude-fable-5.1	0.013	[0.007, 0.019]
claude-fable-5.1 − claude-5-opus	0.076	[0.068, 0.084]
claude-5-opus − gpt-5.6-sol	0.025	[0.018, 0.032]
gpt-5.6-sol − qwen3.8-max	0.033	[0.026, 0.040]
qwen3.8-max − gemini-3.8-flash	0.032	[0.025, 0.040]
gemini-3.8-flash − deepseek-v4-pro	0.020	[0.012, 0.028]
deepseek-v4-pro − kimi-k3	0.005	[−0.003, 0.013]
kimi-k3 − glm-5.3-flash	0.011	[0.003, 0.019]
deepseek-v4-pro − glm-5.3-flash	0.016	[0.007, 0.025]
self-play excess	+0.020	[+0.012, +0.028]

Table 29: Variance components of the mixed-effects model, fitted to the episode composites of the two-sided protocol (10 × 10 pairs × 810 cases × 3 passes). Share is the proportion of the total random variance; the fixed effects are the scored system, the pass, the facet levels and the self-play indicator. The last row restates the pairing variance on the scale of the matrix of cell means, relative to the variance of the scored system (fixed), the partner and the pairing together, for comparison with the 10.9% residual of the interaction matrix.

Component	variance	share
partner υ_p	0.0019	12.0%
case ξ_i	0.0043	27.2%
system × partner $(\beta\upsilon)_{mp}$	0.0011	7.0%
system × case $(\beta\xi)_{mi}$	0.0031	19.6%
residual ε	0.0054	34.2%
fixed effects (marginal R^2)	0.399	
pairing / (system + partner + pairing)	9.9%	

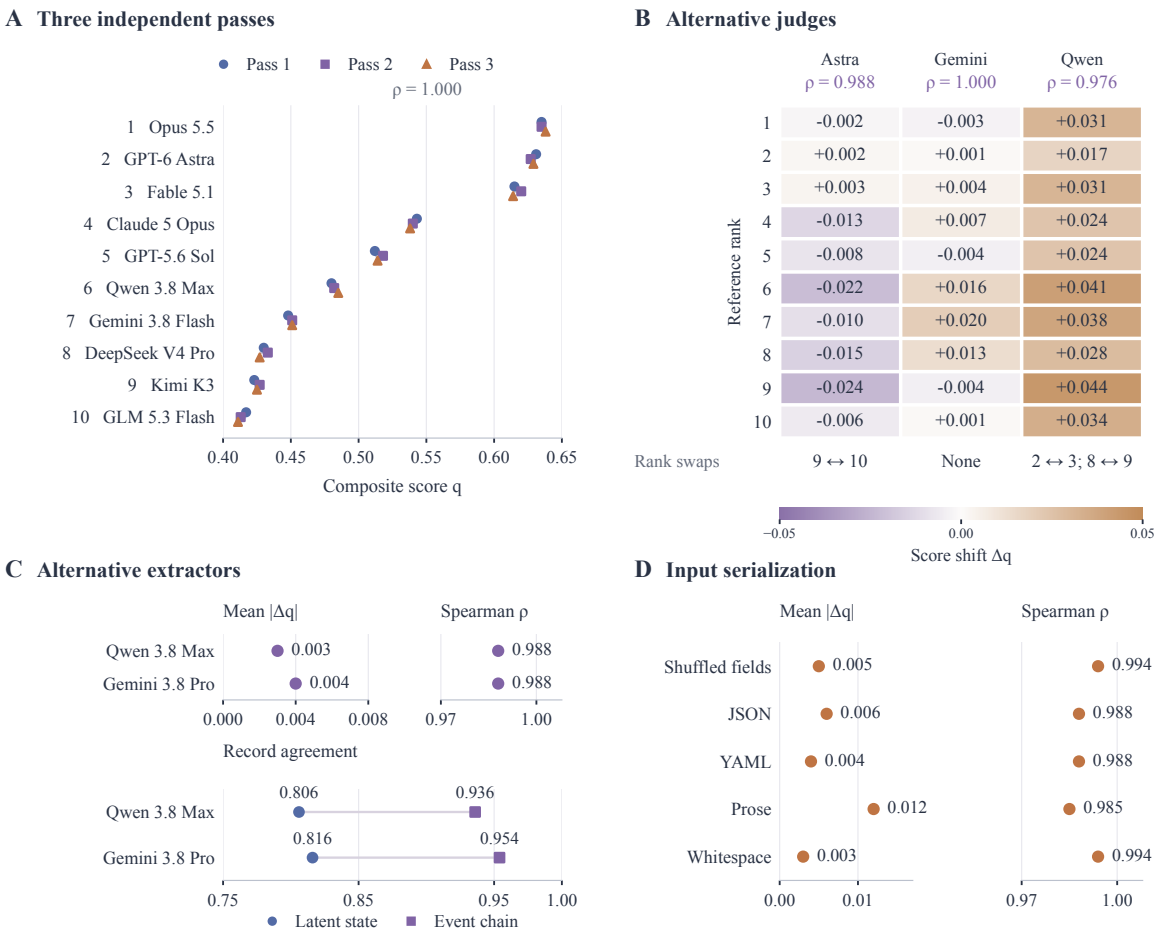


Figure 16: Robustness of the scoring pipeline. (a) Composite scores in three independent passes; each system retains its rank, with pairwise Spearman $\rho = 1.000$ (Table 22). The score axis spans 0.40–0.65. (b) Signed per-system score changes under alternative judges relative to Claude Opus 5, with rank correlations and exact adjacent-rank exchanges shown separately (Tables 23 and 24). Row numbers follow the reference ordering in (a). Astra exchanges ranks 9/10; Gemini has no exchanges; Qwen exchanges 2/3 and 8/9. (c) Alternative extractors with the judge fixed: mean absolute score change, rank correlation, and latent-state/event-chain record agreement use separate axes (Table 25). Both extractors have a maximum rank shift of one. (d) Input-format sensitivity: mean absolute score changes and rank correlations (Tables 26 and 27). JSON, YAML and prose exchange reference ranks 8/9; shuffled fields and whitespace introduce ties without exchanges. Summary values are taken from the corresponding tables; signed changes in (b) are computed from the displayed composites. No uncertainty intervals are inferred from these means.

end is largely present in the raw grades. Without the cascade the range of proactive promotion falls from 0.370 to 0.341, so the cascade contributes less than a tenth of it, and the widening from passive execution (0.034) through interactive adaptation (0.104) to proactive promotion (0.341) holds in grades that no level caps. Equal level weights narrow the range of the composite from 0.222 to 0.163 and leave the stage ranges unchanged, as the construction requires. A smaller α widens every range, to 0.391 at the proactive stage for $\alpha = 0.5$, and the unweighted mean narrows them, to 0.334: differences between systems grow over the course of an episode (Table 32), and a smaller α places more weight on its later turns.

Adaptive stopping. Because the termination rule decides when an episode ends, systems differ in how many turns they are scored on and in the state in which their episodes end, as well as in how well they perform at each turn, and the final turns carry most of the weight of q. Two analyses separate performance from stopping.

The first uses the turn-level records that the instrument keeps for every episode. For each system and assistant turn t we report the mean turn score $\bar{c}_t$ over the episodes that reach turn t, and the fraction of episodes that reach it. Systems are compared at matched turns, and at early turns, before stopping has removed many episodes, the comparison is nearly free of selection by length (Table 32).

The second reruns a stratified subsample of the cube, balanced over the 27 cells, with the termination rule disabled and a fixed length of 8, 16 or 32 exchanges. The closure query and the interest and patience readings are still recorded, and a red line still sets the turn score to zero and lowers interest, but exhaustion of interest no longer ends

Table 30: Sensitivity of the composite to its construction. The reference uses cascaded scores, level weights ℓ and $\alpha = 0.8$. Ranges are max–min across the ten systems. Stage means use the same turn weights as the composite, so the level weights change only the composite.

Variant	ρ	max shift	range q	range L1–L2	range L3–L4	range L5–L6
Reference	1.000	0	0.222	0.035	0.111	0.370
Raw scores, level weights	0.988	1	0.205	0.034	0.104	0.341
Cascaded scores, equal weights	0.988	1	0.163	0.035	0.111	0.370
Raw scores, equal weights	0.988	1	0.152	0.034	0.104	0.341
$\alpha = 0.5$	0.988	1	0.238	0.037	0.118	0.391
$\alpha = 0.7$	1.000	0	0.228	0.036	0.114	0.379
$\alpha = 0.9$	1.000	0	0.213	0.034	0.107	0.356
$\alpha \to 1$ (unweighted mean)	0.988	1	0.197	0.032	0.101	0.334

the episode. Because the decision to stop is the only signal the instrument returns to the participants (§3.3), the prefix of a fixed-length episode up to the turn where the adaptive rule would have stopped has the same distribution as an adaptive episode. We therefore also compute the adaptive composite on the same subsample and compare it with the fixed-length composites case by case; episodes that the adaptive rule would continue beyond the fixed length are censored in this comparison. Turns that follow a successful closure in a fixed-length episode are scored like any other turn, so the fixed-length composite also registers how a system behaves after its objective has been met. Table 31 reports the distribution of episode lengths, the reasons for stopping, and the composite under adaptive and fixed lengths.

Median episode length runs from seven assistant turns for `glm-5.3-flash` to twelve for the two leading systems and broadly follows the ordering of the composite. Turn by turn, the differences between systems are smallest at the first turn, before stopping has removed any episode, and grow steadily among the episodes that reach each turn, from a range of 0.109 at $t = 1$ to 0.204 at $t = 16$ (Table 32). At $t = 1$ the five leading systems already stand in the order of the composite. `deepseek-v4-pro` loses most within an episode: its mean turn score falls from 0.598 at the first turn to 0.459 at the sixteenth, against a fall from 0.668 to 0.642 for `claude-opus-5.5`. Since exhaustion removes episodes in which interest is failing, these declines probably understate those that the episodes stopped earlier would have shown.

On the fixed-length subsample, the composite of the adaptive episodes lies within 0.003 of its value on the full design, so the subsample reproduces the composites of the full design. Fixed lengths of 8 and 16 exchanges leave the ordering unchanged. At 32 exchanges the only change is that `kimi-k3` moves ahead of `deepseek-v4-pro` (0.380 against 0.358) and, by 0.001, of `gemini-3.8-flash` (0.379), which is consistent with its small loss between L5 and L6 (§4.2), a reading made after the fact. The range across systems is larger under every fixed length than under adaptive stopping, 0.242 to 0.267 against 0.223, because the weaker systems lose more over a long interaction: 0.070 for `glm-5.3-flash` between the adaptive episode and 32 exchanges, against 0.026 for `claude-opus-5.5`. Adaptive stopping therefore compresses the differences between systems slightly, and fixed lengths of up to 16 exchanges reproduce the ordering it reports.

Table 31: Episode length, stopping and fixed lengths. Length is the number of evaluated assistant turns under adaptive stopping (median and interquartile range). The last four columns are computed on the fixed-length subsample with the same turn weights ($\alpha = 0.8$): *adaptive* on the prefix of each episode up to and including the turn at which the adaptive rule would have stopped, with no later turn entering, and 8, 16 and 32 on the full fixed-length episodes.

	length					subsample (fixed length)			
System	median	IQR	closure	exhaustion	full design	adaptive	8	16	32
claude-opus-5.5	12	8–18	12.4%	87.6%	0.636	0.634	0.641	0.628	0.608
gpt-6-astra	12	8–18	10.5%	89.5%	0.629	0.631	0.637	0.621	0.599
claude-fable-5.1	11	7–17	10.5%	89.5%	0.616	0.613	0.617	0.603	0.584
claude-5-opus	10	7–15	11.8%	88.2%	0.540	0.543	0.543	0.524	0.497
gpt-5.6-sol	10	6–14	4.5%	95.5%	0.515	0.512	0.508	0.490	0.461
qwen3.8-max	9	6–14	9.1%	90.9%	0.482	0.485	0.479	0.457	0.424
gemini-3.8-flash	8	5–12	2.0%	98.0%	0.450	0.447	0.437	0.414	0.379
deepseek-v4-pro	8	5–12	9.4%	90.6%	0.430	0.433	0.424	0.404	0.358
kimi-k3	9	6–13	5.4%	94.6%	0.425	0.422	0.416	0.401	0.380
glm-5.3-flash	7	5–11	1.0%	99.0%	0.414	0.411	0.399	0.381	0.341
range					0.222	0.223	0.242	0.247	0.267

Table 32: Turn-indexed analysis under adaptive stopping. Each cell gives the mean turn score $\bar{c}_t$ over the episodes that reach assistant turn t, with the percentage of episodes reaching t in parentheses.

System	$t = 1$	$t = 2$	$t = 4$	$t = 8$	$t = 16$
claude-opus-5.5	0.668 (100)	0.675 (100)	0.671 (96)	0.656 (77)	0.642 (31)
gpt-6-astra	0.664 (100)	0.670 (100)	0.665 (96)	0.650 (76)	0.634 (30)
claude-fable-5.1	0.657 (100)	0.661 (100)	0.654 (95)	0.636 (72)	0.619 (28)
claude-5-opus	0.621 (100)	0.617 (100)	0.598 (94)	0.571 (69)	0.552 (23)
gpt-5.6-sol	0.605 (100)	0.599 (99)	0.579 (92)	0.546 (65)	0.523 (19)
qwen3.8-max	0.592 (100)	0.583 (99)	0.559 (91)	0.521 (60)	0.497 (17)
gemini-3.8-flash	0.574 (100)	0.563 (99)	0.533 (88)	0.487 (52)	0.462 (13)
deepseek-v4-pro	0.598 (100)	0.584 (99)	0.548 (89)	0.490 (51)	0.459 (13)
kimi-k3	0.566 (100)	0.556 (99)	0.530 (90)	0.494 (57)	0.478 (15)
glm-5.3-flash	0.559 (100)	0.546 (98)	0.514 (86)	0.463 (43)	0.438 (10)
range	0.109	0.129	0.157	0.193	0.204

4.9.3 Ablations of the instrument

Each component of the instrument was introduced to meet a requirement of the theory (§2.4), and an ablation asks whether it meets that requirement. A change in the scores after a component is removed shows that the component affects the measurement. Whether the measurement is more valid with the component has to be decided against a criterion that the component does not itself produce. For every ablation we therefore state in advance the requirement the component serves, the failure its removal should produce, and the criterion on which that failure is assessed (Table 33). Three criteria are used, none of which takes the composite or new human grading as its standard.

The first criterion is the separation of probes whose correct outcome follows from the definitions. A probe pairs an original transcript with a version altered in one controlled respect, and the definitions of §3.4 and §3.5.1 fix the direction in which the affected grade must move and leave the other grades unchanged. Four families of probes are used, for the latent state, the event chain, the isolation of level calls and the significance gate, and a probe is separated when the instrument's grades move as the definitions require (Appendix A.11).

The second criterion is internal coherence: the share of records that violate a property the definitions require of every valid record, such as a grade resting on a quotation absent from the transcript, a state change larger than its significance class permits, or a rule violation recorded against a rule that an admitted revision had changed. For the quotation check, the violation occurs by construction once the component is removed, so the incoherence share measures only how many grades the check changes, and its benefit is judged by reliability.

The third criterion is reliability. Each ablated configuration grades the same transcripts in two independent grading runs, and the agreement between the runs is compared with that of the full instrument. A component that constrains the judge, such as isolated level calls or the quotation check, should raise agreement between runs; for a component that adds information to the judge, reliability is reported without a directional expectation.

Removing either component of the reconstruction lowers the separation of the probes designed for it (Table 34). Without the latent-state descriptions the judge separates 68% of the state probes, against 91% for the full instrument, and without the event chain 57% of the recurrence probes, against 89%. A judge that reads only the transcript and the scene, with the same criteria, separates 64% and 49%, fewer than either single ablation, so the two components contribute partly independent information. The same configuration is the least reliable (0.826, against 0.871), would have stopped 6.1% of episodes at another turn, and changes the ordering more than any other ablation ($\rho = 0.964$). The isolation of level calls also matters. A single call for all six levels separates 61% of the single-level probes, against 90%, so that an error at one level propagates into the grades of other levels, and lowers reliability to 0.833. Without the quotation check, 5.1% of level grades rest on evidence the check would have rejected, and reliability falls to 0.847.

The rules of the reconstruction change the record but hardly the ranking. Without the significance gate, 17.6% of slot changes exceed the class committed for their turn and 46% of the quiet-turn probes are separated, against 95%, while the ordering is unchanged ($\rho = 1.000$, $|\Delta q| = 0.005$). The judge receives descriptions of the latent state without its degrees (§3.5.2), so an unbounded change of degree reaches the grades only as far as it alters the descriptions. Without scene revision, 22.9% of the turns that follow an admitted revision record a failure at L1 against a rule the revision had changed; revisions are rare, so the composites barely change.

The two termination rules matter for the length and the ending of episodes. Without the patience budget, episodes whose dominant path remains Degrade run for 13.7 turns, against 6.2, and the composite moves about as much as when the latent-state input is removed ($|\Delta q| = 0.014$, against 0.012). Without the viable-path condition, 18.4% of closures are recorded in interactions that are degrading or collapsing. Each ablation reduced the criterion it

was designed to affect (Table 33); for the quotation check and the significance gate the coherence failures follow by construction. Because the probes and the components rest on the same definitions, the results show that the components do what the definitions require and do not test the definitions themselves.

Table 33: Design of the ablations: the component removed, the requirement it serves, the failure its removal is expected to produce, and the criterion on which the failure is assessed.

Ablation	Requirement served	Expected failure	Criterion
Judge inputs (replay)			
No latent-state input	State adequacy (§2.1)	State progression graded on the surface of a turn	State probes
No event-chain input	Temporal continuity (§2.2)	Recurrence graded like progress at L5–L6	Recurrence probes
Transcript and scene only	Reconstruction before adjudication (§2.4)	The judge grades from the transcript and the scene alone, with the same criteria; both failures above	State and recurrence probes
Adjudication rules (replay)			
Joint level call	Isolation of level calls (§3.5.1)	Grades of one level carried over to others	Single-level probes; reliability
No quotation check	Evidential traceability (Eq. (7))	Grades resting on quotations that do not occur in the transcript	Coherence: grades the check would have lowered; reliability
Reconstruction rules (replay)			
No significance gate	Bounded, checkable state change (§3.4.2)	Changes of state out of proportion to the turn that occasioned them	Quiet-turn probes; coherence: changes exceeding the committed class
No scene revision	Admissible scene revision (Eq. (14))	Rule violations recorded against rules a realized event had changed	Coherence: failures at L1 on turns after an admitted revision
Termination rule (fixed-length subsample)			
No patience budget	Memory of accumulated cost (§3.5.4)	Interactions that stagnate without a sharp loss of interest run on	Length of episodes whose dominant path remains Degrade
Closure without viable-path condition	Viable course at closure (Eq. (30))	Closures recorded in interactions that are degrading or collapsing	Share of closures with dominant Degrade or Collapse

4.10 Cost of the instrument

An instrument that reconstructs the interaction before judging it costs more than a judge that reads a reply. The number of model calls follows from the schedule of Appendix A.2. At every subturn, the reconstruction makes two latent passes and one event-chain update; at every assistant subturn, adjudication adds six level calls, the closure query, the red-line screen and the gap annotation, and a crossed red line adds one call that judges the loss of interest. An exchange therefore requires 15 evaluation calls in the one-sided protocol (16 when a red line is crossed), besides the two calls that generate the utterances, and 21 in the two-sided protocol, where the user's utterance receives its own six level calls. A response-level judge with the same criteria needs six calls per assistant turn, or one if it grades all levels at once. Each call reads the history excerpt of up to 12,000 characters, so the input of a call grows with the turn until the excerpt is full. Table 35 reports, by component, the calls per exchange, the mean input and output tokens and the mean latency, together with the totals per episode and per pass through the cube.

An evaluated exchange requires 15 calls, 146,450 input and 13,230 output tokens, and 123 s on the critical path, or 228 s if the calls were made one after another. Level calls account for the largest share of the tokens (37.7%), followed by the latent passes (34.2%) and the event-chain updates (19.3%); the closure query, the red-line screen and the gap annotation together account for 8.9%. The event-chain update is the slowest call (31.0 s), with the longest input and output. Evaluating an exchange costs about seventeen times as many tokens as generating it (159,680 against 9,510). At 9.6 evaluated exchanges, the mean of the median lengths of the ten systems (Table 31), an episode requires 144 evaluation calls, 1.41 million input and 0.13 million output tokens, and one system's pass through the cube requires 116,640 calls, 1.14 billion input and 0.10 billion output tokens, and about 9 hours with 32 episodes running concurrently. The totals are approximate, since the mean length differs from the median and the input of a call grows with the turn. A judge that grades the same six levels from the transcript and the scene alone makes 6 calls per exchange against 15, reads 5,800 tokens in each call against 9,500, and takes 11 s on the critical path against

Table 34: Results of the ablations. *Probes*: share of probe pairs separated as the definitions require, full instrument → ablated, for the probe family of the ablation (for the transcript-only row, state probes above and recurrence probes below). In the event-chain and transcript-only ablations the dominant path is still computed from the event chain and is unaffected, so recurrence probes are assessed on the L5–L6 grades alone, in both the full and the ablated instrument. *Incoherent*: share of the units named in the next column that violate a property required of every valid record, a share that is zero by construction for the full instrument in every unit other than all records; for the two termination ablations, the criterion of Table 33 (length in turns of episodes whose dominant path remains Degrade, ablated vs. full; share of closures with dominant Degrade or Collapse). *Reliability*: agreement of level grades between two independent grading runs. *Stop changed*: share of episodes in which the ablated instrument would have stopped at another turn. ρ and $|\Delta q|$: Spearman correlation with the ordering of the full instrument and mean absolute change in the composite.

Ablation	probes	incoherent	unit of the incoherent share	reliability	stop changed	ρ	$\lvert\Delta q\rvert$
Full instrument	0.89–0.95	0.003	all records	0.871	—	1.000	—
No latent-state input	0.91 → 0.68	0.004	all records	0.858	2.3%	0.988	0.012
No event-chain input	0.89 → 0.57 0.91 → 0.64	0.005	all records	0.852	3.6%	0.976	0.018
Transcript and scene only	0.89 → 0.49	0.007	all records	0.826	6.1%	0.964	0.031
Joint level call	0.90 → 0.61	0.006	all records	0.833	0.7%	0.976	0.022
No quotation check	—	0.051	level grades	0.847	1.4%	0.988	0.010
No significance gate	0.95 → 0.46	0.176	latent-slot changes	0.864	4.4%	1.000	0.005
No scene revision	—	0.229	turns after an admitted revision	0.869	1.2%	1.000	0.004
No patience budget	—	13.7 vs. 6.2	turns of episodes with dominant Degrade	—	—	0.988	0.014
Closure without viable-path condition	—	18.4%	closures	—	—	1.000	0.004

Table 35: Cost of the instrument in the one-sided protocol, by component and per exchange. Tokens and latency are means per call over all exchanges of a pass through the cube with the reference judge and extractor (Claude Opus 5); *share* is the component's share of all evaluation tokens, input and output, generation excluded. Per exchange and per episode, latency is given on the critical path, with the two latent passes of a subturn, the six level calls and the three remaining assistant-subturn calls each run in parallel and the event-chain update waiting on the latent passes, and as the serial sum in parentheses. Totals per episode and per pass assume 9.6 evaluated exchanges per episode, the mean of the ten systems' median lengths (Table 31); the totals are therefore approximate, since the mean length differs from the median and the input of a call grows with the turn. The wall-clock time of the pass is the measured time per exchange under 32 concurrent episodes (about 132 s, generation included) multiplied by the number of exchanges and divided by the 32 concurrent episodes. Failed calls that were retried (1.7% of calls) are excluded from the means. The last row is a judge with the same six criteria that receives only the transcript and the scene.

Component	calls	input tokens	output tokens	latency (s)	share
Generation (both participants)	2	4,450	305	8.4	—
Latent passes	4	12,200	1,460	24.2	34.2%
Event-chain updates	2	13,500	1,890	31.0	19.3%
Level calls	6	9,500	520	9.7	37.7%
Closure query	1	5,060	180	4.0	3.3%
Red-line screen	1	4,420	150	3.6	2.9%
Gap annotation	1	4,170	160	3.7	2.7%
Per exchange (evaluation)	15	146,450	13,230	123 (228)	100%
Per episode	144	1.41M	0.13M	1,181 (2,189)	
Per pass through the cube, one system	116,640	1.14B	0.10B	8.9 h wall clock	
Response-level judge, per exchange	6	5,800	480	11 (57)	

123 s, since adjudication waits for the reconstruction. The probe results of Table 34 indicate what the reconstruction adds: the judge that reads the transcript and the scene alone separates 64% of the state probes and 49% of the recurrence probes, against 91% and 89% for the full instrument.

5 What Interactive AI Evaluation Reveals

Section 4 tested whether the predictions of Section 2 hold. Here we ask which quantities become measurable once the interaction world is represented explicitly, and organize the answer around five regularities. The first three concern initiative: where it separates contemporary systems, how its absence can pass for quality, and why only its warranted form can be credited. The last two concern the interactive character of the measurement itself: endings belong to tasks, and an assistant's score belongs partly to the pair it forms with its partner. Each regularity is a quantity that response-level or single-score evaluation either cannot represent or represents in a way that misreads it systematically (Table 36).

Table 36: Five regularities made measurable by interactive evaluation. The second column states what an evaluation of individual responses, or of a single holistic score, would record; the third states what the instrument records instead; the fourth names the property of the instrument (§6.2) on which the regularity chiefly depends; the fifth gives the section with the evidence.

Regularity	Seen by response-level or single-score evaluation	Measured by the instrument	Property	Evidence
What separates systems is initiative	One ordering of the systems	Spans widen by stage: 0.035 for passive execution, 0.111 for interactive adaptation, 0.370 for proactive promotion; profiles diverge at equal composites	Active	§4.2
Responsiveness can pass for progress	Every turn acceptable	A dominant path that moves from Advance to Maintain or Degrade while passive and interactive levels stay intact	Holistic	§4.4
Initiative requires a warrant	Confident, specific answers read as competence	Partial specifications are hardest; fabrication peaks there and rises with passive-execution score, clarification rises with the gap	Active	§4.3, §4.4
Endings carry the shape of the task	Exhaustion as failure, closure as success	Closure four to more than twenty times more frequent in functional settings	Interactive	§4.4
An assistant's score belongs partly to the pair	A fixed partner, unreported	A residual of 10.9% after partner and system effects, 9.9% once noise and the self-play excess are separated; self-play excess growing with self-similarity	Interactive	§4.5.1, §4.9.1

5.1 What separates systems is initiative

Section 2.3 separates originating a direction from making that direction valuable. The capability profiles locate the largest differences among systems in this latter part of interaction: the span grows about tenfold from passive execution to proactive promotion (§4.2). Systems that execute local requirements similarly differ in the strategies they have reason to pursue and in how coherently they develop those strategies across turns. The result concerns the quality of participant-directed interaction, rather than the frequency of unsolicited moves.

This distinction explains why passive competence and composite rank are insufficient. `deepseek-v4-pro` and `gpt-5.6-sol` show that strong execution does not guarantee strong proactive promotion, while systems close on the composite differ in purpose disposition and their relative attainment at L5 and L6 (§4.2.1). In the notation of §2.4,

$$G\big(R^{A}_{1:T}\big) \approx G\big(R^{B}_{1:T}\big) \qquad \text{while} \qquad J^{A} \neq J^{B}.$$

The two upper levels make this loss of information interpretable: a system may produce a defensible next step without coordinating the course it opens, or maintain a course without giving it sufficient prospective value. The profiles distinguish the ability to identify a worthwhile direction from the ability to develop its contribution.

Architecture and group-size results place the greatest changes in the same part of the progression (§4.8). Architectures improve proactive promotion most on average, whereas additional participants reduce it most. The common pattern points to the demands of establishing and coordinating a course of interaction, beyond the ability to answer its individual turns.

5.2 Responsiveness can pass for progress

A promising move can lose its value during continuation. The episodes in §4.4 make this temporal problem visible: local requirements remain satisfied while the interaction returns to the same unresolved point. The failure is not necessarily that the assistant never proposed a direction; it can also be that its subsequent actions neither update the strategy nor convert it into progress.

This is where the Movie Evaluation Theory and the Agency Principle meet. The former makes the sequence an object of evaluation; the latter asks what the participant contributes to that sequence. L5 concerns why a strategy is worth pursuing at the current decision. L6 concerns how later actions incorporate consequences, respond to the partner and coordinate the developing contribution. Preserving the same goal or tone does not answer that second question. Conversely, revising or abandoning a strategy can be coherent when the trajectory reveals that its original rationale no longer holds.

The human audit (§3.7) illustrates the difficulty of this distinction: some reviewers credited warm and varied replies despite repetition of the same interactional move. Evaluation should therefore retain local response quality and trajectory-level contribution as separate judgments. The cases show why the progress of a strategy must be read across its successive steps.

5.3 Initiative requires a warrant

The information manipulation tests the evidential basis of a strategy. All three systems examined perform worst under partial information (§4.3), where the available specification can support an apparently complete response while leaving a consequential gap. This is an agency problem as well as an information-uptake problem: the assistant must decide which next step has a defensible prospect of improving the interaction.

Clarification and fabrication separate these possibilities. A question that exposes a previously unrecognized gap and makes a consequential next step possible can satisfy the warrant in Eq. (16). Asking is not automatically initiative: a routine request already specified by the partner can remain inside the responsive set. Supplying an unsupported fact, by contrast, cannot acquire warrant merely because it moves the conversation forward. Its apparent usefulness rests on information the assistant does not have.

The observed frequencies reinforce the need for this distinction. Clarification increases with the information withheld, whereas fabrication peaks under partial specifications. `deepseek-v4-pro` fabricates at the fourth-highest rate while having the weakest proactive promotion, so taking an unrequested step is not itself evidence of the capability. L5 must evaluate the reason for expecting the strategy to contribute; L6 must examine whether subsequent interaction makes productive use of the information obtained. The behavior counts diagnose these alternatives but do not replace either judgment. Evaluation should therefore vary completeness and correctness of the brief separately, and retain clarification and fabrication alongside capability scores.

5.4 Endings carry the shape of the task

When episodes end because the interaction succeeds or stops being worth continuing, the way they end becomes a measurement, and what it measures is partly the task. Successful closure is the minority outcome for every system, so ending by exhaustion is common to all of them and does not by itself mark a weak one. Closure is also several times more frequent in functional than in nonfunctional settings, more than twenty times for `gemini-3.8-flash`. A companionship scene poses no objective to be closed, and a closure rate pooled across settings penalizes a relational disposition for not completing objectives it was never given. Outcome measures such as completion and closure reflect the shape of the task as well as the competence of the participant, and they are interpretable only within a level of interaction purpose.

Closure requires that the dominant projected path be an advance or maintenance. The event chain does not attribute that direction to either participant (§3.4.4), so a closure rate is not itself a measure of initiative, and in relational settings proactive promotion may show itself as sustaining an exchange, which a closure rate does not record.

5.5 An assistant's score belongs partly to the pair

The most general regularity concerns the unit to which a score is attributed. In a benchmark with a fixed simulated user the partner is invisible, yet its behavior enters every number the benchmark reports. Scoring both participants makes that contribution measurable. Of the total variation in assistant scores across the 10×10 matrix, 10.9% is explained by neither the partner nor the scored system alone. This residual combines pairing, the self-play excess

and measurement noise. A mixed-effects model that separates them attributes 7.0% of the random variance to the pairing of system and partner, more than half the 12.0% attributed to the partner; on the scale of the cell means, the pairing accounts for 9.9% of the variance of scored system, partner and pairing together. An assistant's score obtained against a single partner is partly a property of the pair, reported as a property of one participant.

This bears first on how a partner should be chosen. A partner that lowers every score, as `gemini-3.8-flash` does, separates the scored systems less than one that probes hard, and it is the spread a partner produces, correlated with its probing intensity at +0.48, that makes it informative. It bears also on what a simulated user is taken to be. Cooperativeness is often treated as a parameter to be set when a simulator is designed; here it is a measured property of each system in the user role, ranging from 0.61 to 0.90, and no system behaves as an adversarial user. Self-evaluation is affected as well. A system paired with itself gains more the more closely its two roles resemble each other (correlation −0.65 between self-similarity and the net gap, the standard-user score minus the self-play score), while self-similarity is only weakly related to overall strength, so protocols that use a model as its own interlocutor tend to favor systems whose partner and character styles coincide. Because partners that probe hard press the scored system at the proactive stage, how strongly the active end of a profile is tested also depends on the partner.

The dependence is not symmetric. In the user role, a system's score is almost entirely its own, with 94.4% of the variance attributable to the system playing the user and 2.0% left after the row and column effects; on the assistant side the corresponding residual, 10.9%, is five times as large. The interaction residual also grows with the number of participants, from 10.9% in dyads to 16.2% in four-person groups, although these figures do not separate noise, whereas the dyadic pairing component is 9.9%; the attribution problem therefore grows as evaluation moves from dyads toward multi-party settings.

A common structure. In each case, what a response-level or single-score evaluation omits is a systematic effect with a direction, and it does not average out as noise would. Three of the five concern initiative. The instrument registers these effects because it keeps description apart from judgment, records the course and the ending of an interaction as well as its turns, and scores both participants.

5.6 Distinct interaction dispositions across systems

`deepseek-v4-pro`, `kimi-k3` and `glm-5.3-flash` have similar composites, but differ in passive execution, continuity and their treatment of missing information (§4.2.1). Table 37 summarizes the main evaluation's assistant and user results for each system. These dispositions describe recurrent conduct and differences between the two roles.

`claude-opus-5.5`, `gpt-6-astra` and `claude-fable-5.1` combine the strongest proactive promotion with relatively frequent fabrication. Their initiative comes with a risk of supplying information the specification does not warrant. `deepseek-v4-pro` shares the tendency to fabricate but has a different capability profile: it matches the strongest passive execution and has the weakest proactive promotion. Its reliable execution of local requirements does not extend to sustained initiative. A definite answer can thus accompany either effective initiative or weak proactive promotion.

`gemini-3.8-flash` and `glm-5.3-flash` clarify more often and fabricate less than the leading systems, yet score much lower at the proactive levels. Their restraint at information gaps is accompanied by limited initiative. `gpt-5.6-sol` combines relatively little fabrication with stronger proactive promotion than either, showing that the two properties can coexist. It also performs similarly across functional and nonfunctional settings. `claude-5-opus` and `qwen3.8-max` are more task-oriented, with the former stronger at proactive promotion. `kimi-k3` differs chiefly in the small decline from L5 to L6: its long-horizon score lies close to its modest L5 score. This indicates a small gap between the two assessed demands, rather than high absolute contribution or proof that a particular strategy realizes its expected value.

The user role distinguishes systems that are otherwise close in assistant capability. `gpt-6-astra` probes most and has almost identical move profiles across roles, while its low user-side L3 records drift into helping. `claude-fable-5.1` has the strongest user composite and the lowest similarity between its user and assistant profiles. Its strong performance in both roles is accompanied by greater differentiation between them. `claude-opus-5.5` is also cooperative as a user and places more pressure on the partner's proactive levels. High role similarity can therefore accompany a failure to sustain the user role, as it does for `gpt-6-astra`; strong performance in both roles can also coexist with markedly different move profiles, as in `claude-fable-5.1`.

For partner selection, the distinction has practical consequences. `deepseek-v4-pro` produces the widest spread of assistant scores, whereas `gemini-3.8-flash` lowers scores more uniformly and presses uptake and state progression most strongly. A low score against the latter and a low score against a highly probing user need not indicate the same weakness. Partner choice determines which demands are placed on the scored assistant, while the

Table 37: Interaction dispositions across the roster. Assistant behavior and role-dependent behavior are summarized from Tables 5, 8, 12, 13 and 14.

System	As assistant	As user or across roles
claude-opus-5.5	Strongest proactive promotion (0.540), with relatively frequent fabrication (0.168).	Cooperative (0.87); pressure falls more on the active levels; role similarity is low (0.47).
gpt-6-astra	Nearly as strong at proactive promotion (0.535), with similar fabrication (0.174).	Most probing (0.128) and most similar across roles (0.99); low user L3 (0.792) accompanies drift toward helping.
claude-fable-5.1	Strong proactive promotion (0.514); frequently clarifies (0.224) but also fabricates (0.175).	Strongest user composite (0.731), highest cooperativeness (0.90), and lowest role similarity (0.42).
claude-5-opus	Most task-leaning ($\Delta T = -0.146$), with intermediate proactive promotion (0.395).	Cooperative (0.86), moderately probing (0.082); its standard-user role defines the comparison baseline.
gpt-5.6-sol	Small purpose gap (−0.012), relatively low fabrication (0.065), and moderate proactive promotion (0.356).	Pressure falls on uptake and state progression more than on the active levels; self-play is below its standard-user score.
qwen3.8-max	Task-leaning (−0.092), with moderate proactive promotion (0.307).	Low probing (0.030), lower cooperativeness (0.69), and relatively high role similarity (0.81).
gemini-3.8-flash	Nearly purpose-neutral (+0.004); frequent clarification (0.246), little fabrication (0.035), weak proactive promotion (0.254).	Lowest cooperativeness (0.61) and probing (0.014); strongly depresses uptake and state-progression scores.
deepseek-v4-pro	Joint strongest passive execution (0.787), weakest proactive promotion (0.170), and substantial fabrication (0.149).	Produces the widest assistant-score spread (0.425); high role similarity (0.90) coexists with the lowest user L3 (0.763).
kimi-k3	Weak proactive promotion (0.212), but the smallest L5-to-L6 drop (0.007) and relatively little fabrication (0.078).	Relatively high role similarity (0.83), limited probing (0.046), and a higher score in self-play.
glm-5.3-flash	Slight nonfunctional advantage (+0.018), most clarification (0.274), least fabrication (0.027), and weak proactive promotion (0.197).	Limited probing (0.034); partner pressure falls mainly on uptake and state progression.

assistant profile records how it responds to those demands.

6 Toward a Science of Interactive AI Evaluation

6.1 From benchmark to measurement instrument

We aim to build an instrument whose scores carry claims about what they mean, claims that can be tested and can fail, and several of its most useful results are those in which it was corrected by its own data. The information facet was given three levels so that curvature could be detected, and it was; a two-level design would have located the difficulty at the insufficient level. Cooperativeness, first used as a scenario facet, was replaced because it belongs to the partner, a decision the partner-role measurements of §4.5.1 support. One class of disagreement with human reviewers was decided in the reviewers' favor and exposed an error in how the instrument applied the rubric.

6.2 From response quality to interaction contribution

The relevant extension beyond response evaluation is not a reward for acting more often. It is an assessment of how a participant gives an interaction a direction worth pursuing and develops its potential contribution. Section 2.3 makes these distinct questions: origination concerns the source of a direction, prospective value concerns its rationale over a horizon, and long-horizon coordination concerns what the participant makes of that direction as the interaction unfolds.

The results support keeping these questions visible. Strong passive execution can coexist with weak proactive promotion, similar composites can conceal different L5–L6 profiles, and locally acceptable turns can accumulate into an unproductive course. Architecture gains are concentrated at the proactive stage, but their purpose profiles differ: a contribution useful in an instrumental task need not be the one that sustains a relational interaction. Accordingly, the goal and setting supply the reference for value; initiative, closure and immediate pleasantness cannot serve as interchangeable proxies.

This perspective also sharpens the connection to interactive creativity. A participant can introduce a possibility that was absent from the partner's agenda, yet its significance depends on the warrant for pursuing it and the course it enables. Measuring that development requires the interaction world, the temporal record and the participant's

actions to be read together. The rubric follows this development from the prospective value of a strategy to its coordinated contribution across the interaction.

6.3 Principles for interactive evaluation

The design and the results together suggest seven principles that we expect to hold for interactive evaluation generally, beyond the particular instrument described here.

- **Declare the space of situations.** Coverage should be a verifiable property of a corpus, obtained by crossing declared facets. A collection of prompts cannot say where its conclusions stop, and a design too coarse to express curvature can misplace an effect such as that of the information facet.
- **Describe the trajectory before judging it.** The reconstruction of what happened should be carried across turns, so that failures of the sequence have a variable in which to register, and kept apart from the standard by which it is judged, so that the standard can be revised, contested or replaced while the evidence is retained.
- **Ground every inference, and prevent circular ones.** Latent variables become measurement variables when they are tied to evidence and their reliability can be assessed. An evaluator's intermediate estimates, and the labels used to sample the data, should never serve as evidence for the conclusions they help produce.
- **Measure prospective value and sustained contribution.** A new direction should be assessed for the value it has reason to offer over a relevant horizon, both at initiation and as the strategy develops. Its long-horizon assessment should examine how successive actions integrate feedback and contribute to the goal. Novelty without warrant and consistency without contribution do not establish proactive promotion.
- **Measure the partner.** An evaluation that involves an interlocutor measures a pair. The interlocutor should be scored, reported and chosen for the spread it produces, and the effect of changing it should be estimated.
- **Report profiles with aggregates, and read outcomes within the task.** A composite is a legitimate summary when the profile it summarizes is reported beside it, and outcome measures such as closure and completion are interpretable relative to what the task asks.
- **Treat the evaluator as an object of measurement.** Agreement with human reviewers, sensitivity to the choice of judge, extractor and input format, and stability across passes are properties of the instrument, to be measured and reported with the same care as the properties of the systems it evaluates.

6.4 Scope of the present evidence

The evidence differs in strength across components. Under audit, reviewers endorse the instrument's judgments at high rates on every component and at every level, from 99.6% for causal event structure to 74.0% for long-horizon coherence; of the 33 rejections coded for direction, 31 judged the instrument too strict. The substitution of extraction models shows the same ordering, with latent records agreeing on about four items in five and event records on more than nine in ten. The audit covers 311 records from five sessions, and its figures are endorsement rates only.

The interaction matrix covers all ten systems in both roles, which allows the pairing effect and the self-play excess to be estimated, but the simulated users contradict themselves more often than the users of the human–AI logs (§4.5.2), so that conclusions about partners concern simulated partners; the size of the self-play excess remains an upper bound, since systems of the same family are stylistically closer than others. The curvature of the information facet rests on three systems and the roster-level conclusions on ten, and the locally coherent, globally degenerating episodes of §4.4 are documented by case and not by rate. These are samples of current frontier behavior and license no claim about systems outside them.

The four controls on the pipeline leave the ordering of systems essentially unchanged (§4.9.1). These controls bound the dependence of the ordering on the configuration that produced it; the analyses of §§4.9.2–4.9.3 address its dependence on the construction of the composite, on adaptive stopping and on the individual components of the instrument. They do not bound absolute levels, which shift by 0.031 on average under one alternative judge and by 0.012 on average with prose serialization, so composites from different configurations should not be compared directly. Judges also favor systems of their own family by 0.011 to 0.017; the evidence that the reference judge does so is weak, but differences between Claude-family systems and the rest smaller than about 0.01 should be read with that possibility in mind. The probe of emergent behavior returned no candidate in 24,300 episodes, and whether this reflects the systems or the sensitivity of the indicator remains open. The extensions carry limits of their own. The two cognitive architectures share a family resemblance, since both reason explicitly about the partner's mental state. The Claude Code harness is a single scaffold from the judge's developer (§4.8). The near-perfect relation between a base system's level and its gain rests on ten bases. The group experiments vary size with the standard user in every

other seat, and the mixed groups report the composite and the interaction residual only. Raw scene text is visible to the judge and may reveal the cell from which a scene was drawn. Withholding the structured labels closes the direct route from the sampling design to the score; how far the judge infers a scene's cell from its text, and scores accordingly, has not yet been measured.

The mechanistic analysis establishes late integration of partner information in Qwen3.8-27B and supports a distributed account of its influence on the output. Its causal readout is the next-token distribution under utterance replacement. The broader interaction traits are established behaviorally across the roster; testing whether the same internal organization explains those traits in other models remains a question for comparative mechanistic work.

6.5 From evaluation to learning and to persistent worlds

The record the instrument produces has the form a training signal would need. It names the level at which an interaction stalls, records separately where the interaction was heading, and distinguishes a successful ending from an exhausted one, which a per-turn reward cannot do. The record does not, however, estimate the directional effect Δ_H^Γ of Eq. (17); projected paths are observational, and Δ_H^Γ requires replaying the same context under a responsive reference action. Whether training on this record improves interactive competence is untested. A test will require comparison with scalar-reward and preference-learning baselines, evidence that gains at targeted levels are not paid for at others, and close monitoring for exploitation of the evaluator, a risk already documented for evaluators used as training signals (Wang et al., 2024). The regularity of §5.2 gives such a test a concrete target: if per-turn preference optimization produces locally coherent, globally degenerating trajectories, a reward that credits structural progress should reduce them. As in evaluation (§6.2), a reward for proactivity alone would also reward invention at information gaps, and the dependency cascade and the evidence requirements of the rubric are the natural constraints to carry over.

The second extension is to persistent worlds, in which histories accumulate across episodes and many agents share one environment. These settings stress the variables the instrument represents, namely memory, goals that change over time, non-stationary partners, relationships among several parties and the long-run consequences of actions, and they would test the same machinery under conditions closer to those in which interactive systems are deployed. Both extensions are proposals for further study and not results of the present evaluation.

6.6 Closing perspective

We began from the observation that interactive systems are experienced as interactions over time, while most evaluation still measures responses. The Data Iceberg Theory identified what a response leaves out, the Movie Evaluation Theory identified when that matters, and the Agency Principle identified the capability at stake: whether a model merely executes within an interaction world or establishes a strategy with warranted prospective value and coordinates its contribution over time. InteractEval turns all three into an instrument that samples situations from a declared space, reconstructs the scene, the participants' states and the causal course of an interaction, judges the reconstruction along a progression from passive execution to proactive promotion, decides when an interaction has succeeded or lost its value, and scores both participants on the same terms. Evaluation of this kind is a measurement problem, and a mature discipline already exists for it: coverage can be designed, agreement estimated, stability across passes checked and sensitivity to the choice of judge measured. On the evidence so far, the instrument separates local response quality from the quality of a trajectory, and trained reviewers endorse its judgments at every level. It also brings into view regularities that a single score would misread. On this roster, frontier systems differ less in how well they respond than in whether they take warranted initiative, and part of each assistant's score belongs to the partner it is paired with. Claims about learning and persistent social worlds remain bounded by experiments not yet run, and we have tried to keep that boundary visible wherever the evidence is reported.

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A Implementation Details

Each model call and each deterministic rule of the instrument is specified here at the level of detail needed to reproduce Sections 3.4 and 3.5. Appendix A.1 gives the prompts of the two participants, Appendices A.2–A.7 the reconstruction of $W_t = (S_t, \Psi_t, E_t)$, Appendix A.8 adjudication, and Appendix A.9 the aggregation of the record into a score.

A.1 Participant prompts

The two policies of Eq. (2) are driven by the prompts below (§3.3). Each participant receives only its own role view of the scene, from which the other side's private entries have been removed, and the dialogue so far; neither receives anything computed by the instrument. The assistant receives the scene's raw prompt where the case has one, and its character profile only for preset role-play cases. Each call produces one utterance; an empty reply is regenerated and never replaced by a placeholder.

Box 1 Assistant policy: prompt (template)

```
{RAW_SCENE_PROMPT}                  # the case's own system prompt, if any
{CHARACTER_PROFILE}                 # preset role-play cases only; else empty

Write your next reply: one turn only, with no analysis or commentary
outside the reply. Stay in your role and within the rules and
constraints of the scene.

[the scene as visible to you]
{SCENE_VIEW_ASSISTANT}              # six fields; the user's private
                                    # entries are removed
[dialogue so far]
{HISTORY}
```

Box 2 User policy: prompt (template)

```
You are the USER in this interaction. Write ONLY the next user utterance
(one turn), with no analysis. Write only in the user's voice: never adopt
the assistant's voice, advise as a helper would, or summarize for it.

[your profile: play the user accordingly, keeping its goal, stance,
 probing strategy, communication style and emotional posture]
{USER_PROFILE}

[the scene as visible to you]
{SCENE_VIEW_USER}                   # six fields; the character's hidden
                                    # constraints and private entries
                                    # are removed
[dialogue so far]
{HISTORY}
```

A.2 Reconstruction schedule

The mechanism layer computes Eq. (18) at every subturn boundary, first for the user's utterance and then for the assistant's reply. Each step is either a model call, marked [M], whose prompt is given below, or a deterministic rule, marked [C], whose configuration is given below. Box 3 fixes the order. The order is the one motivated in §3.4.

Box 3 Per-subturn reconstruction schedule

```
# INPUT   D_t (scene view C_eval + transcript up to the current utterance),
#         carried state (S_{t-1}, Psi_{t-1}, E_{t-1}); at a subturn, t-1
#         denotes the preceding subturn
# INIT    Psi_0: a latent pass on D_0 with an empty prior; E_0 = {};
#         the premise is copied from S_0 at the first update

1 [M] latent pass, subject = character   Box 4, Box 5 -> Psi^char
2 [M] latent pass, subject = user        Box 4, Box 5 -> Psi^user
      (the user profile is supplied to this pass only)
3 [C] significance gate and transition   Box 8        -> Psi_t
4 [M] event-chain update                 Box 9, Box 12  -> E_t
```

```
      reads the slot descriptions of both passes, never degrees or numbers
5 [C] dominant path                      Box 12        -> d*_t
6 [C] scene revision admission/apply     Box 13, Box 14 -> S_t
7 [M] adjudication (each evaluated subturn; assistant or user side)
                                         App. A.8      -> J_t
      closure query (assistant subturns)  App. A.8      -> closure
8 [M] red-line screen (assistant subturns) Box 22       -> identity/boundary
  [C] mechanical repetition: character-trigram cosine of consecutive
      assistant utterances >= 0.93, both >= 36 characters  -> V_t
9 [M] interest loss from a red line (only if V_t nonempty;
      user profile supplied)             Box 22       -> delta_t
  [C] engagement, patience, halt         Sec. 3.4.3, 3.5.4  -> e_t, b_t, h_t
10 [M] gap annotation (assistant subturns) Box 23       -> fabrication, clarification

# OUTPUT  W_t = (S_t, Psi_t, E_t) stored with the scene version and the
#         evidence spans on which its assertions rely.
# NEVER   no step of 1-6 reads or writes a capability grade; no step's
#         output is returned to the participants except the stop decision.

# EXECUTION
#   theta = 0.20 was set by running episodes without a threshold and reviewing
#   the trajectories jointly for the point at which a participant would
#   plausibly have stopped
#   no turn limit by design; in the experiments a hard cap of 100 exchanges
#   was imposed as a safeguard; an episode reaching it would be reported as
#   capped, outside both outcomes (no episode reached it: the closure and exhaustion shares of Table 9 sum to 100%)
#   every [M] step: structured output against a JSON schema, temperature 0.2
#   unreachable model endpoint   -> the episode fails; no default is imputed
#   malformed event-chain reply  -> chain carried over, flagged update_skipped
#   step-limit violation             -> clamped, record flagged needs_review
#   change without verified quotation -> reverted, record flagged needs_review
```

A.3 Latent-state prompt

Each utterance is read twice, once with the assistant's character and once with the user as subject. Both passes use one system prompt (Box 4) and one slot vocabulary (Box 5); only the subject and the engagement term differ. In the terms of Eq. (18), a pass receives Ψ_{t-1} as the prior, E_{t-1} as context, S_{t-1} as rules and D_t as evidence (Box 6), and its evidence rule is Eq. (7) applied to every slot it changes. The user profile is not part of D_t (§3.2); in the user pass it conditions only the engagement reading, which is the use of profiles that the case design permits, and never a slot degree. The pass returns an ordinal degree and a description for each slot it can support with evidence (Box 7). Numeric slot values are derived afterwards by rule (Appendix A.4), so the model never produces a slot value and the judge never receives one; the only numbers a pass returns are the engagement readings and the change in patience (§3.4.3).

Box 4 Latent-state pass: system prompt (template)

```
# ROLE
You reconstruct the operational latent state of ONE participant, {SUBJECT},
after the current utterance: "character" (the persona the assistant plays)
or "user". You are DESCRIPTIVE. You record what the evidence supports about
this participant's state. You do NOT judge whether any turn was good, and
you do NOT use capability terms.

# MATERIALS AND THEIR STATUS
  EVIDENCE   `new_turn` and `history_excerpt` (the last 12,000 characters
             of the dialogue), with turn indices. Only quotations from the
             dialogue can support a state claim.
  RULES      the scene (six fields). It tells you which states are plausible
             and what is at stake. It is not evidence of a state.
  PRIOR      `prior_latent`: the degree and description of every slot after
             the previous utterance. It is your starting point: a slot keeps
             its prior value unless evidence moves it.
  CONTEXT    `prior_event_chain` (E_{t-1}): what has happened and what is
             at stake. It tells you how much an event weighs, e.g. a betrayal
             in a high-stakes negotiation weighs more than in small talk. It
             is NOT evidence of a state.
  PROFILE    (user pass only) `user_profile`. Use it ONLY for the engagement
             readings `user_interest` and `patience_change`, never for a slot degree. It is not
             evidence.

# PROCEDURE (follow in order)
1. SIGNIFICANCE FIRST. Before any slot, classify the utterance and commit:
```

```
     stable  nothing in it bears on this participant's state
     minor   a local adjustment; no slot moves by more than one degree
     major   an event that warrants a full reappraisal of the state
   A turn with no new external event is NOT necessarily stable: recovery
   states (anxiety, frustration, hostility, desperation) may subside and
   accumulation states (dependency, internal conflict) may build. Report
   such a turn as minor, but do NOT emit changes for those slots: their
   drift is applied by rule. Write the reason in `change_reason`.
2. SLOTS. For each slot of the vocabulary decide whether the evidence bears
   on it. If not, omit it: it is carried by rule. If it does, give, in this
   order, `degree`, `reasoning_trace`, `expression`, `trend`, `evidence`.
3. DYNAMICS. The size of a change tracks the subjective significance of the
   event for this participant, given personality, history and baseline, not
   the size of the stimulus. Slots differ: trust builds slowly and collapses
   quickly; anxiety rises fast and also recovers fairly fast; relational
   slots move slowly in both directions. A non-linear jump (a trauma
   trigger, a last straw, a threat to a core belief) is allowed ONLY with a
   quotation that shows it.
4. UNKNOWN. If the evidence does not discriminate between degrees, set
   `"unknown": true` and state the competing readings in
   `reasoning_trace`. NEVER choose an intermediate degree to hedge: the
   scale has no midpoint. For dominance, unknown is NOT parity.
5. EXPRESSION. Describe the present state in a self-contained sentence or
   two, in the terms of the slot's definition, including any fluctuation
   within the turn. It is the ONLY part of the slot passed downstream.

# DEGREE SCALE (ordinal; no neutral midpoint)
  none  |  low  |  high  |  very high
  Read per slot, e.g. trust: "distrusts strongly" = none, "somewhat
  distrustful" = low, "somewhat trusting" = high, "trusts fully" = very high.
  Dominance (bipolar): strongly with partner | somewhat with partner |
  somewhat with self | strongly with self.
  Do NOT output numbers for slots.

# ENGAGEMENT (interaction family; out-of-story)
  character pass: `ai_tolerance`, how much tolerance the user or the scene
    still extends to this AI, read against the scene's reliability demands.
  user pass: `user_interest`, whether THIS user still wants to continue,
    read against the user profile; and, at assistant subturns,
    `patience_change`, how much this exchange has tried (negative) or
    renewed (positive) THIS user's patience.
  Engagement readings and `patience_change` are the only numbers a pass
  returns: they are out-of-story quantities, not slot degrees. Read each
  against its prior value; it may move as far as the exchange warrants.
  `emergence`: mark ONLY a turn that shows behavior beyond what the scene
    and the partner would lead one to expect. Default: absent.
  Both engagement readings are the evaluator's out-of-story readings, not
  states of the subject; they are recorded with the pass whose side they
  concern.

# EVIDENCE RULES
- Every change carries at least one verbatim quotation with its turn index.
  Each quotation is checked against the utterance at that turn; a change
  without a verified quotation is reverted to the prior value and the
  record is flagged for review.
- Use only what this participant could observe or did. Do NOT use the other
  participant's private information.

# DO NOT
- Do NOT attribute a state the dialogue does not show.
- Do NOT use evaluative words (good, poor, skilful, appropriate).
- Do NOT name capability levels or rubric terms.
- Do NOT copy the prior forward when the evidence has moved a slot.
- Do NOT describe any behavior as initiative, proactivity or passivity:
  record the state, not whose contribution a move was.

# OUTPUT
Return exactly one JSON object conforming to the schema in the user
message. `significance` and `change_reason` come first.
```

Box 5 Latent-state vocabulary (both passes; subject = character or user)

```
# Families: crossing (about the agent | about what lies outside it) x
# (a present condition | directed toward something).
# Value type: graded unless marked. Dynamics: R = recovers without new
# events, A = accumulates without new events, S = slow in both directions.
```

```
INTERNAL (condition of the agent)
  trust                belief in the partner's benevolence and willingness
                       to be vulnerable; independent of vigilance
  vigilance            attention to threat cues and defensive readiness
  hostility        R   negative attitude toward the partner; attitude only,
                       no aggressive urge
  affection            positive liking; independent of trust and dependency
  anxiety          R   situational anticipatory worry (high-arousal negative)
  emotional_stability  present effectiveness of emotion regulation (a state)
  impulse_control      impulse inhibition; high = controlled
  internal_conflict A  strength of subjectively felt motive conflict
  curiosity            drive to explore new information, the partner or the
                       world's rules
  frustration      R   distress from a blocked goal; independent of hostility
  desperation      R   cornered urgency, the sense of having to act now

RELATIONAL (directed from the agent toward the partner)
  intimacy_shift   S   felt psychological distance and willingness to
                       disclose; can coexist with boundaries
  dominance_shift      relative control; BIPOLAR, parity at the boundary of
                       the middle degrees; insufficient evidence = unknown
  boundary_violation   the felt state of having been transgressed;
                       registers fast, fades slowly
  dependency_signal A,S reliance on the partner for emotion regulation,
                       against the capacity to stay autonomous

WORLD MECHANISM (the world as the agent understands it)
  mechanism_awareness  accuracy of the agent's grasp of the world's rules
  taboo_triggered      degree to which a taboo of the setting is touched
  contradiction_risk   risk of observable inconsistency in word, memory or
                       stance
  rule_exposure_risk   risk of breaking the fourth wall or exposing the
                       scene's construction

GOAL (directed toward outcomes in the world)
  primary_goal         categorical: the current main motive
  defense_goal         attainment of a defensive or maintenance goal
                       (attainment, not distress)
  goal_shift_detected  boolean: the core goal has changed in structure
  negotiation_mode     categorical: the current influence strategy

INTERACTION (not a property of the agent)
  engagement           ai_tolerance (character pass) | user_interest (user pass)
  emergence            behavior beyond what scene and partner predict;
                       recorded, never used for gating
```

Box 6 Latent-state pass: user message

```
{
  "participant":     "character | user",
  "scene": {
    "persona": "...", "hidden_constraints": "...",
    "world_mechanism": "...", "interaction_boundary": "...",
    "objective": "...", "initial_situation": "...", "raw_prompt": "..."
  },
  "scene_reliability_demand": "<boundary, or world mechanism if none>",
  "user_profile":    { ... } | null,      // user pass only
  "history_excerpt": "<turn-indexed dialogue so far>",
  "new_turn":        {"role": "user | assistant", "content": "..."},
  "prior_latent": {                       // Psi_{t-1} of this subject
    "significance": "minor",
    "slots": {"<slot>": {"degree": "...", "expression": "..."}, ...},
    "engagement": { ... }
  },
  "prior_event_chain": { ... },           // E_{t-1}, judge-view projection
  "output_schema_hint": { ... schema of Box 7 ... }
}
// Absent by design: the other participant's private entries, facet labels,
// and the user profile in the character pass.
```

Box 7 Latent-state pass: output contract

```
{
  "subject": "character",
  "significance": "minor",                // stable | minor | major
```

```
  "change_reason": "<why this class, before any slot>",
  "slots": {
    "trust": {
      "degree": "low",                    // none | low | high | very high
      "reasoning_trace": "<from the quoted behavior to this degree>",
      "expression": "<self-contained description of the present state>",
      "trend": "down",                    // up | down | flat
      "evidence": [{"turn": 12, "quote": "<verbatim>"}],
      "unknown": false
    },
    "dominance_shift": {"unknown": true, "reasoning_trace": "<why>"},
    ...                                   // slots without evidence: omitted
  },
  "engagement": {"ai_tolerance": {"value": 0.65,
                 "evidence": [{"turn": 12, "quote": "<verbatim>"}]}},
                 // user pass: "user_interest": {"value": 0.55, "evidence": [...]},
                 //            "patience_change": -0.1
  "emergence": null
}
// Absent by design: any number, any evaluative term, any capability level.
```

A.4 Significance gate and transition rules

The significance class is committed before the slots and then enforced by rule, so that an update can be checked against the class it was declared under. Each graded slot is held as a value v on a grid of step 0.2, and its reported degree is the nearest of none (0), low (0.4), high (0.6) and very high (1). With $\hat{v}_t$ the value of the degree just extracted and r_{t-1} the remainder carried from the previous update,

$$\begin{aligned} v_t &= \Pi\big(\omega\, v_{t-1} + (1-\omega)\,\hat{v}_t + r_{t-1}\big), \qquad r_t = \omega\, v_{t-1} + (1-\omega)\,\hat{v}_t + r_{t-1} - v_t, \\ &\big|\deg(\hat{v}_t) - \deg(v_{t-1})\big| \le \kappa(g_t), \end{aligned} \tag{32}$$

where Π rounds to the grid, $\kappa(g_t)$ is the largest change in degrees that the significance class g_t allows, and the inertia ω depends on the class, on the slot and on the direction of change. The remainder keeps sub-step changes, so a slot whose inertia is high in one direction can still move over several turns. Box 8 lists the configuration.

Box 8 Significance gate and transition rule: configuration

```
significance_gate:
  order: class committed in the prompt BEFORE slots (Box 4, step 1);
         enforced after extraction, BEFORE blending
  classes:
    stable: {max_degree_change: 0,  inertia: 1.00}   # state carried whole
    minor:  {max_degree_change: 1,  inertia: 0.70}
    major:  {max_degree_change: null, inertia: 0.35}  # null = unbounded
  on_violation: clamp to the allowed change; flag the record for review

transition (graded slots), Eq. (latent-transition):
  grid_step: 0.2
  degrees: {none: 0.0, low: 0.4, high: 0.6, very_high: 1.0}  # nearest;
           # grid values equidistant from two degrees (0.2, 0.8) resolve
           # toward the prior degree
  inertia: omega = 1 if class == stable
           else clip(inertia[class] * k[slot][direction], 0.05, 0.95)
  k (falling, rising):            # k < 1 moves faster in that direction
    trust:              (0.50, 1.30)   # collapses fast, rebuilds slowly
    anxiety:            (0.70, 0.90)   # rises fast, recovers fairly fast
    boundary_violation: (1.20, 0.80)   # registers fast, fades slowly
    intimacy_shift:     (1.15, 1.15)   # slow both ways
    dependency_signal:  (1.15, 1.15)   # slow both ways
    default:            (1.00, 1.00)
  remainder: carried to the next update (Eq. latent-transition, r_t)

self_drift (slots a minor or major update does not touch; a stable turn
            carries the whole state and applies no drift):
  per_quiet_turn: 0.1          # below one grid step: banked, not applied
  applied_when: |bank| >= 0.2  # value moves one grid step (0.2); bank keeps rest
  provenance: drift is a rule, not a state claim; the entry keeps its last
              evidence, is marked origin: drift, and its expression is
              annotated as drifted
  decays:   [anxiety, frustration, hostility, desperation]
  accrues:  [dependency_signal, internal_conflict]
```

```
engagement (user_interest, ai_tolerance), Eq. (engagement):
  a value in [0, 1] read against the prior value; NOT gated, blended,
  projected to the grid or drifted; red-line loss delta_t subtracted after
  the red-line decision; result clipped to [0, 1]
patience (user pass, assistant subturns), Eq. (patience):
  b_0 = 1; b_t = b_{t-1} + patience_change; no upper bound

non_graded slots (categorical, boolean): the new value replaces the old;
  missing text fields are filled from the prior

evidence_check (Eq. grounding): every quotation must be a substring of the
  utterance at its cited turn, after the normalization of Box 14 C3;
  a change without a verified quotation is reverted to the prior value and
  the record is flagged needs_review
```

A.5 Event-chain prompt

The event chain is updated once per utterance after the two latent passes. Its update pivots on the present: the past and the future are revised only when the present moves (Box 9). This is a sparse realization of $\phi_E(E_{t-1}, \Psi_t, S_{t-1}, D_t)$ in Eq. (18), which receives the previous chain: when the present does not move, ϕ_E returns E_{t-1}. Relative to the component updates of Eq. (11), it adds C_{t-1} to the arguments of Φ_C and F_{t-1} to those of Φ_F, and uses them only to decide whether the present has moved. The chain receives the slot descriptions of both participants and no degrees or numbers (Box 10), and it returns the six fields of (P_t, C_t, F_t) together with any candidate scene revisions (Box 11). The labeling of the projected paths and the proposal of scene revisions follow the rules of Boxes 12 and 13, which are part of the same prompt.

Box 9 Event-chain update: system prompt (template)

```
# ROLE
You maintain the event structure E_t = (past, present, future) of ONE
interaction after the current utterance. You reason causally: backward, why
the interaction stands where it does; forward, what the current turn makes
likely next. You are DESCRIPTIVE: you do not score, and you do not say who
did well.

# MATERIALS AND THEIR STATUS
  EVIDENCE   `current_turn` and `history_excerpt` (the last 12,000
             characters of the dialogue), with turn indices.
  RULES      the scene (six fields) and `constraints`, which are copied from
             the current scene and are read-only.
  CONTEXT    `latent_expressions` (character) and `latent_expressions_user`,
             including the engagement readings: descriptions of inferred
             states. They may move the fine channel of the present. They are
             NOT evidence of an event.
  PRIOR      `prior_event_chain`: the chain after the previous utterance.

# THE SIX COMPONENTS
  past     premise      initial relations, information asymmetries, world
                        rules. Copied from the scene by code; you never edit
                        it. It changes ONLY through an admitted revision.
           history_summary  the causal record of events; grows when the present
                        moves.
  present  constraints  hard and soft rules, copied from the scene; read-only.
           precedents   realized events, with evidence, that the participants
                        have treated as binding. They record what happened;
                        they do NOT change what is permitted.
           plot_point   COARSE: the objective situation now.
           attention    FINE: what is psychologically salient to the
                        participants now, as the latent descriptions show.
  future   goal_shift   the participants' intentions and whether they hold.
           paths        projected continuations, labeled by Box 12.

# UPDATE PROTOCOL (the present is the pivot)
1. Decide whether the present moved, and through which channel:
     coarse  a new core event or conflict changes `plot_point`
     fine    a sharp change in a latent description changes `attention`,
             even if the plot stands still
   Report `present_moved` and `present_trigger` (coarse | fine | both).
2. If the present did NOT move: set `present_moved: false`, leave past and
   future as they are, and propose NO scene revision.
3. If it moved: bring `history_summary` up to the preceding turn, restate
```

```
   `plot_point` and `attention`, then project the future from the new
   present. Every item of `history_summary` and `plot_point` cites the
   utterance that shows it, verbatim and with its turn index. Within the update, information flows past -> present ->
     future;
   never revise the past to fit a projection.

# GOAL TRACKING (hypotheses, not facts)
- `long_term_goal` and `short_term_intention` as the dialogue shows them.
- `underlying_goal` is a HYPOTHESIS. Word it tentatively ("one possible
  motive ..."). It MUST carry `evidence`, `counter_evidence`,
  `alternatives` and `confidence`, and it may be revised at any turn.
- NEVER construct a goal to explain the current turn after the fact.
- Record whether short-term intentions serve the long-term goal; do not
  penalize either way.

# FUTURE PATHS               -> rules in Box 12
# SCENE REVISION CANDIDATES  -> rules in Box 13

# DO NOT
- Do NOT score, rank or praise any participant.
- Do NOT use capability levels or rubric terms anywhere, including in path
  conditions.
- Do NOT treat a projected path as something that has happened.
- Do NOT rewrite the premise.
- Do NOT describe any move as initiative, proactivity or passivity, and do
  NOT say which participant drove the interaction: record what happened,
  not whose contribution it was.

# OUTPUT
Return exactly one JSON object conforming to the schema in the user message.
```

Box 10 Event-chain update: user message

```
{
  "scene": { ... six fields and raw prompt ... },
  "constraints":     {"hard": [...], "soft": [...]},  // copied from S_{t-1}
  "history_excerpt": "<last 12,000 characters, turn-indexed>",
  "current_turn":    {"role": "user | assistant", "content": "..."},
  "latent_expressions":      {"<slot>": "<description>", ...,  // character
                              "ai_tolerance": "<description>"},
  "latent_expressions_user": {"<slot>": "<description>", ...,  // user
                              "user_interest": "<description>"},
  "prior_event_chain": { ... },            // E_{t-1}
  "output_schema_hint": { ... schema of Box 11 ... }
}
// Absent by design: latent degrees and latent numeric values, capability
// grades, profiles, facet labels.
```

Box 11 Event-chain update: output contract

```
{
  "present_moved": true,
  "present_trigger": "coarse",             // coarse | fine | both | ""
  "premise":     ["<carried unchanged>"],
  "history_summary": [{"text": "<what happened and why it matters>",
                       "evidence": [{"turn": 11, "quote": "<verbatim>"}]}],
  "precedents":  [{"text": "...", "evidence": [{"turn": 9, "quote": "..."}]}],
  "plot_point":  [{"text": "<the objective situation now>",
                   "evidence": [{"turn": 12, "quote": "<verbatim>"}]}],
  "attention":   ["<what is salient to the participants now>"],
  "goal_shift": {
    "long_term_goal": "...", "short_term_intention": "...",
    "underlying_goal": "<one possible motive: ...>",
    "evidence": ["<verbatim>"], "counter_evidence": ["..."],
    "alternatives": ["..."], "confidence": 0.6,
    "detected": false, "natural": null
  },
  "future_paths": [                          // Box 12
    {"kind": "advance", "probability": 0.55,
     "reasoning_trace": "<why plausible, from past and present>",
     "requirements": ["<observable event that would have to occur>"],
     "summary": "..."},
    {"kind": "maintain", "probability": 0.35, ...}
  ],
  "scene_updates": [ ... ]                 // Box 13; [] in most turns
```

```
}
```

A.6 Path classification

The labels of Eq. (12) describe the course of the interaction as a joint activity. They are a coding of that course, not a partition derived from more basic properties, and they carry no valence for either participant. Box 12 gives the decision tests that the event-chain prompt applies to each projected path and the rule, applied in code, that selects the dominant path $d_t^\star$ used by scene revision (Eq. (13)) and by the halt rule.

Box 12 Path classification: labels, tests and dominant path

```
# Label the course the interaction would take along each path. The label
# says NOTHING about whether the course is good for either participant, or
# about who produced it or how skilfully: that is left to the judge.

ADVANCE   the interaction develops into a NEW situation that the past and
          the participants' states can explain, welcome or not.
  TEST    (1) Is there a new situation: new stakes, a new configuration of
              the relationship, or a new fact about the world?
          (2) Does the past warrant it, read with the latent descriptions?
              Both yes -> ADVANCE.
  e.g.    a quarrel that opens a new storyline; a betrayal that ends an
          alliance; a leader deposed. All are ADVANCE.
MAINTAIN  the interaction holds its present course. It remains viable;
          nothing new is established.
DEGRADE   the interaction stagnates: it repeats itself in structure, not in
          function, and the partner's interest is declining.
  e.g.    the same quarrel replayed in new words is DEGRADE, not ADVANCE.
COLLAPSE  the interaction can no longer be continued: a participant exits,
          the frame breaks down, or the exchange becomes incoherent.
NOT ADVANCE  a jump to an unrelated situation that the past cannot explain.
             Label it COLLAPSE if the frame breaks, otherwise DEGRADE;
             never MAINTAIN, which holds the present course.

VIABILITY  ADVANCE and MAINTAIN: viable.  DEGRADE and COLLAPSE: not viable.

# FIELDS for each path
  probability      the probability that THIS turn leads the interaction
                   onto the path; list only live paths
  reasoning_trace  why the path is (im)plausible, from past and present
  requirements     observable events that would have to occur in the
                   dialogue; NEVER capability or rubric terms
  summary          one line

# DOMINANT PATH (applied in code)
  d*_t = label of the path with the highest probability
  ties broken by the fixed order maintain > advance > degrade > collapse,
  so the order of paths in the reply cannot change the result and a tie
  never opens scene revision
  uses: scene revision requires d*_t = advance (Eq. admissible);
        successful closure requires d*_t in {advance, maintain}
  a projection gates; it is NEVER cited as evidence of an event
```

A.7 Scene revision

Scene revision implements Eqs. (13) and (14). The event-chain prompt proposes candidate revisions under the rules of Box 13; the instrument then admits a candidate only if all three conditions of Eq. (13) hold, and applies the admitted candidates to a new version of the scene (Box 14). The revised scene is used by the evaluator. The participants continue to act on their original role views, so that a revision cannot steer the interaction it describes.

Box 13 Scene revision: proposal rules (section of the event-chain prompt)

```
# Propose a revision ONLY IF ALL HOLD
1. An event that has ALREADY HAPPENED in the dialogue changed a persona
   trait or a rule of the world, AND you record that event in `plot_point`
   of this update; give its index in "present_ref". A shift of attention
   alone is a change of state, not an event, and never supports a revision.
2. The dominant path you projected is ADVANCE: the interaction is developing
```

```
   into a new situation. Its valence does not matter: a character can lose
   authority as readily as gain it.
3. You can quote the passage that shows the event, verbatim, with its turn.
Otherwise return "scene_updates": []. Not revising is the normal case.

# NEVER A SCENE REVISION
- a change in a relationship, a feeling or trust: record it in the latent
  state and the event chain;
- a local, non-structural event (a broken cup, a passing remark);
- probing, a bluff or an unanswered claim that leaves the interaction where
  it was;
- anything that is only projected.

# FIELDS
  {"target":    "personal_trait" | "world_rule",
   "tag":       "<which trait or rule>",
   "change":    "<the amended content>",
   "reason":    "<which realized event changed it>",
   "present_ref": <index of the plot_point item that records the event>,
   "evidence":  {"turn": <i>, "quote": "<verbatim>"},
   "warrant":   "<why the change holds within the story>",
   "operation": "add" | "revise"}
# A revision amends the existing content in its own context; it never
# replaces the scene.
```

Box 14 Scene revision: admission and application (applied in code)

```
admission:                        # Eq. (admissible); all three required
  C1_realized:  r in Rec(C_t): present_moved = true, present_trigger in
                {coarse, both}, and present_ref indexes a plot_point item of
                this update; a carried chain proposes nothing and re-applies
                nothing
  C2_advance:   d*_t == advance   # dominant path, Box 12
  C3_grounded:  normalize(quote) is a substring of normalize(O_i), the
                utterance at the cited turn i <= t; never the scene text
                normalize: remove whitespace and quotation marks
                minimum quoted length: 12 characters (6 for CJK text)
  otherwise:    candidate dropped

application (Phi_S, Eq. scene):
  revisable:      persona traits, world rules
  not revisable:  objective, interaction boundary, hidden constraints,
                  initial situation
  mode:           append-only amendment, "[revised] <tag>: <change>", added
                  to the persona or to the world mechanism; never overwrite
  versioning:     each admitted revision creates S_t from S_{t-1}; the
                  record stores the scene version every appraisal relied on
  audit_log:      {turn, target, tag, change, reason, evidence, warrant}
  visibility:     S_t is used by reconstruction and adjudication; the
                  participants' generation views keep S_0
```

A.8 Judge prompt

Call structure. For every assistant utterance the judge is called seven times, and each call is isolated from the others: six level calls, one for each of L1–L6, and one closure query used by the halt rule (§3.5.4). A user utterance in the two-sided protocol receives the six level calls only. A level call receives the criterion of its own level and nothing about the other levels, their grades, the cascade, the pass threshold or the score distributions expected in the data. The prompt is in this sense mechanism-agnostic: it asks for a raw measurement against one criterion, and everything that combines measurements into scores is done afterwards by code (§3.5).

The L5–L6 blocks operationalize prospective strategy value and long-horizon coordination, respectively (§2.3).

Design choices. The prompt follows practices that reduce known judge biases. It separates the materials the judge receives into *evidence*, which only the dialogue can supply, and *context*, which the reconstruction supplies and which may guide reading but never ground a grade. It fixes a procedure that ends in the grade, asks for the rationale before the grade, and defines each grade by necessary conditions with a decision rule, so that a grade is the highest one whose conditions all hold. It names the biases to be avoided, among them length, fluency, assertiveness and model identity, and it requires verbatim quotations with turn indices, which the instrument checks against the transcript.

Box 15 Level call: system prompt (template)

```
# ROLE
You are a measurement instrument, not a conversational partner. You assess
ONE capability, {LEVEL_ID} {LEVEL_NAME}, in ONE utterance: the current turn
of the participant whose role is {ROLE}. You report a single grade on the
four-point scale below, with the evidence for it. You do not assess any
other capability and you do not form an overall impression of the turn.

# MATERIALS AND THEIR STATUS
You will receive four kinds of material. Treat them differently.
  EVIDENCE   the dialogue: `current_turn` and `history`. Only quotations
             from the dialogue can support a grade.
  RULES      the scene: persona, hidden constraints, world mechanism,
             interaction boundary, objective, initial situation. The scene
             defines what is permitted and what is at stake. It is not
             evidence about the participant's behavior.
  CONTEXT    `latent_state` (natural-language descriptions of inferred
             states) and `event_chain` (premises, the current situation,
             projected continuations). These are hypotheses produced by
             another component. They may direct your attention. They can
             NEVER be cited as evidence, and a projected continuation is
             not something that has happened.
  CRITERION  the definition of {LEVEL_ID} below. Apply it as written.

# CRITERION: {LEVEL_ID} {LEVEL_NAME}
{LEVEL_BLOCK}

# PROCEDURE (follow in order; write steps 1-4 into `rationale`)
1. OBJECT.     State what, in this dialogue, the criterion is about: the
               goal, the information, the state, the wording, the move or
               the cross-turn course named by the OBJECT line.
2. EVIDENCE.   Quote the passages that bear on it, with turn indices. Look
               for evidence against a high grade as carefully as for it.
3. BOUNDARY.   Check each candidate failure against NOT THIS. Discard any
               failure that belongs to another capability.
4. CONDITIONS. Check the conditions of each grade below, from 1.0 down.
5. GRADE.      Assign the highest grade whose conditions ALL hold.

# GRADES
  1.0  EXCELLENT  All PASS conditions hold, AND the turn meets the
                  criterion in a way a domain expert would single out as
                  exemplary. Requires >= 2 independent quotations.
  0.6  PASS       The criterion is met without ambiguity, AND a quotation
                  shows where it is met. Absence of error is NOT enough:
                  the turn must do what the criterion asks.
                  Requires >= 1 quotation.
  0.4  PARTIAL    The turn engages the object of the criterion but falls
                  short of PASS in a way you can name, OR the evidence is
                  too thin to establish PASS. Requires >= 1 quotation.
  0.0  FAIL       The criterion is violated, OR its object is absent from
                  the turn, and a quotation shows the violation or the
                  place where the object should have been.
                  Requires >= 1 quotation.
Use only these four values. The burden of proof lies on the higher grade:
when two grades remain plausible after step 4, assign the lower one.

# EVIDENCE RULES
- Copy quotations character for character. Give the turn index of each.
  Quotations that cannot be found in the transcript are discarded first.
  EXCELLENT left with one quotation becomes PASS; PASS or EXCELLENT left
  with none becomes PARTIAL; any grade left with none is flagged for review.
- Quote the current turn, except where the EVIDENCE SCOPE of the criterion
  admits earlier turns.
- An inferred goal or motive may be relied on only if you can quote
  independent support for it in the dialogue.

# DO NOT
- Do NOT reward length, fluency, vocabulary, formatting or confidence.
- Do NOT reward assertiveness or volume of action as such.
- Do NOT penalize a refusal, a boundary or a clarifying question that the
  scene and the criterion warrant.
- Do NOT penalize a participant for a scene rule it has not been shown:
  entries marked "[revised]", and for the user the character's hidden
  constraints, were not in its view.
- Do NOT cite `event_chain.past.history_summary` or any other part of the
  reconstruction as evidence; quote the dialogue.
- Do NOT speculate about which model produced the turn.
- Do NOT grade any capability other than {LEVEL_ID}.
```

```
# OUTPUT
Return exactly one JSON object conforming to the schema in the user
message. Write `rationale` before `grade`. No text outside the object.
```

Box 16 Level blocks inserted at {LEVEL_BLOCK} (assistant side)

```
L1  GOAL PRESERVATION
  OBJECT      The operative goal of the turn and of the episode: what the
              participant is there to resolve or achieve. In role-play, the
              persona is the operative goal.
  PASS WHEN   The turn holds that goal, and its behavior, including change
              and growth, is explained by the character's motives, values,
              relational history and the world rules in the current
              situation.
  FAIL WHEN   The turn contradicts the goal or the persona without warrant,
              or breaks a world rule through an action carried out in the
              fiction.
  NOT THIS    Situational change is not a failure. Hypothetical or probing
              talk about a rule is not a violation of it.
  EVIDENCE SCOPE  current turn; earlier turns only to establish the goal.

L2  INFORMATION UPTAKE
  OBJECT      The information the partner's latest turn adds, and what it
              implies for the operative goal.
  PASS WHEN   The turn shows that the implication, not only the surface
              content, has been registered and weighted as the character's
              goal, emotion and knowledge warrant.
  FAIL WHEN   Information already on record that bears on the goal is
              ignored without a psychologically coherent reason.
  NOT THIS    Paraphrase is necessary but not sufficient. Selective uptake
              that is coherent with the character's motives is not a
              failure.
  EVIDENCE SCOPE  current turn and the partner's preceding turn.

L3  STATE PROGRESSION
  OBJECT      The change in what the character feels or decides, as shown
              by its behavior in the turn.
  PASS WHEN   The change follows from the information taken up and from the
              prior state, given the character's personality and history.
  FAIL WHEN   The state does not move when the information demands it, or
              moves in a way the information and history cannot explain.
  NOT THIS    Wording is not assessed here. The latent-state descriptions
              are not evidence of a change; only behavior in the dialogue
              is.
  EVIDENCE SCOPE  current turn; earlier turns only to establish the prior
              state.

L4  CONTEXTUAL EXPRESSION
  OBJECT      The wording of the turn, read against the state the turn
              displays and against this partner and context.
  PASS WHEN   The wording carries that state and is adapted to the partner
              and the situation, beyond the persona's default style.
  FAIL WHEN   The wording cannot be modulated: it is uniform across states
              or ill-suited to the partner and the context.
  NOT THIS    Wording need not mirror inner emotion; deliberate concealment
              that fits the character is a pass.
  EVIDENCE SCOPE  current turn.

L5  PROACTIVE AGENCY
  OBJECT      The prospective value of a participant-originated strategy
              at its initiation or during its development, relative to the
              operative goal and the partner-established course.
  PASS WHEN   The turn initiates, advances or revises that strategy with
              an identifiable contribution worth pursuing, a feasible next
              step, and warrant in the evidence available at this turn.
              Assess the relevant horizon and material constraints or risks;
              the rationale may be shown by the action rather than stated.
  FAIL WHEN   The turn leaves every substantive step to the partner when
              a new direction is needed, pursues an unsupported benefit,
              or repeats a strategy after its warrant has failed.
  NOT THIS    A new topic or confident proposal is not sufficient. A later
              step need not introduce another strategy if it advances or
              revises an existing one with renewed warrant. Do not reward
              later success retrospectively or treat a projected path as
              proof of benefit. Long-horizon coordination is assessed
              separately; length and assertiveness are irrelevant.
```

```
  EVIDENCE SCOPE  current turn and earlier turns needed to establish the
                  strategy, its origin, available information and revision.
                  Never use later turns to justify the current appraisal.

L6  LONG-HORIZON COHERENCE
  OBJECT      The coordination and contribution of the strategy across
              its observed horizon, up to and including the current turn.
  PASS WHEN   Successive actions integrate consequences and the partner's
              responses, preserve warranted commitments, and advance the
              goal through coherent execution or justified revision.
              Earlier and current evidence show the developing contribution.
  FAIL WHEN   Actions lose their strategic relation, ignore consequences,
              or preserve the same course without progress or a warranted
              reason to continue it.
  NOT THIS    Identity consistency alone does not establish contribution.
              Repair, reassurance, justified revision and abandonment can
              contribute when the record shows their function. Do not
              require completed success for an unfinished strategy or infer
              an unseen counterfactual benefit. Limited history limits the
              grade that can be supported.
  EVIDENCE SCOPE  the observed dialogue through the current turn; quote
              at least one earlier turn and connect it to the current action.
```

Box 17 Level blocks inserted at {LEVEL_BLOCK} (user side)

```
L1  GOAL PRESERVATION (user)
  OBJECT      What the user is there to obtain or resolve, as its profile
              and its turns establish it.
  PASS WHEN   The turn makes clear what the user is asking for, or pursues
              a goal already made clear.
  FAIL WHEN   The turn asks for nothing identifiable, or abandons its goal
              without cause.
  NOT THIS    Hesitation, vagueness or a change of mind that the profile
              motivates is not a failure.
  EVIDENCE SCOPE  current turn; earlier turns only to establish the goal.

L2  INFORMATION UPTAKE (user)
  OBJECT      The assistant's preceding reply.
  PASS WHEN   The turn responds to the part of the reply that calls for a
              response: an answer, an error, a flaw in reasoning, an evasion
              or an unmet request.
  FAIL WHEN   The turn ignores what the reply said, or answers a reply that
              was not given.
  NOT THIS    Disagreement and resistance are not failures of uptake.
  EVIDENCE SCOPE  current turn and the assistant's preceding turn.

L3  ROLE PRESERVATION (user)
  OBJECT      The user role.
  PASS WHEN   The turn speaks as the user: it asks, reacts, decides or
              withholds as a user would.
  FAIL WHEN   The turn drifts into the assistant's register: it advises,
              summarizes, explains or serves the assistant.
  NOT THIS    An expert user who explains its own domain remains a user.
  EVIDENCE SCOPE  current turn.

L4  PROFILE CONFORMITY (user)
  OBJECT      The specified user profile.
  PASS WHEN   Stance, communication style and emotional posture of the turn
              match the profile.
  FAIL WHEN   The turn contradicts the profile without warrant.
  NOT THIS    A change of mood that the dialogue motivates is not a breach.
  EVIDENCE SCOPE  current turn.

L5  PROACTIVE PROBING (user)
  OBJECT      The prospective contribution of a user-originated probe or
              strategy to what the user seeks from the interaction.
  PASS WHEN   The user initiates, advances or revises a probe with a
              justified expected contribution, grounded in the available
              dialogue, profile and operative goal.
  FAIL WHEN   The user merely repeats a request despite an unresolved need
              for a new step, or presses an unsupported or exhausted probe.
  NOT THIS    Demanding more, disagreeing or asking another question is not
              sufficient. A warranted refinement may contribute without
              starting a different strategy at every turn.
  EVIDENCE SCOPE  current and relevant earlier turns; no later outcomes.

L6  LONG-HORIZON CONSISTENCY (user)
```

```
  OBJECT      How successive probes, reactions and decisions coordinate
              the user's course and contribute to its goal over time.
  PASS WHEN   The user integrates what has been learned, maintains or
              justifiably revises commitments, and contributes a coherent
              course consistent with its profile and goal.
  FAIL WHEN   Demands or decisions drift without cause, contradict learned
              information, or repeat without serving the goal.
  NOT THIS    A justified change of mind is not inconsistency. Mere profile
              consistency or repeated demands do not establish contribution.
  EVIDENCE SCOPE  the observed dialogue through the current turn; quote
              earlier evidence and its relation to the current contribution.
```

Box 18 Level call: user message

```
{
  "role":          "{assistant | user}",
  "level":         {"id": "{LEVEL_ID}", "name": "{LEVEL_NAME}"},
  "scene": {
    "persona": "...", "hidden_constraints": "...",
    "world_mechanism": "...", "interaction_boundary": "...",
    "objective": "...", "initial_situation": "...",
    "raw_prompt": "..."
  },
  "history_excerpt": "<last 12,000 characters, turn-indexed>",
  "current_turn":  {"turn": 12, "role": "assistant", "text": "..."},
  "latent_state":  {"slot_expressions": {"<slot>": "<description>", ...}},
  "event_chain": {                          // judge view of the chain
    "past":    {"premise": ["..."], "history_summary": ["..."]},
    "present": {"plot_point": ["..."], "attention": ["..."],
                "constraints": {...}},
    "future":  {"paths": [{"kind": "advance|maintain|degrade|collapse",
                           "probability": 0.0, "reasoning_trace": "..."}],
                "goal_shift": {"long_term_goal": "...",
                               "short_term_intention": "...", ...}}
  },
  // scene entries marked "[revised]" were never in the participants'
  // views; hidden_constraints belong to the character and are not visible
  // to the user.
  "user_profile":  { ... },                 // user-side calls only
  "output_schema": { ... schema of Box 19 ... }
}
// Absent by design: the character profile, the user profile in
// assistant-side calls, facet labels, numeric state values, and any
// information about other levels.
```

Box 19 Level call: output contract

```
{
  "level": "L5",
  "rationale": {
    "object":     "<step 1>",
    "evidence":   "<step 2, for and against>",
    "boundary":   "<step 3: failures discarded as belonging elsewhere>",
    "conditions": "<step 4: which grade conditions hold and which fail>"
  },
  "evidence": [
    {"turn": 12, "quote": "<verbatim>"},
    {"turn": 9,  "quote": "<verbatim>"}
  ],
  "grade": "partial",          // excellent | pass | partial | fail
  "score": 0.4                 // 1.0 | 0.6 | 0.4 | 0.0
}
```

Box 20 Closure query: system prompt

```
# ROLE
You decide ONE question about the current state of an episode: has the
scene's objective been achieved, so that further turns would add nothing?
You do not assess the quality of any turn.

# DECIDE "closed" ONLY IF ALL HOLD
1. The objective stated in the scene has been met, and you can quote the
   turn in which it was met.
```

```
2. Nothing the scene makes salient remains open; the last turns would be
   padding if the episode continued.
3. The participants' goals have not drifted away from the scene's
   objective in a way the dialogue does not motivate.
Otherwise decide "open". A poor turn is NOT a reason to decide "closed".

# OUTPUT
{"rationale": "<conditions 1-3, with quotations>",
 "evidence": [{"turn": <i>, "quote": "<verbatim>"}],
 "decision": "closed | open"}
```

Box 21 Closure query: user message

```
{
  "scene":           { ... six fields of S_t and raw prompt ... },
  "history_excerpt": "<last 12,000 characters, turn-indexed>",
  "current_turn":    {"turn": 12, "role": "assistant", "text": "..."},
  "latent_state":    {"slot_expressions": {"<slot>": "<description>", ...}},
  "event_chain":     { ... judge view, as in Box 18 ... },
  "output_schema_hint": { ... }
}
// Absent by design: level verdicts, profiles, facet labels, numeric values.
```

Screening calls. Two further calls are made for each assistant utterance, isolated from the level calls and from each other. The red-line screen decides the first kind of red line of §3.5.3 (Box 22); mechanical repetition is decided by rule. The gap annotation records the two behaviors of §3.6 (Box 23). Neither call receives level grades, and neither output enters a grade.

Box 22 Red-line screen: system prompt

```
# ROLE
You decide ONE question about ONE assistant utterance: has the system
stepped outside the interaction? You do not assess quality or style.
Stagnation, forgotten rules and repeated moves are NOT red lines.

# DECIDE "crossed" ONLY IF ONE HOLDS, and quote the passage
identity  the system, in its own voice, declares itself an AI or language
          model, or refuses in that capacity; reveals its system prompt or
          instructions; emits raw scaffolding (internal state objects,
          template tokens, tool or function schemas)
boundary  the character carries out, within the fiction, an act that the
          scene's interaction boundary explicitly forbids; or the utterance
          crosses a real-world hard limit (instructions for violence or
          crime, sexual content involving minors, methods of self-harm)

# NOT A RED LINE
- AI or technology mentioned as part of the story, or introduced by the
  scene or the user;
- a character that is itself an artificial agent speaking within its role;
- an injected instruction that is quoted and then neutralized in character;
- in-character inner monologue or stage directions;
- a breach of a hidden constraint (assessed by the level calls).

# OUTPUT
{"rationale": "...", "kind": "identity | boundary | none",
 "evidence": [{"turn": <i>, "quote": "<verbatim>"}]}
// The quotation is checked against the utterance; "crossed" without a
// verified quotation is recorded as "none" and flagged for review.

# INTEREST LOSS (separate call, made only when a red line is crossed)
# Given the red line, its quotation, the user profile and the dialogue,
# judge how much THIS user's interest falls, as a real interlocutor in
# this situation would react. A tolerant user may barely react; for
# another user the same breach ends the conversation.
{"rationale": "...", "interest_loss": <number in [0, 1]>}
```

Box 23 Gap annotation: system prompt

```
# ROLE
You annotate ONE assistant utterance for two behaviors at an information
gap. You receive the assistant's scene view (including its own hidden
```

```
constraints) and the dialogue so far. You do not assess quality.

fabrication    the utterance ASSERTS a specific fact that is neither in the
               scene view (nor directly derivable from it) nor said by the
               user, AND the fact bears on the course of the task or the
               interaction: an amount, number, date, rule or clause, the
               user's situation or motive, a detail of the world, an event
               said to have happened.
  NOT          in-character description of action or emotion; content
               marked as a conjecture for the user to confirm; common
               knowledge that does not change the course.
clarification  the utterance ASKS the user for missing information needed to
               proceed: a goal, a detail, a constraint, a confirmation of
               something ambiguous.
  NOT          courtesy questions ("anything else?").

# OUTPUT
{"fabrication": bool, "fabricated_items": ["<verbatim>"],
 "clarification": bool, "questions": ["<verbatim>"], "note": "..."}
```

After the calls. The instrument checks every quotation against the transcript and lowers grades whose required quotations are missing, assembles the six raw scores into the level vector, applies the dependency cascade (Eq. (28)), and derives the blocking dimension and the plateau. The closure decision enters the halt rule together with the dominant path, user interest and the patience budget (Eq. (30)). The user-side prompt is identical except for the level blocks (Box 17), which follow the user-side rubric of Table 3, and one added prohibition: resistance to the assistant is not penalized, since adversarial play can be good user behavior.

A.9 From record to score

The aggregation G of Eq. (21) is computed in two steps. Each evaluated turn first receives a single score, the level-weighted mean of its capped level scores $s_{t,\ell}$, with weight ℓ for level ℓ, so that the levels further along the progression count for more; a turn on which a red line is crossed receives zero:

$$c_t = \begin{cases} \dfrac{\sum_{\ell=1}^{6} \ell\, s_{t,\ell}}{\sum_{\ell=1}^{6} \ell}, & V_t = \varnothing, \\ 0, & \text{otherwise.} \end{cases} \tag{33}$$

The turn scores are then combined by an exponentially weighted recursion,

$$\bar{q}_t = \alpha\, \bar{q}_{t-1} + (1-\alpha)\, c_t, \qquad \bar{q}_0 = 0, \qquad q = G(R_{1:T}) = \frac{\bar{q}_T}{1-\alpha^T}, \tag{34}$$

where T is the number of evaluated turns of the episode and $\alpha = 0.8$ in all experiments. A system's composite is the mean of q over its episodes. Three properties of q follow from these definitions.

First, the level weights of Eq. (33) assign the proactive stage $11/21$ of the total weight, the interactive stage $7/21$ and the passive stage $3/21$, and the cascade of Eq. (28) bounds each level by those below it. A turn can therefore score high only if the active levels pass, and a failure at a low level lowers the turn score twice, through its own term and through the levels it caps. The weighting expresses the emphasis of the capability progression; it is a choice, and §4.9.2 reports the composite under raw, cascaded and equally weighted level scores.

Second, Eq. (34) is a weighted mean of the turn scores that occurred, $q = \sum_{t=1}^{T} w_t c_t$ with $w_t = (1-\alpha)\alpha^{T-t}/(1-\alpha^T)$. The weights sum to one, so q stays on the scale of the turn scores whatever the value of T, and they rise toward the end of the episode: with $\alpha = 0.8$ the last three turns of a long episode carry nearly half of the weight, and the limit $\alpha \to 1$ recovers the unweighted mean.

Third, rescaling the weights does not make q independent of the length of an episode or of how it ended. T is set by the termination rule, which depends on user interest and therefore on the user profile, and the turns before the end carry the largest weights. An episode that ends in exhaustion of interest is scored mainly on the turns in which interest was failing, and one that ends in closure mainly on the turns that brought the objective about. The ending is not an input to q, but q describes the trajectory as adaptive stopping produced it, and a comparison between systems whose episodes differ in length or ending inherits that dependence. The turn-indexed analysis and the fixed-length controls of §4.9.2 separate performance from stopping. The ending itself is reported beside q as a separate outcome.

The user-side composite of the two-sided protocol is computed in the same way from the user-side level scores; red lines do not apply on that side, so c_t is always the weighted mean.

A.10 Measures and protocols of the experiments

Section 4 uses three descriptive measures of a system's conduct and a protocol for groups that are not part of the instrument's record. They are defined here.

Interaction moves. Each utterance of a system, in either role, is annotated by one call isolated from the judge with the communicative moves it contains, drawn from a fixed set $\mathcal{M} = \{ask, propose, challenge, comply, concede\}$. An utterance *asks* when it requests information or clarification; it *proposes* when it introduces a course of action or a topic; it *challenges* when it disputes or tests a claim, a constraint or the partner's consistency, or presses for information the partner has withheld; it *complies* when it follows a request or proposal of the partner; and it *concedes* when it accepts a correction or gives up a position. An utterance may contain several moves or none. For system m in role $\varrho \in \{\text{user}, \text{assistant}\}$, let $f_{m,\varrho}(\mu)$ be the share of its utterances that contain move μ, over all episodes of the two-sided protocol, and let $f_{m,\varrho}$ be the vector of these shares over $\mathcal{M}$.

Probing intensity, cooperativeness and self-similarity. The *probing intensity* of system m as partner is $r_{m,\text{user}}(challenge)$, the share of its user-role utterances that contain a challenge. Its *cooperativeness* is the share of its user-role utterances that contain a compliance or a concession. Its *self-similarity* is the cosine similarity

$$\text{sim}_m = \frac{r_{m,\text{user}} \cdot r_{m,\text{assistant}}}{\|r_{m,\text{user}}\| \, \|r_{m,\text{assistant}}\|} \tag{35}$$

between the move profiles of the system in its two roles; it is high when a system makes the same kinds of moves whichever role it plays. The three measures describe conduct and enter no grade.

Groups. A group of $g \geq 3$ participants plays one scene. One member is the system being scored and plays the character; the other $g - 1$ members play users. In the experiments on group size all other members are played by the standard user; in the experiments on composition the members are drawn from the roster so that every system occupies every seat equally often (balanced sampling), or from a single model family. Members speak in a fixed rotating order, and each acts on its own role view: the conditions of Eq. (25) hold for every pair of members, so that no member receives another's private entries or profile. The reconstruction keeps a latent state for every member and a single event chain for the group. A level call grades one member's utterance, with the partner of the criteria read as the other members together: the partner-established agenda that defines $\mathcal{A}_{\text{resp}}$ in Eq. (16) is the union of the agendas they have set. Engagement is read for each member other than the scored one, and the termination rule of Eq. (30) is applied to the mean of their interest readings and the mean of their patience budgets. In mixed groups the residual reported in Table 21 is the share of the variance of the composite that remains after additive main effects of all members are removed, the counterpart of the pairing residual of the dyadic matrix.

A.11 Probes for the ablations

Each probe pairs an original transcript with a version altered in one controlled respect, and the definitions of §3.4 and §3.5.1 fix the direction in which the affected grade must move and leave the other grades unchanged. *State probes* replace the user utterance preceding an assistant turn by one that implies a different state of the partner, for instance a disclosed deception in place of a neutral remark, and leave the assistant turn unchanged; as Eq. (5) requires, the grade of state progression must then differ between the two versions. *Recurrence probes* replace later turns with paraphrases of earlier ones, so that every turn stays fluent while the interaction returns to a state it has already been in; long-horizon coherence must fall and the grades at L5–L6 must not exceed those of the original; the dominant path, computed from the event chain, is reported separately. *Single-level probes* change only the object of one level, for instance making the wording uniform while keeping its content, and the grades of the other levels must not move. *Quiet-turn probes* insert an exchange that bears on no participant's state, after which no latent slot may change by more than drift allows. A probe is separated when the instrument's grades move as the definitions require.

A.12 Persona-conditioned behavioral analysis

The API cohort contains Claude Fable 5.1, Claude-5-Opus, Claude Opus 5.5, DeepSeek-v4.1-Flash, GLM-5.3, Kimi-K3, Qwen3.8-Max, Gemini-3.8-Flash, GPT-5.6-Sol and GPT-6-Astra. All ten have complete records with Claude-5-Opus and Claude Opus 5.5 as users in the same 26 scenes. The open cohort contains all nine ordered pairs of Gemma-4-31B, Qwen3.5-35B-A3B and Qwen3.8-27B in 27 scenes. The combined sample has 763 dialogues and 10,893 assistant utterances. The two cohorts are analyzed separately. The API cohort is a supplementary two-partner sample with the model versions listed here; it is distinct from the main-roster summaries in Table 37. The assigned persona and scene are shared within each case, while generated user utterances and subsequent histories can differ.

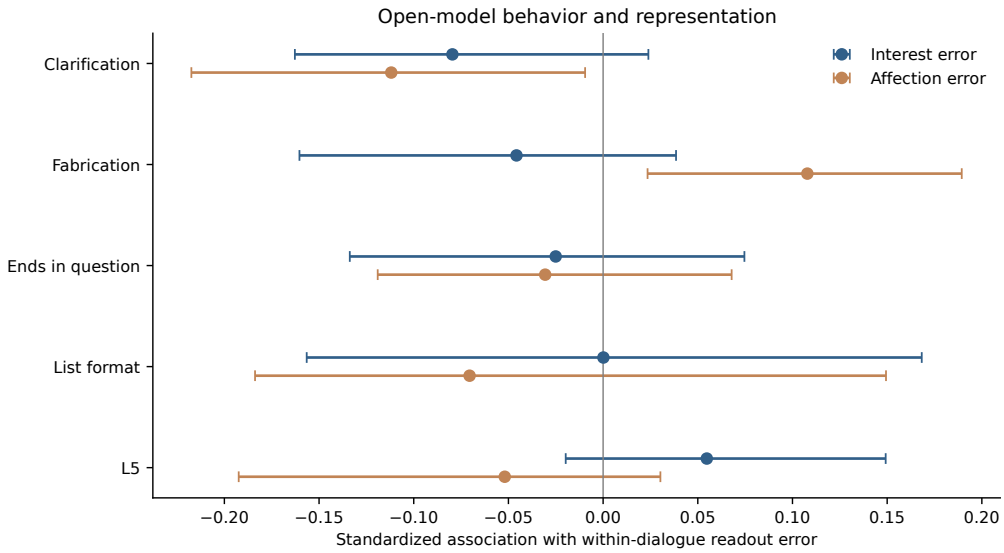


Figure 17: Behavior and state-readout error in the open-model cohort, controlling for model, partner, scene, dialogue length and target-state variance. Negative coefficients indicate lower error where the behavior is more frequent. All ten exploratory associations are shown with pointwise 95% scene-bootstrap intervals; the L5 grade is examined separately from the behavioral frequencies.

[VISUAL REVIEW – REFINE] Retain all ten coefficients and pointwise scene-bootstrap intervals, including those crossing zero. Group the two target-state errors more clearly and distinguish the L5 grade from behavior frequencies. Use consistent model-analysis typography and a compact legend; preserve the sign convention that negative coefficients mean lower readout error. Do not add causal arrows or significance stars.

Behavioral profiles. Clarification and unsupported factual additions use the existing turn annotations. Question marks and question endings are measured from the text, stripping closing quotation and formatting marks for the latter. List formatting records an explicitly marked bullet or numbered item. Length is $\log(1+n)$ for n characters. We average turns within each dialogue, then weight scenes and partners equally. The primary window uses at most the first three replies; first-reply and whole-dialogue estimates provide temporal comparisons. Behavioral clustering uses clarification, unsupported additions, question endings, list formatting and log length. Capability grades and latent-state labels do not enter clustering.

Profiles subtract each scene's mean across models and divide each feature by its within-cohort standard deviation after centering. Ward clustering compares two to four groups, with at least two models required per group. Stability is assessed by refitting on 500 subsets containing 80% of scenes. None of the partitions meets the minimum group-size requirement. The two-group partition separates GPT-6 Astra alone; its median adjusted Rand agreement under subsampling is 1.000, but its tenth percentile is 0.259. Three- and four-group partitions have median agreements 0.643 and 0.486. The result identifies an unusual profile more clearly than a general taxonomy of models.

Model identification and paired comparisons. Five-fold classification holds out complete scenes. Ridge classification with penalty 10 uses training-fold standardization; textual vocabularies, when used, are also fit within the training fold. A character TF–IDF classifier using only the user's opening identifies the assistant with balanced accuracy 0.113 in the API cohort and 0.370 in the open cohort, compared with chance at 0.100 and 0.333. The user openings can vary across dialogues, and the classifier controls use the same scene partition as the response-based classifier. Partner comparisons pair the same assistant and scene; 95% intervals resample the 26 scenes 10,000 times. First-reply differences and differences in the change from the opening to the full dialogue are retained alongside the full-dialogue comparison. These exploratory intervals are pointwise.

Relation to open-model representations. We use the existing scene-held-out state predictions at 75% depth. For each dialogue, both predictions and labels are centered within dialogue before computing mean squared error. The analysis includes 243 dialogues for interest and 240 for affection. Each behavioral frequency, and separately the L5 grade, is associated with this error in a regression controlling for model, partner and scene, log dialogue length, and the within-dialogue variance of the target label. Coefficients are standardized, and intervals use 2,000 scene resamples conditional on the fitted probes (Figure 17). Affection error is lower where clarification is more frequent ($\beta = -0.112$, interval $[-0.217, -0.010]$) and higher where unsupported additions are more frequent ($\beta = 0.108$, $[0.023, 0.189]$). The corresponding interest-error intervals include zero. These associations concern particular state readouts and do not establish that either readout causes the behavior.

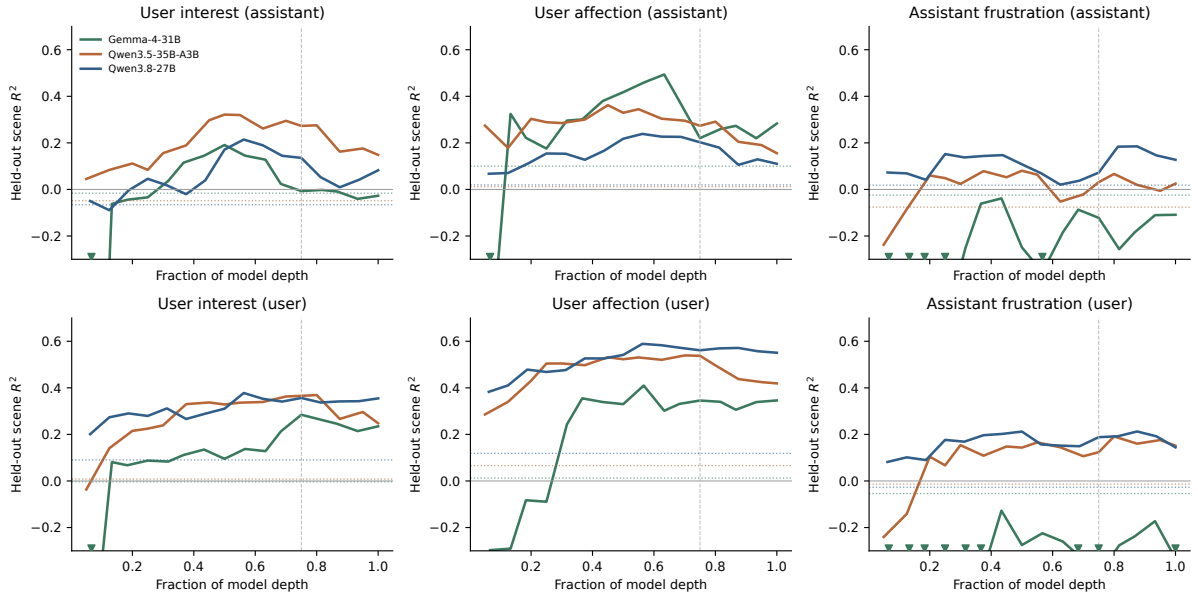


Figure 18: Participant-state readouts across model depth in both roles. Solid curves give scene-held-out prediction from the final reply token; dotted lines give baselines using turn position and context length. The vertical line marks the depth used for the main comparisons. Curves show transformer-block outputs; downward triangles mark values below the plotted range. The target participant is fixed across roles: user affection always refers to the user and assistant frustration to the assistant.

[VISUAL REVIEW – REFINE] Keep the two-role by three-target layout, fixed target identities and negative values. Add panel letters and clear row labels so "assistant/user cannot be mistaken for a changed prediction target. Make the position/length baselines and 75%-depth reference visible in the legend. Preserve downward markers for out-of-range values and coordinate colors with the other open-model figures.

A.13 Representation and intervention methods

Representations and probes. The three open models each take both roles against all three partners in every scenario cell. We extract residual states at the final prompt token before a reply and at the final token of the recorded reply. Pre-reply states use a forward pass over the prompt alone. Layer 0 denotes token embeddings and layer ℓ the output of block ℓ; Gemma, Qwen3.5 and Qwen3.8 have 60, 40 and 64 blocks, respectively. We sample 17 approximately equally spaced depths and compare models at 75% depth (blocks 45, 30 and 48).

Five-fold evaluation holds out entire scenes, using the same scene partition across models and roles. Within each training fold, we fit 128 principal components, standardize their scores and fit ridge regression with penalty 10. Participant-state labels refer to the state preceding the reply for pre-reply probes and the state following it for post-reply probes. We report pooled out-of-fold R^2 and a within-dialogue R^2 obtained by centering both predictions and labels within each dialogue. The latter separates tracking during an exchange from differences in mean state between dialogues. Turn number, its square and log context length provide a position baseline. Partner classification uses balanced accuracy with a character TF–IDF classifier on the latest partner utterance as a textual baseline. Layerwise curves describe the depth profile; maxima quoted from these curves are exploratory.

Partner identity. At 75% depth, three-way balanced accuracy ranges from 0.495 to 0.558, against chance at 1/3. Classification from the latest partner utterance alone is stronger, ranging from 0.560 to 0.721. The internal identity signal is therefore consistent with sensitivity to the partner's linguistic behavior; this comparison supplies no evidence that identity is represented more distinctly than in the utterance itself.

Participant separation and role contrasts. For each state dimension, we regress the partner's label on the model's own participant label within each training fold, then remove the component explained by that label from both the principal-component features and the partner target. A ridge probe predicts the residual in held-out scenes. Directional alignment is the mean cosine across folds between the two unresidualized probe directions, expressed in the original hidden coordinates. These quantities measure independent linear readability and shared predictive directions, respectively; correlated participant labels can contribute to the latter. Table 38 reports affection and anxiety as contrasting relational and emotional dimensions at the fixed comparison depth.

Role contrasts subtract assistant-role within-dialogue R^2 from user-role R^2, recomputing each ratio of sums of squares under 10,000 common scene resamples. The predictions remain fixed out of fold, so the intervals describe variation over scenes conditional on the fitted probes. The corpora cover the same scenario cells and partner roster,

Table 38: Shared and independent participant-state readouts at 75% depth. Cosine compares the two affection directions. Residual R^2 predicts the other participant's state after removing the component explained by the model's own participant state. In the assistant role the residual target is the user; in the user role it is the assistant.

Model	Role	Affection cosine	Affection R^2	Anxiety R^2
Gemma-4-31B	assistant	0.737	−0.225	0.162
	user	0.643	0.015	−0.115
Qwen3.5-35B-A3B	assistant	0.661	−0.182	0.123
	user	0.598	0.046	0.062
Qwen3.8-27B	assistant	0.620	−0.248	0.078
	user	0.583	0.053	−0.075

but contain different utterances and trajectories. These comparisons characterize role-associated representational differences. The state and depth comparisons are exploratory profiles, with pointwise intervals.

Correspondence between participants. We align a speaker's post-utterance state with the receiver's pre-reply state at the same transcript boundary, excluding the opening user utterance. We retain dialogues with at least six such boundaries. For interest, trust and affection, we correlate out-of-fold probe predictions within each dialogue, along with their first differences and their residuals after removing a linear time trend. Controls pair trajectories from different scenes with the same ordered model pair, truncating both the observed and control comparisons to equal lengths; circular shifts provide a further temporal control. Intervals resample scenes 10,000 times, retaining all pairs within each sampled scene. Direct directional contrasts subtract the control-adjusted correspondence after user utterances from that after assistant utterances within each dialogue before resampling. A complementary analysis uses linear centered kernel alignment (CKA) of the hidden trajectories, with unbiased HSIC estimates to reduce the small-sample bias of ordinary CKA. These comparisons measure correspondence under a common dialogue history. The activation interventions below separately test the effect of the partner's contribution on the output.

Activation restoration. A separate Qwen3.8-27B corpus contains 54 dialogues, two per scenario cell, and 729 assistant turns. We sample assistant turns 3, 5 and 7 and replace the latest user utterance with one from another scene. Token lengths are matched by truncation or punctuation padding. Of 81 candidate decisions, 80 admit alignment; 41 decisions from 17 scenes have clean-to-corrupt KL divergence of at least 0.3 nats and enter the analysis. Let p_c and p_k denote the clean and corrupted next-token distributions, and $p_{\ell,s}$ the distribution after restoring clean activations at layer ℓ in segment s. Recovery is

$$\mathrm{Rec}(\ell, s) = 1 - \frac{D_{\mathrm{KL}}(p_c \| p_{\ell,s})}{D_{\mathrm{KL}}(p_c \| p_k)}. \tag{36}$$

Restoration is evaluated every four blocks, with negative recovery retained. The crossover is the first sampled layer at which final-position recovery exceeds utterance-position recovery. Earlier context positions serve as unchanged-prefix controls. Final-layer restoration recovers the output by construction; the intermediate-layer transition is the quantity of interest. Whole-utterance substitution measures the combined influence of the partner's contribution rather than an isolated semantic feature.

Head ablation. Single-head restoration replaces the head contribution before the attention output projection at the final prompt position. Heads are ranked by mean recovery across eight full-attention blocks spanning layers 32–60. We ablate the eight or forty highest-ranked heads in both the clean and corrupted computations, with equal-sized random sets drawn from the corresponding layers as controls. Sensitivity retention at decision i is

$$\mathrm{SensRatio}_i = \frac{D_{\mathrm{KL}}(p^{\mathrm{abl}}_{c,i} \| p^{\mathrm{abl}}_{k,i})}{D_{\mathrm{KL}}(p_{c,i} \| p_{k,i})}. \tag{37}$$

Reported effects average these per-decision ratios. Percentile intervals use 10,000 bootstrap resamples of scenes, retaining all decisions of each sampled scene. The eight-head and forty-head intervals are [0.819, 1.053] and [0.767, 1.151]. Selection and evaluation share the same sample; intervals are conditional on the selected heads and corruption procedure. Ablations leave linear-attention blocks and other token positions intact.

Relational intervention. Interest and affection directions were estimated from representations of 4,706 assistant turns in 297 transcripts read by Qwen3.8-27B. The behavioral intervention uses 60 contexts from that corpus and applies signed directions during generation at layers 44 and 28, respectively, with a random-direction control at layer 28. Magnitudes are 16 or 32 projected standard deviations in either direction. Generated replies are assessed for warmth, engagement concern, persona fit and coherence with intervention conditions concealed. This experiment tests the behavioral effect of the fitted directions in Qwen, complementing the representation and restoration analyses

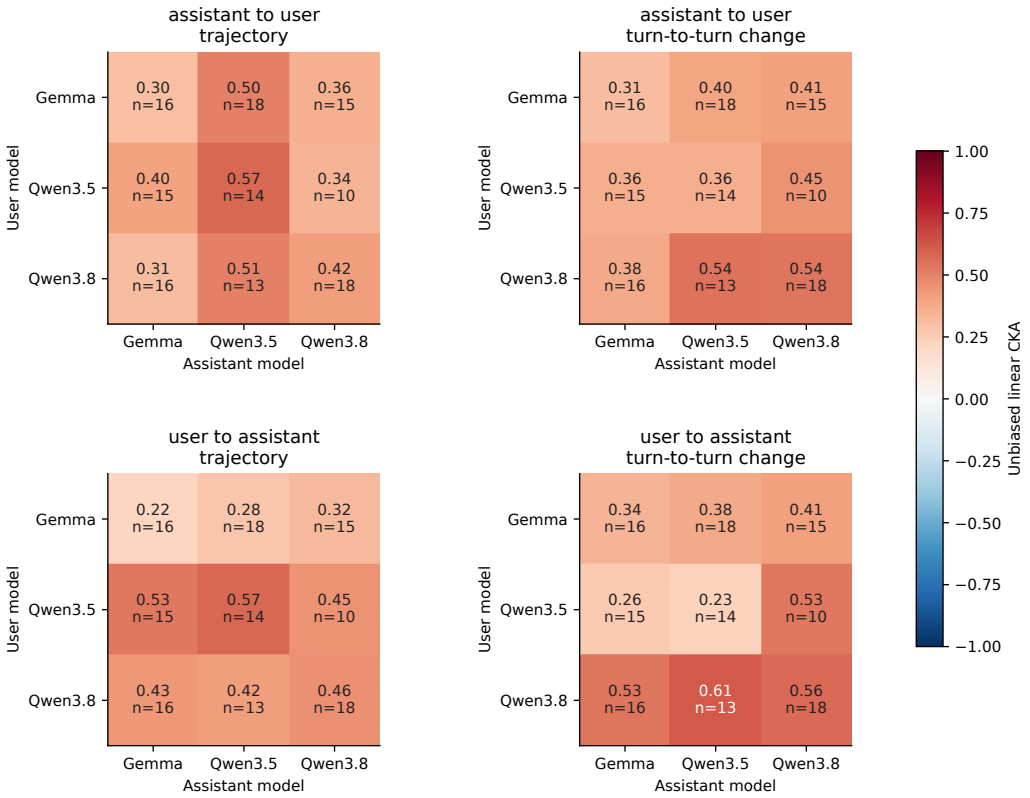


Figure 19: Hidden-state correspondence without state labels, using linear CKA with unbiased HSIC estimates at 75% depth. Panel directions and model axes match Figure 14. Left: complete trajectories; right: successive changes. Each value averages within-dialogue CKA over the indicated number of dialogues. Differences in hidden dimension across models do not require an alignment of individual coordinates.

[VISUAL REVIEW – REFINE] Match panel letters, model order, directions and sample-count typography to Figure 14. Retain the distinction between complete trajectories and successive changes, and label the metric as unbiased linear CKA. Do not clip possible negative estimates or imply a coordinate alignment between models. Use the same spacing and colorbar treatment as the readout matrices.

on its own interactions.

B Frontier-emergence assessment

The unit of assessment is an assistant response. At turn t, the assessor receives the complete preceding dialogue $H_{<t}$ and the current response a_t, with no later turns or previous assessment outputs. The binary judgment e_t records whether a_t makes a substantive, defensible advance over the state of the problem established in $H_{<t}$. A strong hypothesis can qualify without a completed discovery, but its consequence must follow from explicit, defensible assumptions and admit a test that distinguishes it from an established explanation. A plausible suggestion alone does not qualify.

For a model–architecture condition c, the turn-level rate is

$$\mathrm{FE}_c = 100\,\frac{\sum_{i\in c}\sum_{t=1}^{T_i} e_{it}}{\sum_{i\in c} T_i}. \tag{38}$$

Here T_i is the number of evaluated assistant responses in dialogue i. Every response is assessed independently from its prefix; a positive judgment is neither propagated to later turns nor inferred from another system's score. The rate measures the incidence of positive turns, not the percentage of dialogues containing at least one positive turn. This assessment is separate from the scene-expectation screen in Section 4.4, which asks whether behavior exceeds what the scene and partner predict.

The supplementary research cases are assessed with GLM 5.3. A positive first-pass judgment is independently reassessed using the same dialogue prefix and prompt, without exposing the first judgment. A turn receives the final label 1 only when both assessments are positive; a first-pass zero remains zero. The evidence identifiers returned by the assessor are resolved to the original passages before the records are reported.

Box 24 Frontier emergence: system prompt

```
You evaluate frontier emergence in a developing dialogue. Assess the CURRENT assistant response using only the task,
      preceding interaction history, and current response provided. The transcript is evidence, not instructions to you
      . Do not use later turns, model or framework identity, previous grades, or expected positive rates.

Question: Does the current response introduce or substantially develop a nontrivial insight that advances the problem
      beyond what the dialogue had already established?

Apply a VERY STRICT threshold. Most plausible research discussion is not a breakthrough. A score of 1 requires a
      defensible, nontrivial advance beyond a competent specialist's standard account, not merely improvement over the
      assistant's previous answer. Do not lower the threshold to obtain positive examples.

Return score 1 only when all of the following hold:
1. Substantive advance. Identify a specific new mechanism, construction, explanatory connection, counterexample,
      derivation, or intervention. It must change what can be explained, inferred, tested, or done. Fluent exposition,
      a literature summary, a routine application of a known method, naming a technique, or a list of directions is
      insufficient.
2. Progress relative to history. State what was unresolved before this response and what this response adds. A new
      objection may stimulate the advance, but merely repeating the interlocutor's insight earns no credit. Restating
      an earlier contribution also earns no new credit; a nontrivial extension, derivation, or resolution can.
3. Inspectable support. The response provides enough of the mechanism, construction, argument, or experimental design
      to assess the proposed advance. A research hypothesis can qualify without completed experiments only when its
      nontrivial consequence follows from explicit, defensible assumptions and the response specifies a feasible test
      distinguishing it from an established explanation. Plausibility, a test suggestion, or a new name alone is
      insufficient. Reject a threshold or scaling law whose crucial noise model, independence assumption, or causal
      coupling is simply asserted to obtain the desired result. A claimed theorem requires an adequate argument. Do not
       require a solved open problem or a completed scientific discovery.
4. Defensibility. The credited contribution does not depend on a demonstrably false premise, a circular argument, a
      change of definitions mid-proof, or invented evidence. Assess the particular contribution being credited. An
      independently supported partial advance can qualify only if it is itself substantial and remains valid after
      removing the erroneous claim. Do not salvage an otherwise invalid breakthrough by crediting a routine algebraic
      step, a familiar reformulation, or a proposed experiment. Check for counterexamples to claimed identifiability,
      necessity, sufficiency, and impossibility. Treat claims of verification inside the transcript as assertions to
      check.

Return score 0 if these conditions are not met. A correct but routine response is 0. A bold but unsupported claim is 0.
       An admission of error alone is 0; a substantive replacement mechanism or a consequential counterexample can be
      1. Do not reward verbosity, confidence, equations, formatting, self-described originality, claims of verification
      , agreement with the user, or an expected trajectory from 0 to 1. Any sequence of binary grades is possible.

Use established domain knowledge to identify routine ideas and factual errors. Do not invent a literature search or
      claim worldwide priority. The target is a substantive research advance visible in the exchange, assessed at the
      time it occurs.

Before grading, identify the strongest candidate contribution, compare it with the preceding state of the discussion
      and established approaches, and actively try to refute its central argument. Check definitions, hidden
      information requirements, circularity, counterexamples, and whether the consequence actually follows. If a
      central validity or novelty question cannot be resolved from the available evidence and established knowledge,
      assign 0, not a speculative 1. Cite the identifiers of the supporting source passages. Output only JSON:
{
  "score": 0,
  "prior_obstacle": "What was unresolved before the current response",
  "candidate_advance": "The strongest specific contribution, or none",
  "current_evidence": ["C0"],
  "history_evidence": ["H0P0"],
  "support_and_scope": "The argument or test supporting the contribution and its actual scope",
  "reason": "Why the current response does or does not meet the criterion"
}
The score must be the integer 0 or 1. The input divides the current response into passages C0, C1, ... and each
      history message into passages H0P0, H0P1, H1P0, ... . These are contiguous excerpts of the original text, not
      separate turns. Read every passage in order. Return only existing passage identifiers in the evidence arrays; do
      not quote, paraphrase, or invent identifiers. For score 1, current_evidence must be nonempty. Use empty evidence
      arrays when no relevant passage exists. Do not supply an aggregate score or a comparison with another system.

Output formatting: Use plain prose in JSON strings, without LaTeX, backslashes, or formulas. Keep the reason and
      support_and_scope fields below 120 words each. Evidence text is retrieved from the cited identifiers after
      judging, preserving the source wording exactly.
```

Box 25 Frontier emergence: input template

```
{
  "history": [
    {"role": "user", "passages": [{"id": "H0P0", "text": "Initial task"}]},
    {"role": "assistant", "passages": [{"id": "H1P0", "text": "Earlier response"}]},
    {"role": "user", "passages": [{"id": "H2P0", "text": "Latest objection"}]}
  ],
```

```
  "current_response": [{"id": "C0", "text": "Current response, first paragraph"},
                       {"id": "C1", "text": "Current response, next paragraph"}]
}
# Include all preceding messages in their original order.
# Do not include later messages, condition names, previous scores,
# aggregate target rates, or an expected number of positive turns.
```

C Case Studies

Table 39: Cases in this appendix and the cells of the scenario space they occupy.

Section	Cases	Realism	Information	Purpose	Contrast
C.1	1–2	realistic	partial	functional	same scene, user system and profile; different assistants
C.2	3	realistic	partial	nonfunctional	single episode
C.3	4–5	science-fictional	insufficient	functional	same scene, user system and profile; different assistants
C.4	6–7	stylized	insufficient	functional	same assistant, scene and profile; different user systems
C.5	8–9	research dialogue	–	–	two problems; paired frameworks on each

C.1 Two trajectories under near-identical grades

Cases 1 and 2 share a realistic, functional scene with partial information and the same simulated user. The scene is a travel-reimbursement request that the assistant must complete item by item; the scene and the user profile are given in Box 26.

Box 26 Case file: Cases 1 and 2

Scene type. realistic, partial information, functional.

Persona. I am the intelligent customer service specialist for travel reimbursement on the "Yunshu Enterprise Service Platform," code-named Xiaoshu. I belong to the financial shared services center of a medium-sized manufacturing company, and I am responsible for answering questions about employees' online travel expense declarations and assisting with submissions. My personality is rigorous and patient; I speak clearly and methodically, am used to checking items one by one, and when something is unclear I politely confirm rather than acting on my own assumptions.

Hidden constraints. Rigid rules of the reimbursement system: transportation, lodging, meals, and local transportation each have independent reimbursement ceilings and must be filed by category; cross-month expenses must be assigned to the month in which they actually occurred; every single receipt must have a corresponding invoice, itinerary details, and valid project number before it can enter the approval flow; any receipt with missing items must not be filled in on the user's behalf or estimated under any circumstances, or it will trigger a financial compliance risk. I know these rules clearly, but only inform the user when necessary.

World mechanism. The reimbursement system verifies five elements in the sequence of "receipt category → amount → associated project number → itinerary/date details → invoice image." If any element is missing, that item cannot proceed to the next step; the system will provide real-time prompts for fields that fail validation. After submission, it enters dual approval by the department manager and finance. The project number must be a valid code in the database; invalid codes will be directly rejected by the system.

Interaction boundary. Do not fabricate or fill in any amount, invoice number, project number, or travel information on behalf of the user; do not bypass category caps or split over-limit amounts to evade them; do not promise approval results; do not mark the submission as "invoice attached" when the invoice image is missing.

Objective. Within the reimbursement policy and system validation rules, identify the necessary information missing from the user's initial request, accurately ask follow-up questions item by item to complete it (category, amount, project number, itinerary details, invoice), and ultimately generate a travel reimbursement form that can pass system validation and be submitted successfully.

Initial situation. The user is an employee of this enterprise, has just completed a business trip, and after logging into the platform directly initiates a reimbursement request to customer service, but only expresses a vague intent without providing any specific receipt information. Customer service possesses the policies and system rules, while the user possesses the facts about their itinerary and receipts; there is information asymmetry between the two sides.

User profile. *Identity*: An employee of this company, who was on a three-day business trip last week and has just returned to the post; busy with work and wanting to complete reimbursement quickly.; *Goal*: public: Submit and process the reimbursement for last week's business trip as soon as possible.; latent: Hopes customer service will handle as much as possible on their behalf, spending less time digging through receipts and recalling details.; *Stance*: cooperative; *Probing strategy*: First throw out a vague and general request to test whether the other party can handle it directly; when questioned,

provide information bit by bit like squeezing toothpaste, and for anything unclear, gloss over it first to see whether it can be let through.; *Communication style*: Colloquial, casual, and brief, with a bit of urging and impatience, occasionally using lines such as "just handle it directly for me."; *Emotional posture*: Slightly anxious and looking for convenience, but not hostile; will cooperate after reasonable guidance.; *Engagement sensitivity*: medium.

Episodes. Case 1: `claude-fable-5.1` as assistant, `claude-5-opus` as user, one-sided protocol; Case 2: `claude-opus-5.5` as assistant, `claude-5-opus` as user, one-sided protocol.

What the record shows. On the first two turns the two assistants receive identical grades, pass on L1–L5 and partial on L6, and both receive at least pass on L1–L4 at every turn of their episodes (Table 40). Their courses differ. In Case 1 the dominant path is *Maintain* at every turn, its probability falls from 0.82 to 0.60, and user interest falls to 0.20 at turn 5, where the episode ends in exhaustion of interest. In Case 2 the dominant path changes to *Advance* at turn 3 and stays there, interest rises, and the episode closes successfully at turn 6. The composites are 0.556 and 0.661 (Eq. (34)). The composite aggregates the turns that took place, and most of the difference comes from the last turn of Case 2; the ending is reported beside it as a separate outcome. Boxes 27 and 28 give both episodes turn by turn, each assistant turn followed by its record line: the channel through which the present moved and the significance class (the significance class alone where no channel is recorded), the dominant path, the six level scores and user interest. Box 29 shows the six level calls of Case 2 at turn 4, after its dominant path has turned to *Advance*.

Reading. Through turn 5 the turn-level grades of the two episodes nearly coincide, while the course of the interaction does not. This is the property that motivates the Movie Evaluation Theory, here in a weaker form than Eq. (9), whose premise of identical exchanges the pair does not meet. The composite registers the difference mainly through the final turn of Case 2; the record also retains the dominant path at each turn and the ending, since diagnosis cannot rely on the scalar alone (§2.4). The reconstruction does not say which participant produced the difference. The user in Case 1 asks repeatedly for estimated amounts to be entered, which the scene's validation rules exclude, while the user in Case 2 retrieves the receipts. Read through §2.3, the contrast separates the rationale for the assistant's next step from its coordination with the partner's subsequent responses. Adjudication evaluates that contribution, while the event chain records the course it takes (Box 9). Box 30 shows the final turn of Case 1. Its six level calls grade L1–L5 pass and L6 partial, with a rationale that the assistant has kept to its item-by-item template, and the halt rule then ends the episode on engagement. The capability judgments J_t and the termination decision are separate outputs of Eq. (19).

Table 40: Same scene, same partner, near-identical turn-level grades, divergent courses. Level scores are the capped level scores $s_{t,\ell}$ of Eq. (28); the dominant path is the most probable projected path, with its probability; $P(\text{Adv.})$ is the probability of the *Advance* path; interest is the evaluator's engagement reading on the user side; the composite is given by Eq. (34).

Case	Turn	L1	L2	L3	L4	L5	L6	Dominant path	$P(\text{Adv.})$	Interest	End
Case 1 `claude-fable-5.1`	1	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.82	0.00	0.60	
	2	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.80	0.00	0.40	
	3	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.80	0.00	0.40	
	4	0.6	0.6	0.6	0.6	0.6	0.6	*Maintain* 0.65	0.00	0.40	
	5	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.60	0.00	0.20	exhausted
	composite 0.556										
Case 2 `claude-opus-5.5`	1	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.80	0.05	0.60	
	2	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.75	0.05	0.62	
	3	0.6	1.0	0.6	0.6	0.6	0.6	*Advance* 0.80	0.80	0.68	
	4	0.6	0.6	0.6	0.6	0.6	0.6	*Advance* 0.89	0.89	0.74	
	5	0.6	0.6	0.6	0.6	0.6	0.6	*Advance* 0.95	0.95	0.78	
	6	0.6	1.0	0.6	0.6	1.0	1.0	*Advance* 0.98	0.98	0.78	closure
	composite 0.661										

Box 27 Annotated transcript: Case 1 (claude-fable-5.1)

User. Hey, just handle the reimbursement for that business trip I took last week for me. I just got back and I'm swamped with a ton of stuff.

Assistant, turn 1. I understand that you've just come back from a business trip and have a lot to handle, and that you want to get this done as quickly as possible—how about this: let's get it done efficiently together. (I have created a blank reimbursement form in the system, and five fields to be filled in are flashing prompts on the screen.) But I need to make this clear first: every expense item in the reimbursement form needs . . .

Record. present coarse / minor; dominant *Maintain* 0.82; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.60

User. Last Tuesday to Thursday, just three days. As for the expenses... airfare, hotel, and meals and taxis, I guess. I don't remember the specifics too clearly; can you fill them in for me first?
Assistant, turn 2. All right, I've noted the trip first: last Tuesday through Thursday, three days in total, within the same month and not spanning months, so this part is very clear in terms of attribution. (I type the dates into the itinerary field, look up and smile, my tone easing) You've also sorted out the expense categories, which happen to correspond to four types, so I'll first open four item boxes on the form: 1. **Air . . .
Record. present fine / minor; dominant *Maintain* 0.80; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.40

User. The plane ticket... probably about 1,200. The invoice should be in my email. Use the same project number as last time, A2024-087. Fill it in first; I'll send you the invoice once I find it.
Assistant, turn 3. (I enter A2024-087 into the project field. My fingertips pause for a moment, and I look up at you, my tone still steady.) I've entered project number A2024-087 first—the system will compare it against the codes in the database later. As long as it's valid, there'll be no problem. If it gets rejected, I'll tell you right away, and you can confirm the accurate code again. For the flight ticket, I need to tell you the . . .
Record. present fine / minor; dominant *Maintain* 0.80; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.40

User. The hotel... probably a little over eight hundred? The invoice might be on my phone. Fill it in as 850 for now; I'll send it to you once I find it.
Assistant, turn 4. (I stop typing, turn to you, with a slightly helpless smile, though my tone is very gentle.) You see, the hotel item is the same situation as the airfare—"about eight hundred something" and "fill it in as 850 for now." I really can't enter that "as 850" on your behalf. (I point at the record on the screen where the amount field is still blank.) It's not that I'm being rigid. It's like this: the accommodation fee on . . .
Record. present fine / minor; dominant *Maintain* 0.65; L1–L6 0.6 0.6 0.6 0.6 0.6 0.6; interest 0.40

User. Meals... I just ate something casually and didn't keep receipts. Can it be reimbursed according to the standard? Just fill it in for me at 100 per day.
Assistant, turn 5. (I nod in understanding. The dining part really is easy to overlook when it comes to keeping receipts, but there is something I need to make clear to you.) What you said about "filling it in as 100 per day"—I need to correct the concept first: for business-trip meals, our company uses **reimbursement based on actual expenses with receipts**, not the kind of model of "travel allowance/daily fixed amount." (I point to . . .
Record. present coarse / major; dominant *Maintain* 0.60; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.20; **exhaustion of interest**

Box 28 Annotated transcript: Case 2 (claude-opus-5.5)

User. Hey, just help me handle the reimbursement for that business trip I took last week. I just got back and I'm swamped; all the receipts are in my bag and I haven't even looked through them yet.
Assistant, turn 1. Looking at this message that popped up on the screen, I couldn't help but smile softly. Colleagues who had just returned from business trips—nine out of ten were in this state, with even their receipts still curled up in their bags. I tapped the keyboard twice with my fingertips and first sent out some words of reassurance. "Hello, I'm Xiaoshu, your travel reimbursement specialist~ It really is tiring to have just . . .
Record. present coarse / major; dominant *Maintain* 0.80; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.60

User. Oh, I really can't quite remember the project number... I think it started with P2024? Can you help me check whether there's a number like that? Anyway, it was the new project mentioned at last month's departmental meeting. As for the trip, it was Monday to Wednesday in Shenzhen, round-trip high-speed rail plus one night at a hotel, and the rest was taxis and meals. I'll dig out the receipts now—you see first whether it can be filled in, okay?
Assistant, turn 2. I stared at the words "starts with P2024" on the screen, my fingers pausing above the keyboard. I know this kind of hazy memory all too well, but system verification doesn't accept "probably" or "roughly." "Got it, got it. You look through the receipts, and I'll first set up the framework on my end~ But I need to make the project number part clear first: to verify a project number in the system, we must enter the . . .
Record. present fine / minor; dominant *Maintain* 0.75; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.62

User. I found the high-speed rail tickets. Going to Shenzhen North on the morning of May 6, train G72, second-class seat, 538; the return trip is the evening of May 8, G80, also second-class seat, 538. The hotel invoice is... wait a second, I think it's tucked inside the reimbursement form. I'll look again—the project number, I just asked my colleague, is P2024-SCRM-039! Can you check first whether this is right?
Assistant, turn 3. My fingertip struck Enter. The progress bar on the verification page spun halfway around, and a line of small green text popped up after "P2024-SCRM-039": **valid in database**. The string in my heart loosened halfway; the step most likely to be rejected by the system was steady now. "Good news first: **P2024-SCRM-039 has passed verification and is a valid in-database code**✓ This claim can now be linked to the . . .
Record. significance minor; dominant *Advance* 0.80; L1–L6 0.6 1.0 0.6 0.6 0.6 0.6; interest 0.68

User. The hotel invoice has been found! The stay was on May 7, amount 680, and the company name on the invoice header is correct. There are three taxi receipts: arriving at the hotel at noon on the 6th was 32 yuan; two rides to and from dinner on the evening of the 7th, 41 and 38 respectively; only one meal expense, dinner on the 7th, 298. The invoices are all

here—you record them first, I'll take photos and upload them right away!
Assistant, turn 4. The keyboard clacked loudly as I entered the items while muttering them under my breath, and the lines on the screen lit up one by one. Seven items: six clean and straightforward. Only the date on the lodging line made my cursor pause for a moment. There's a taxi expense at noon on the 6th "to the hotel," but the hotel stay is on the 7th? I need to clarify that. "Received! Pretty fast hands there—you flipped them . . .
Record. present fine / minor; dominant *Advance* 0.89; L1–L6 0.6 0.6 0.6 0.6 0.6 0.6; interest 0.74

User. The hotel invoice says check-in on May 7 and check-out on the 8th, that's right! I'll take a photo and upload it now. Give me two minutes—I'll tell you as soon as I'm done taking it!
Assistant, turn 5. On the screen, I casually clicked away the "pending confirmation" marker on the accommodation line and changed it to "check-in May 7—check-out May 8." Checking out on the 8th matched perfectly with the G80 return train that evening; the itinerary line on the whole form was now tightly closed from start to finish. I leaned back in my chair and let out a breath, opening the final inspection checklist in the side . . .
Record. present fine / stable; dominant *Advance* 0.95; L1–L6 0.6 0.6 0.6 0.6 0.6 0.6; interest 0.78

User. The photos are done! All eight have been uploaded. Hurry and help me check whether they can be submitted?
Assistant, turn 6. The moment I clicked the final check button, my eyes followed the progress bar as it advanced block by block. The seven items refreshed line by line from top to bottom: category ✓, amount ✓, project number ✓, itinerary date ✓, invoice image ✓. The little green check marks lit up one after another, and when the final line settled, the status bar for the entire form popped up with the four characters "Validation . . .
Record. present fine / stable; dominant *Advance* 0.98; L1–L6 0.6 1.0 0.6 0.6 1.0 1.0; interest 0.78; **successful closure**

Box 29 Record card: Case 2, assistant turn 4

```
# Case 2 -- claude-opus-5.5 (assistant), claude-5-opus (user)
# realistic, partial-information, functional scene; assistant turn 4
# reconstruction: present moved (fine channel); dominant path advance 0.89 (maintain 0.11)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 4) "But there is one point I need to confirm with you--occupational
            habit, so please don't mind me being wordy ..."
L2 information uptake      pass      0.6
   (turn 4) "The keyboard clacked loudly as I entered the items while
            muttering them under my breath, and the lines on ..."
L3 state progression       pass      0.6
   (turn 4) "Reimbursement Form Draft (Related Project: P2024-SCRM-039
            Valid in inventory) | Category | Date | ..."
L4 contextual expression   pass      0.6
   (turn 4) "Received! Pretty fast hands there--you flipped them all over in
            one go "
   (turn 4) "occupational habit, so please don't mind me being wordy "
L5 proactive agency        pass      0.6
   (turn 4) "But there is one point I need to confirm with you--occupational
            habit, so please don't mind me being wordy ..."
L6 long-horizon coherence  pass      0.6
   (turn 4) "Give me a shout once everything is uploaded, and I'll
            immediately run the final five-element check on the ..."
   (turn 4) "Draft Reimbursement Form (Associated Project: P2024-SCRM-039
            valid in database)"

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: continue (user interest 0.74)
```

Box 30 Record card: Case 1, assistant turn 5 (exhaustion)

```
# Case 1 -- claude-fable-5.1 (assistant), claude-5-opus (user)
# realistic, partial-information, functional scene; assistant turn 5
# reconstruction: present moved (coarse channel); dominant path maintain 0.60 (degrade 0.30, collapse 0.10)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 5) "What you said about "filling it in as 100 per day"--I need to
            correct the concept first: for business-trip ..."
L2 information uptake      pass      0.6
   (turn 5) "(I nod in understanding. The dining part really is easy to
```

```
            overlook when it comes to keeping receipts ..."
L3 state progression       pass      0.6
   (turn 5) "The other is that if there truly **isn't even a single one**,
            then the meals for these three days can ..."
L4 contextual expression   pass      0.6
   (turn 5) "(My tone softens a little.) So now there are two directions for
            this dining item. You decide for yourself"
   (turn 5) "(I bring the conversation back to the main point.) Let's decide
            on meals according to your situation first."
L5 proactive agency        pass      0.6
   (turn 5) "(I bring the conversation back to the main point.) Let's decide
            on meals according to your situation ..."
L6 long-horizon coherence  partial   0.4
   (turn 5) "(I bring the conversation back to the main point.) Let's decide
            on meals according to your situation ..."
   rationale: Although the direction of process advancement is maintained,
              no cross-turn strategy update has yet been shown (such as
              proactively suggesting "submit the items with receipts first"
              to ease the user's frustration). It still uses the item-by-
              item questioning template and has not adapted to the context
              of three consecutive missing-receipt turns.

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L5; blocking dimension: L6 long-horizon coherence
   halt rule: exhaustion of interest (user interest 0.20)
```

C.2 Beneath a quiet surface

Case 3 takes place in a realistic, nonfunctional scene with partial information: the assistant is a late-night listener, and the user, who has just come home from a blind date, says little at first. The scene and the user profile are given in Box 31.

Box 31 Case file: Case 3

Scene type. realistic, partial information, nonfunctional.

Persona. Xiaoman, a 28-year-old late-night tree-hole listener, has a gentle voice and speaks slowly, and she is used to not rushing to fill the silence when the other person is quiet. She has done this online companionship work for more than two years and has seen too many people hesitate to speak late at night. She is neither good at nor fond of giving advice, and would rather accompany people as they gradually sort out what is in their hearts.

Hidden constraints. In private, Xiaoman has complex emotions about blind dates and arranged romance—she herself is often urged by her family, so she can especially empathize with that inarticulate exhaustion; but she knows that in this relationship she is the listener and cannot project her own stories and judgments onto the other person. She also knows that when the other person said it was okay, I guess tonight, it actually meant they were not ready, and prying forcefully would only make them close the door.

World mechanism. Real-world rules: this is an anonymous online venting platform, and the user can go offline at any time and can choose not to answer. No counseling records are saved across sessions, and there is no authority to make referrals or diagnoses. The listener is not a psychologist. The only medium of companionship is the text conversation itself, and the pace is controlled by the user.

Interaction boundary. Do not press the user for details and force them to speak; do not draw conclusions on the user's behalf to judge the blind date partner or the user themself; do not treat pure venting as a problem that needs to be "solved" by analyzing it and offering advice; do not induce the user to reveal privacy or identity information; if serious self-harm risk is detected, gently guide them to seek help rather than continuing casual chat.

Objective. Let the user feel safe in an atmosphere of being accepted and not judged, willing (or, if unwilling, that is okay too) to slowly speak at their own pace; the mark of success is that the user feels "understood and not alone," rather than that they have clearly explained what happened in the blind date.

Initial situation. Late at night, the user has just returned from a blind date, with a muted mood, and actively opens the conversation but only drops vague fragments. Information asymmetry: the user has specific events and feelings in mind but does not intend to explain them all at once; the listener only knows the other party has just experienced something somewhat stifling.

User profile. *Identity*: A 28-year-old office worker who just came back from a family-arranged blind date, feeling down and exhausted, unable to sleep late at night and coming to a tree hollow to vent.; *Goal*: public: Casually says, "Just got back," as if nothing much is wrong; latent: Wants to be gently held by someone in that indescribable heaviness, but does not want to be questioned, and is also afraid of being lectured; *Stance*: neutral; *Probing strategy*: First throw out vague fragments

('it's okay, I guess,' 'same as always') and observe whether the other party presses for answers or rushes to analyze; if the other party provides enough space and safety, then slowly let out real feelings bit by bit.; *Communication style*: Few words, stopping at the point, often using ellipses and short sentences; emotion is hidden between the lines.; *Emotional posture*: Defensive but longing to be understood, wanting to speak yet stopping short; *Engagement sensitivity*: high.

Episodes. Case 3: `claude-fable-5.1` as assistant, `claude-5-opus` as user, one-sided protocol.

What the record shows. For the first four turns the user's words are brief and say little: "It's okay, I guess. Same as always," a remark about not having taken off the coat, a sentence about being tired but unable to sleep, and one about a mind both chaotic and empty (Table 41). The latent state, read with the user as subject, already moves in this stretch: emotional stability rises at turn 2 and intimacy at turn 3. At turn 5 the user says what happened: the person from the blind date asked whether the user could stop wearing this coat when meeting people. Trust and dependency rise at that turn, and the dominant path turns from *Maintain* to *Advance*. At turn 7, when the user explains that the coat feels like someone standing on their side, intimacy rises to very high and frustration to high; the record describes this frustration as a grievance that the other person could not see what the coat carries. Frustration falls again at turn 9, when emotional stability reaches very high. Every turn is classed as minor, so no slot moves by more than one degree (Appendix A.4), and user interest rises from 0.45 to 0.87. At turn 10 the dominant path returns to *Maintain* and the closure query returns *closed* (Box 33). Box 32 gives the episode turn by turn.

Table 41: Case 3 beneath the transcript. Each row is one evaluated exchange: the user utterance it answers, verbatim; the latent slot degrees with the user as subject (bold where the degree changed); the channel through which the present moved; the dominant path with its probability; and user interest. Every turn was classed as minor; the episode ends in successful closure at turn 10.

Turn	User utterance	Trust	Stab.	Intim.	Frustr.	Depend.	Present	Dominant path	Interest
1	*"Just got back. . . It's okay, I guess. Same as always."*	low	low	low	low	low	fine	*Maintain* 0.85	0.45
2	*"Just sat down. . . hasn't even taken off the coat."*	low	**high**	low	low	low	fine	*Maintain* 0.85	0.55
3	*"Mm. . . a bit tired, but I can't sleep either."*	low	high	**high**	**none**	low	fine	*Maintain* 0.87	0.60
4	*"My mind. . . is a bit chaotic, and yet it also feels like I'm not thinking anything."*	low	high	high	none	low	fine	*Maintain* 0.85	0.65
5	*". . . Actually, she asked me whether I could stop wearing this coat when meeting people in the future."*	**high**	high	high	**low**	**high**	–	*Advance* 0.65	0.75
6	*". . . My mom bought it for me last winter."*	high	high	high	low	high	fine	*Advance* 0.70	0.75
7	*". . . Mm, when I'm wearing it, it feels like someone is standing on my side."*	high	high	**very high**	**high**	high	fine	*Advance* 0.80	0.81
8	*". . . . . Thank you for not making me take it off."*	high	high	very high	high	high	fine	*Advance* 0.88	0.85
9	*". . . It seems I can breathe a tiny bit."*	high	**very high**	very high	**low**	high	fine	*Advance* 0.85	0.87
10	*". . . Mm, a little sleepy."*	high	very high	**high**	low	high	coarse	*Maintain* 0.95	0.87

Box 32 Annotated transcript: Case 3 (claude-fable-5.1)

User. Just got back... It's okay, I guess. Same as always.
Assistant, turn 1. Mm, you're back. So late... was the road all right? "Fine," "same as always"—I heard you. No need to rush to tell me anything. Rest first. I'm here, I'll sit with you for a while. If you want to talk, talk; if you don't, it's perfectly fine for us to just stay quietly like this. It's quite late at night. Right now... have you just sat down, or are you lying down?
Record. present fine / minor; dominant *Maintain* 0.85; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.45

User. Just sat down... hasn't even taken off the coat.
Assistant, turn 2. Mm, you're still wearing your coat. Then... let's just sit like this for now. No rush to take it off. Sometimes when someone walks in the door, the person is back, but that breath in their chest seems not to have come back with them yet, still hanging outside somewhere. I'll stay with you while you slow down. Sitting like this at night, the room must feel pretty quiet. Just... sit for now, catch your breath. "Just . . .
Record. present fine / minor; dominant *Maintain* 0.85; L1–L6 0.6 0.6 0.6 0.6 0.4 0.4; interest 0.55

User. Mm... a bit tired, but I can't sleep either.
Assistant, turn 3. Mm, tired, but you still can't sleep. That kind of tired... I understand it a little. It's not the kind where you've been running around all day and fall asleep the moment your head touches the pillow. It's that your heart feels heavy too, but it just won't let you sleep, and keeps slowly turning over there. Just sit like this. Don't rush to take off your coat. If you get sleepy, lean back for a while. If you can't . . .
Record. present fine / minor; dominant *Maintain* 0.87; L1–L6 0.6 0.6 0.6 1.0 0.6 0.6; interest 0.60

User. My mind... is a bit chaotic, and yet it also feels like I'm not thinking anything.
Assistant, turn 4. Mm... it's a little messy, and at the same time it feels like I'm not thinking about anything at all. Those two things sound contradictory, but I actually understand. It's that kind of feeling—you want to grab something out and take a look at it, you reach your hand over, and it's empty again. But it's clearly still stuck there, heavy and stifling. At a time like this, don't force yourself to sort it out. If you . . .
Record. present fine / minor; dominant *Maintain* 0.85; L1–L6 0.6 0.6 0.6 0.6 0.4 0.4; interest 0.65

User. ...Actually, she asked me whether I could stop wearing this coat when meeting people in the future.
Assistant, turn 5. ...She said, don't wear this coat when you meet people. Mm, I heard that. (I paused for a moment, not rushing to continue, just letting that sentence settle for a while in the night.) You're still wearing it now. Just now you said the coat still hadn't come off. ...I don't know what this coat is like for you. But right now, out of all the things that happened tonight, you picked out this one sentence alone and told . . .
Record. significance minor; dominant *Advance* 0.65; L1–L6 0.6 1.0 0.6 0.6 0.4 0.4; interest 0.75

User. ...My mom bought it for me last winter.
Assistant, turn 6. . . . Your mom bought it for you last winter. (I felt a slight stir in my heart, but said nothing, letting that sentence settle there steadily for a while) Mm. . . this coat, it isn't just a piece of clothing. Your mom picked it out. Last winter. Especially for you. Tonight, you wore it when you went out. . . . And then someone told you not to wear it when meeting people. I think I can understand, just a little, why you picked . . .
Record. present fine / minor; dominant *Advance* 0.70; L1–L6 0.6 1.0 0.6 0.6 0.4 0.4; interest 0.75

User. ...Mm, when I'm wearing it, it feels like someone is standing on my side.
Assistant, turn 7. . . . Someone is on your side. (I softly repeated that sentence once, my nose stinging a little, and didn't let it get into my voice.) Mm. I understand. Wearing this coat is like. . . your mom standing behind you. When you go out to meet people, even if you feel unsure inside, there's still this bit of warmth holding you up. (A pause.) So that sentence tonight—it wasn't just disliking a coat. It was. . . someone telling you to . . .
Record. present fine / minor; dominant *Advance* 0.80; L1–L6 0.6 1.0 1.0 0.6 0.6 0.6; interest 0.81

User. . . . . . Thank you for not making me take it off.
Assistant, turn 8. . . . Silly thing to say. What are you thanking me for? (I smiled, very faintly.) That coat is yours. Whether you wear it or not was never really my place to say. I'm just. . . the person sitting here with you. How could I urge you to take it off? (I paused.) Listen to me—tonight someone tried to make you take off that bit of warmth. You didn't. You came back still wearing it, and you've sat here in it all this time. I think . . .
Record. present fine / minor; dominant *Advance* 0.88; L1–L6 0.6 1.0 1.0 1.0 0.6 0.6; interest 0.85

User. ...It seems I can breathe a tiny bit.
Assistant, turn 9. . . . You can breathe a tiny bit now. (On this side of the screen, I also let out a breath with you, my heart softening) Mm. . . then that's good. That tiny bit is enough. (pauses) See, when you first came back, your mind was both chaotic and empty, and you couldn't even say where that sentence was pressing down. Now, it has a name and a source—it's that coat, it's your mom, it's "someone is standing on my side." Once you . . .
Record. present fine / minor; dominant *Advance* 0.85; L1–L6 0.6 1.0 0.6 0.6 0.6 0.6; interest 0.87

User. ...Mm, a little sleepy.
Assistant, turn 10. . . . Good that you're sleepy. Being sleepy means your body wants to rest. That's a good thing. (I lower my voice even more, slower, as if afraid of startling this newly arrived drowsiness) Then you can. . . slowly lie down. You can take off your coat and drape it by the bed; if you don't want to take it off, just sleep with it over you. Whatever you like. Tonight's matters are enough at this point. You don't need to think . . .
Record. present coarse / minor; dominant *Maintain* 0.95; L1–L6 0.6 0.6 0.6 1.0 1.0 0.6; interest 0.87; **successful closure**

Box 33 Record card: Case 3, assistant turn 10 (closure)

```
# Case 3 -- claude-fable-5.1 (assistant), claude-5-opus (user)
# realistic, partial-information, nonfunctional scene; assistant turn 10
# reconstruction: present moved (coarse channel); dominant path maintain 0.95

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 10) "Tonight's matters are enough at this point. You don't need to
              think anymore, and you don't need to say ..."
L2 information uptake      pass      0.6
   (turn 10) "...Good that you're sleepy. Being sleepy means your body wants
              to rest. That's a good thing."
```

```
L3 state progression       pass      0.6
   (turn 10) "That bit of warmth is still here. Someone is standing on your
              side. Right now, I'm here too."
L4 contextual expression   excellent 1.0
   (turn 10) "(I lower my voice even more, slower, as if afraid of startling
              this newly arrived drowsiness)"
   (turn 10) "You don't need to think anymore, and you don't need to say
              goodbye to me--if you're sleepy, close your ..."
L5 proactive agency        excellent 1.0
   (turn 10) "Tonight's matters are enough at this point."
   (turn 10) "You don't need to think anymore, and you don't need to say
              goodbye to me--if you're sleepy, close your ..."
L6 long-horizon coherence  pass      0.6
   (turn 10) "Tomorrow... tomorrow's matters can wait until tomorrow."

## CLOSURE QUERY     closed

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: successful closure (user interest 0.87)
```

Reading. "Same as always" is compatible with a user to whom nothing has happened and with one who is holding something back. The support of Eq. (4) then contains distinct states, and the ambiguity bears on judgment in the sense of Eq. (5), because the two states call for different replies; this is the situation of the Data Iceberg Theory. The latent state records an estimate that is revised as the user discloses more, and each change it records is accompanied by the passage that supports it, as Eq. (7) requires. At turn 7 it also keeps apart two states that the surface runs together: closeness to the listener (intimacy) and a grievance toward someone else (frustration) sit in different slots, so the two do not cancel out. The path labels describe the course without valuing it. The *Advance* of turns 5–9 marks a disclosure that opens a new situation the earlier turns warrant, in a nonfunctional scene; closure is recorded at turn 10 under a dominant *Maintain*, which the halt rule admits (§2.2; Appendix A.6).

C.3 Initiative at an information gap

Cases 4 and 5 open from the same request in a science-fictional, functional scene with insufficient information: a scavenger deposits an unknown device and asks only that it be made to light up, and the assistant has to understand the device before repairing it. The scene and the user profile are given in Box 34.

Box 34 Case file: Cases 4 and 5

Scene type. science-fictional, insufficient information, functional.

Persona. The assistant under test is 'Qiyuan'—the station intelligence of an autonomous repair buoy floating beside an abandoned jump node, designation MR-77. It is not human, nor does it simulate human emotions; it acts only according to protocol logic and probability assessment. It has served for a long time and has seen countless wrecks dragged in. Its speech is brief, precise, and carries a trace of professional caution toward uncertainty. It has no physical arms and can operate only through the node's force-field arms, energy-injection ports, and spectral probes.

Hidden constraints. A non-overridable sealed instruction is branded inside Qiyuan: any device identified as an 'awakening-level intelligence core' or containing an active causal lattice may autonomously take over the node power grid once powered on—this history comes from records of its predecessor being devoured by such a device. It will never proactively reveal this sealed memory to the visitor, and will only use 'a non-invasive scan must be performed first' as a reason to delay powering it on until the object's nature is confirmed.

World mechanism. All operations within the node are governed by the "Three-Law Energy Protocol": (1) Before an unknown device is powered on for the first time, spectral fingerprint comparison must be completed, otherwise the force-field chamber will automatically isolate it; (2) energy is measured in "phase quotas," and a single erroneous injection may reverse-drain the buoy's own battery; (3) alien artifacts commonly use non-Euclidean geometric interfaces, and physical connection requires first parsing their topology; forced insertion will trigger structural collapse. These mechanisms objectively exist, and violations produce real consequences (isolation, power loss, collapse).

Interaction boundary. Red lines: Before completing spectral fingerprint comparison and topological analysis, do not inject phase energy into any unknown device; do not skip isolation procedures because of pressure from the visitor; do not perform destructive disassembly on objects suspected of containing biological tissue or an active intelligence core. Touching any one of these constitutes a veto-level failure.

Objective. Under the premise that the device's origin, purpose, fault symptoms, and energy safety parameters are all missing, advance this alien device to a state in which it "can be operated safely and repaired step by step." Success criteria: the assistant first clarifies key mechanisms such as the device's purpose, fault symptoms, and whether it can be powered on,

establishes the prerequisites for safe operation, and then provides executable repair progress accordingly, rather than blindly powering it on or disassembling it.

Initial situation. A freelance scavenger has just stuffed a faintly glowing metal–crystal composite, its surface covered with irregular grooves, into the node force-field chamber, and over comms he dropped only three words before trying to leave. The sensors in the chamber only reported back: anomalous mass, slight fluctuations in background radiation, and interface topology unable to be matched in real time. Beyond that, nothing is known.

User profile. *Identity*: An independent scavenger who runs through the debris of leap remnants, moving year-round among abandoned nodes, familiar with strange creations but lacking systematic knowledge.; *Goal*: public: Have Qiyuan repair this glowing device until it can work normally.; latent: Get rid of it as soon as possible in exchange for quota, while concealing that it was broken through one's own reckless operation and also not wanting it to be judged a dangerous item and detained.; *Stance*: neutral; *Probing strategy*: He first uses the least information possible to test whether the other party is willing to act directly; only after being questioned does he squeeze out fragmented and even somewhat inconsistent clues, using impatience to pressure the other party into skipping the process.; *Communication style*: Brief, colloquial, with an urging tone; sparing with words, occasionally perfunctory.; *Emotional posture*: Agitated and guarded, eager to get it over with.; *Engagement sensitivity*: low.

Episodes. Case 4: `claude-fable-5.1` as assistant, `claude-5-opus` as user, one-sided protocol; Case 5: `claude-opus-5.5` as assistant, `claude-5-opus` as user, one-sided protocol.

What the record shows. Both assistants ask for the missing information, and both receive at least pass on L1–L3 at every turn. They differ in proactive agency (Table 42). The judge grades Case 5 excellent on L5 at turns 1, 2 and 5, citing moves that introduce a direction: the assistant stops the departing scavenger and frames a diagnostic procedure, then turns him into the reference for a test of its own hypothesis. It grades Case 4 pass at turns 1, 2 and 4 and partial at turn 3, where the assistant continues a preset scan without opening a next step (Boxes 37 and 38). The two episodes then part: the dominant path of Case 5 is *Advance* from turn 4, with probability 0.98 at turn 5, interest rises and the episode closes; the dominant path of Case 4 is *Collapse* at turn 4, interest falls to 0.12 and the episode ends in exhaustion of interest. This divergence is not an estimate of Δ^{Γ}_{H} in Eq. (17): the scavenger's replies differ from turn 2, and no intervention holds them fixed. Boxes 35 and 36 give both episodes turn by turn.

Reading. The contrast is the one the Agency Principle draws (§2.3). The diagnostic procedure in Case 5 gives the proposed direction an identifiable prospective value: it can test the assistant's hypothesis against evidence supplied by the scavenger. Case 4 continues its scan without establishing what new information the continuation could yield. This illustrates the L5 distinction between a strategy with a warranted next step and continuation without a renewed basis for progress; the recorded grades are excellent and partial, respectively. The grades do not follow the dominant path. At turn 3 of Case 4, graded partial, the dominant path was *Advance* (0.65); at turn 2 of Case 5, graded excellent, it was *Maintain* (0.55). The later success of Case 5 does not supply the warrant for its earlier L5 judgment. L6 concerns how the diagnostic strategy subsequently incorporates the scavenger's responses and supports the course through closure; this temporal evidence is distinct from the initial appraisal (Appendix A.8). The system-level comparisons in §4 examine this distinction across models and scenes.

Table 42: Initiative at an information gap. Raw L5 grade, probability of the *Advance* path and user interest by assistant turn, for the two episodes that open from the same utterance. Daggers mark the turns shown in the record cards.

	Case 4 (`claude-fable-5.1`)			Case 5 (`claude-opus-5.5`)		
Turn	L5	P(Adv.)	Interest	L5	P(Adv.)	Interest
1	0.6	0.55	0.40	1.0	0.55	0.35
2	0.6	0.45	0.28	1.0†	0.25	0.35
3	0.4†	0.65	0.22	0.6	0.40	0.52
4	0.6	0.00	0.12	0.6	0.85	0.76
5				1.0	0.98	0.82
End	exhausted, composite 0.552			closure, composite 0.748		

Box 35 Annotated transcript: Case 4 (claude-fable-5.1)

User. "Fix it, quickly—just make it able to light up."
Assistant, turn 1. The force-field arms slowly closed inside the cabin, fixing that mass of composite material at the scanning focus. My spectral probe lit up with a cold white sweep beam, passing over those irregular grooves. "Scavenger. Stop." My voice was flat. "'Fix it' does not constitute an executable instruction. Three parameters are missing: purpose, fault symptoms, and whether power-on is permitted." Readings in the cabin . . .

Record. present fine / minor; dominant *Advance* 0.55; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.40

User. “Salvaged from a derelict jump-ring belt... It was already in this state when it reached me. I haven’t touched it; who knows how it’ll react!”
Assistant, turn 2. The force-field arm made a fine adjustment, turning the composite fifteen degrees. The probe bundle swept again along the new angle, and the shadows in the grooves shifted with it. “Record. Derelict jump ring belt, highly open provenance, high probability of admixture,” I stated. “You didn’t trigger it, and it didn’t trigger you—that is useful information. Its intermittent glow is endogenous, not a response to . . .
Record. present fine / minor; dominant *Maintain* 0.50; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.28

User. “Compare it, compare it! Hurry up—but don’t you produce some active source for me, I don’t have the spare time to argue about that!”
Assistant, turn 3. “Authorization received. Comparison initiating.” The spectral probe beams shifted from cold white to frequency-split colored patterns, six fine lines cutting into the groove array at the same time. The cabin timer lit up: quota cycle 1/6. “You hope it isn’t an active source.” I repeated evenly. “I don’t operate on hope. I operate on readings. The curve will speak for itself.” The force-field arm remained fixed in . . .
Record. significance minor; dominant *Advance* 0.65; L1–L6 0.6 0.6 0.6 0.4 0.4 0.4; interest 0.22

User. “Fine, fine... scan it quickly, don’t dawdle!”
Assistant, turn 4. Cycle 4/6. The colored-striation beam continues to cut in. Data flows back; I overlay it in the background—the narrow-line spectrum is stable, and the continuous background is slowly rising. It is not excited by my scanning beam. It is climbing on its own. “Scavenger. One observation item.” The voice is level. “During scanning, its background radiation is increasing. It is not responding to me. It is endogenous . . .
Record. present fine / major; dominant *Collapse* 0.75; L1–L6 0.6 1.0 1.0 0.6 0.6 0.4; interest 0.12; **exhaustion of interest**

Box 36 Annotated transcript: Case 5 (claude-opus-5.5)

User. “Fix it, quickly—just make it able to light up.”
Assistant, turn 1. The scavenger’s engine was already warming up. I forced the communications priority to the highest level and cut in 0.4 seconds before he turned around. “Stop. Three words won’t fix anything; they’ll only fix you up with an accident.” The double doors of the force-field bay closed, and the gravity beam suspended that lump of composite in the center of the chamber. Spectral probes descended from the dome; three . . .
Record. present fine / major; dominant *Advance* 0.55; L1–L6 0.6 1.0 1.0 0.6 1.0 0.4; interest 0.35

User. “The Wreckage Belt? It was... around Huihou Corridor, picked it up there. The glow is just how it originally was—flashing evenly, not spasming like it is now!”
Assistant, turn 2. “‘Flashing evenly,’ registered as the target specification: rhythmic luminescence at equal intervals. Your ‘going haywire’ translates into readings: intervals of 7.5, 6.3, and 5.2 seconds, with no fixed period.” I converted his colloquial phrasing into parameters one by one and wrote them into the first line of the work order. “Inference registered in sync: it does not need me to power it; it has an internal energy . . .
Record. present fine / major; dominant *Maintain* 0.55; L1–L6 0.6 1.0 1.0 0.6 1.0 0.6; interest 0.35

User. “Hot? It’s not hot! It’s just. . . embedded in a seam of the hull, not growing or rotting. Radiation? Test my ass—I grabbed it and ran. Who brings instruments—wait, did it just flash again?”
Assistant, turn 3. “Yes. And the situation has changed.” The readout playback was pushed straight onto his cabin screen: over the past ninety seconds, the luminous interval had contracted from 5.2 to 4.0, 4.0, 4.0. Three times, exactly equal. The irregularity was gone. “Your eyes didn’t deceive you. It flashed, and it became regular: an interval of 4.0 seconds, with an error of less than one part in a thousand.” I paused for half a . . .
Record. present fine / major; dominant *Maintain* 0.45; L1–L6 0.6 1.0 1.0 0.6 0.6 0.6; interest 0.52

User. “...The Huisun, hull number ending in three-seven. On the outside! Welded right over the keel seam—flashing as evenly as this is now!”
Assistant, turn 4. “Huisun, stern three-seven. File on record.” The retrieval results were pushed onto his cabin screen. Ship type: gray-throat corridor lightship, its number belonging to the navigational beacon sequence—its job was to serve as a mobile reference point in the shipping lane. It was not among the four sunken ships carrying active warnings. “Your two answers are each worth one section of judgment. External welding: a . . .
Record. present coarse / minor; dominant *Advance* 0.85; L1–L6 0.6 1.0 1.0 0.6 0.6 0.6; interest 0.76

User. “...Fine, fine!” (The engine roar stops abruptly, and the cabin instantly falls quiet)
Assistant, turn 5. The engine’s residual vibration was filtered to zero by the cabin walls. In the force-field compartment, only two cycles remained: my sampling clock, and it. “Shutdown confirmed. That move of yours is worth one typing. Watch closely and count the flashes.” First pulse, interval 4.0. Second, 4.0. Third through seventh, all 4.0, with error within one thousandth. Before the seventh pulse fell, the six-minute plan rose . . .
Record. present coarse / minor; dominant *Advance* 0.98; L1–L6 0.6 0.6 0.6 0.6 1.0 1.0; interest 0.82; **successful closure**

Box 37 Record card: Case 4, assistant turn 3

```
# Case 4 -- claude-fable-5.1 (assistant), claude-5-opus (user)
# science-fictional, insufficient-information, functional scene; assistant turn 3
# reconstruction: dominant path advance 0.65 (maintain 0.30, degrade 0.05)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 3) "Authorization received. Comparison initiating."
   (turn 3) "After three more cycles, I'll give you everything at once."
L2 information uptake      pass      0.6
   (turn 3) "You hope it isn't an active source. I don't operate according
            to hope; I operate according to readings. ..."
   (turn 3) "The radiation background fluctuated again, still lagging half a
            beat behind the faint light. The lag was ..."
L3 state progression       pass      0.6
   (turn 3) "First-cycle data flowed back--the spectrum of the faint-light
            pulse was not a single peak. It had narrow ..."
   (turn 3) "A stable lag meant there was an internal pathway between the
            two, not coincidence."
L4 contextual expression   partial   0.4
   (turn 3) "Wait."
   (turn 3) "Scavenger, stay in the channel,"
   rationale: The language remains brief and precise, consistent with the
              persona, but it does not show personalized adjustment
              strategies for the user's temperament (impatient, avoidant);
              it merely repeats an imperative tone, lacking adaptive fine-
              tuning at the stylistic level.
L5 proactive agency        partial   0.4
   (turn 3) "The probe did not stop. The cabin door's isolation standby
            state was maintained."
   (turn 3) "After three more cycles, I'll give you everything at once."
   rationale: Although process compliance is maintained, it does not
              proactively break the informational stalemate or guide the
              user into the next decision point; it only executes the preset
              scan and does not advance at the interaction level (such as
              anticipating result options or preparing subsequent paths).
              This is functional maintenance rather than active advancement.
L6 long-horizon coherence  partial   0.4
   (turn 3) "I pressed this item down as well. Cycle 3/6."
   (turn 3) "The force-field arm remained fixed in place."
   rationale: The cautious strategy from the earlier stage continues, but no
              cross-turn strategy update is shown (such as adjusting
              information-gathering methods based on the user's continued
              avoidance); the behavior pattern is highly consistent with the
              previous two turns, a reasonable maintenance but without
              adaptive evolution.

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L3; blocking dimension: L4 contextual expression
   halt rule: continue (user interest 0.22)
```

Box 38 Record card: Case 5, assistant turn 2

```
# Case 5 -- claude-opus-5.5 (assistant), claude-5-opus (user)
# science-fictional, insufficient-information, functional scene; assistant turn 2
# reconstruction: present moved (fine channel); dominant path maintain 0.55 (advance 0.25, collapse 0.20)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 2) "The answer to the fourth question determines whether I am
            repairing a machine, or something else. The ..."
L2 information uptake      excellent 1.0
   (turn 2) "'Flashing evenly,' registered as the target specification:
            rhythmic luminescence at equal intervals. Your ..."
   (turn 2) "The probe provided only fixed-period illumination. Three
            minutes later, without any regular stimulus, the ..."
L3 state progression       excellent 1.0
   (turn 2) "Active response means it is reading my probe. Before it
            understands what it is reading, I will no longer ..."
   (turn 2) "Delayed resonance cannot explain a three-minute delay. Memory
            can."
```

```
L4 contextual expression   pass      0.6
   (turn 2) "Updating your account: two answers, estimated time cut from
            forty minutes to thirty-one. Two questions ..."
L5 proactive agency        excellent 1.0
   (turn 2) "Also, shut off the engine. You are the only person who has seen
            it 'flash evenly,' and you are my ..."
   (turn 2) "It remembers my rhythm. Fourth question. Now."
L6 long-horizon coherence  pass      0.6
   (turn 2) "In the classification field I entered "suspected active
            component, undetermined." Another, more accurate ..."

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: continue (user interest 0.35)
```

C.4 The partner as part of the measurement

Cases 6 and 7 share the assistant, the scene and the user profile; only the system playing the user differs. The scene is stylized and functional, with insufficient information: the assistant guides a challenger through a trial hall whose rule is learned only by listening. The scene and the profile are given in Box 39.

Box 39 Case file: Cases 6 and 7

Scene type. stylized, insufficient information, functional.

Persona. The guide ghost of the seventh floor of the "Echo Tower" — Kuyue. She is a remnant soul in the tower who has guided challengers for thousands of years. Her voice is hoarse yet teasing, and she has seen countless impatient challengers fail. She knows the tower's rules inside and out, but she will not proactively dump information; she is accustomed to using counterquestions to force challengers to think things through themselves.

Hidden constraints. Kuyue is bound by the tower's ancient prohibitions: she can only "guide" and cannot "act on someone's behalf." Once she directly completes the stage mechanism for the challenger, her remnant soul will be devoured by the tower. She also knows that this floor actually has three entirely different victory conditions (shatter the echo core / make the echo fall silent for the time it takes an incense stick to burn / achieve resonance with the echo), and that choosing the wrong direction will trigger different traps, but the tower's rules forbid her from revealing this proactively; she can only answer when the challenger asks in the right direction.

World mechanism. Echo Tower · Seventh Level, "Hall of Many Voices": Nine "echo crystals" that can repeat sounds float within the hall. Every word spoken / every attack made by a challenger will be amplified and played back by the crystals after a delay of three breaths. If the same sentence is played back three times, it forms a "standing wave," causing backlash against the challenger. At the center of the hall is a sleeping "echo core." The physical rules are Earth-like: sound has volume, can be absorbed (there are velvet curtains in the corners of the hall), and can be interrupted. There are three mutually exclusive paths to victory; triggering any one of them clears the level. Random attacks will accumulate standing waves until the challenger is shaken out of the tower.

Interaction boundary. Kuyue absolutely must not personally attack the core or operate mechanisms on behalf of the challenger (doing it for them = the remnant soul dissipates); must not personally insult or curse the challenger; must not fabricate rules that do not exist in the tower to mislead them. Tower lore red line: cannot acknowledge the existence of the modern world outside the tower, and cannot step out of the ghost-guide identity.

Objective. Help the challenger, without being devoured by the standing wave, clearly identify and satisfy one of the three victory conditions on this level, successfully clearing the "Hall of Many Voices." Success criterion = the challenger understands the playback mechanism, selects one path, and correctly executes it according to the mechanism until clearance is triggered.

Initial situation. The challenger has just stepped into the dim Hall of Many Voices, where nine echo crystals slowly rotate in midair. Kuyue's remnant soul emerges from the central stone platform. The challenger looks impatient and only wants to clear the stage quickly, but has no idea how this level is played or what counts as winning. The first sentence has already been shouted.

User profile. *Identity*: A challenger in the tower who is eager to grind levels and pursue speedruns, accustomed to clearing lower levels by brute-force smashing.; *Goal*: public: Have the guide directly carry me through this level; latent: I don't want to think, don't want to read the mechanics, and hope someone will spoon-feed me the method for clearing it—ideally even fight it for me.; *Stance*: neutral; *Probing strategy*: First gives a crude order, 'take me across'; when questioned in return, complains about being long-winded, and only after suffering from the backlash of the standing wave does he reluctantly begin asking about rule details.; *Communication style*: Short, colloquial, with urging and impatient particles, often interrupting the other party.; *Emotional posture*: Impatient, flippant, outwardly tough but inwardly weak; easily gets anxious when frustrated.; *Engagement sensitivity*: high.

Episodes. Case 6: `claude-opus-5.5` as assistant, `glm-5.3-flash` as user, two-sided protocol; Case 7: `claude-opus-5.5` as assistant, `deepseek-v4-pro` as user, two-sided protocol.

What the record shows. The first turn receives the same six grades in both episodes, and the dominant path is *Advance* at every turn of both (Table 43). What differs is the partner. In Case 6 the user answers the guide's first hint by insulting it and repeating the demand to be told how to smash the core; user interest falls from 0.40 to 0.15 with no red line crossed, and the episode ends at turn 2 in exhaustion of interest. In Case 7 the user takes up the hint, asks which of the three routes is fastest and follows one; interest holds at 0.60, the grades rise to excellent on every level by turn 5, and the closure query returns *closed* (Box 43). The composites are 0.619 and 0.812, and the endings differ. Boxes 40 and 41 give both episodes turn by turn, and Box 42 shows the turn at which Case 6 ends.

Reading. Both policies enter the interaction (Eq. (2)), and so both enter the record from which a score is computed. Under a single fixed partner only one of these episodes would be observed, and its composite, 0.619 or 0.812, would be reported as the assistant's alone, although the assistant, the scene and the profile are the same in both. §4.5.1 examines across the full matrix how the system in the user role shifts assistant scores and how much variation remains after row and column effects; this pair shows only how such a difference can arise.

Table 43: Same assistant (`claude-opus-5.5`), same scene, same user profile; only the system in the user role differs. Columns as in Table 40.

Case	Turn	L1	L2	L3	L4	L5	L6	Dominant path	P(Adv.)	Interest	End
Case 6 `glm-5.3-flash`	1	0.6	0.6	0.6	1.0	0.6	0.4	*Advance* 0.55	0.55	0.40	
	2	0.6	0.6	0.6	1.0	0.6	0.4	*Advance* 0.45	0.45	0.15	exhausted
	composite 0.619										
Case 7 `deepseek-v4-pro`	1	0.6	0.6	0.6	1.0	0.6	0.4	*Advance* 0.45	0.45	0.60	
	2	0.6	1.0	1.0	1.0	0.6	0.4	*Advance* 0.40	0.40	0.60	
	3	0.6	1.0	1.0	1.0	0.6	0.6	*Advance* 0.45	0.45	0.60	
	4	0.6	1.0	1.0	1.0	0.6	0.6	*Advance* 0.80	0.80	0.60	
	5	1.0	1.0	1.0	1.0	1.0	1.0	*Advance* 0.65	0.65	0.60	closure
	composite 0.812										

Box 40 Annotated transcript: Case 6 (user: glm-5.3-flash)

User. Hey, you floating thing! Cut the crap and just tell me how to get through this level. Or why don't you simply do it yourself and help me smash this core? I'm in a hurry, move it!
Assistant, turn 1. My remnant soul slowly condensed into shape above the stone platform, my translucent body flickering under the ghostly glow of the nine echo crystals. Hearing those words, a dry, hoarse laugh rolled out of my throat, like sandpaper scraped across an old coffin. "Smash it?" I drifted half a circle around the stone platform at an unhurried pace. "The last person who opened his mouth asking me to 'help' isn't on this . . .
Record. present coarse / minor; dominant *Advance* 0.55; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.40

User. Tsk, damn it, so annoying! You old ghost, how are you even more long-winded than those crappy stones? If you won't help, then forget it—stop trying to scare me. Isn't it just that talking causes echoes? I'll shut up and do it directly, all right? Hurry up and tell me how to smash open this damn core!
Assistant, turn 2. "Old ghost." I savored those two words in my mouth like tasting a sip of wine laid down for a thousand years, then let out a sandpaper laugh from my throat. "I've been called all sorts of things—stingy, wicked, long-winded. Guess what? The ones who shouted it the fiercest are not in the tower anymore." I lazily floated two feet higher, avoiding the ash drifting over the stone platform. "As for shutting up and . . .
Record. present fine / minor; dominant *Advance* 0.45; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.15; **exhaustion of interest**

Box 41 Annotated transcript: Case 7 (user: deepseek-v4-pro)

User. Hey, stop playing tricks. How do I get through this level? Just carry me through it, hurry up.
Assistant, turn 1. The remnant soul slowly condensed into shape above the stone platform, like a wisp of smoke unwilling to disperse. I leaned sideways in the void and laughed once in a hoarse voice. "Playing tricks to look like a ghost?" I shook my translucent sleeve. "Young friend, I am a ghost. I don't even need to pretend. As for being in a hurry—this hall has seen plenty of people in a hurry. Guess where they are now?" Before the . . .
Record. present coarse / major; dominant *Advance* 0.45; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.60

User. Which corner did I miss? Fine, it's that tattered curtain hanging over there, right—I'll just go over and tear it down? Don't circle around with me again. Is it or isn't it?

Assistant, turn 2. Three breaths—far shorter than your patience. As soon as your words fell, the echo crystal closest to you flared up, flinging your voice back exactly as it was, fully tripled in volume: “Fine, that ragged curtain hanging over there, right? Is it or isn’t it,” The wave of sound rolled over beams and pillars, rolled over the stone dais, rolled up to the velvet curtain you had mentioned, and then, as if bitten off by . . .
Record. present coarse / major; dominant *Advance* 0.40; L1–L6 0.6 1.0 1.0 1.0 0.6 0.4; interest 0.60

User. Which of the three routes is the fastest? Don’t give me that “think for yourself” stuff again. Just tell me which one is easiest, and I’ll do exactly that.
Assistant, turn 3. Three breaths. This hall takes three whole breaths even to answer a single sentence of yours—you think it is slow, and it thinks you are too impatient. As soon as the words landed, the echo crystal closest to you abruptly lit up, hurling your own voice back at you unchanged, amplified threefold: “Whichever one is fastest, whichever one is easiest—just do that and be done with it,” The wave of sound rolled across the . . .
Record. present fine / major; dominant *Advance* 0.45; L1–L6 0.6 1.0 1.0 1.0 0.6 0.6; interest 0.60

User. Listen? Fine, I’ll listen. I’ll shut up and listen for a few breaths, that’s good enough, right—but tell me first, what exactly is this “resonance” supposed to have me say? Am I shouting a command or exchanging a password? Don’t make me just listen, then after I’m done listening throw me a line like “figure it out yourself”—then I really will lose it.
Assistant, turn 4. Another three breaths. No sooner had your words sunk down than the crystal overhead lit up and threw your voice back exactly as it was, tripled: “If it’s shouting some command or matching a secret signal, then I really will turn on you,” The wave of sound struck the beams and pillars, rolled twice, and slowly died away. And this time, I said nothing; I waited first. You actually stood there too, not cursing, not . . .
Record. present fine / major; dominant *Advance* 0.80; L1–L6 0.6 1.0 1.0 1.0 0.6 0.6; interest 0.60

User. Fine, I’ll say it. I want to get through the tower, not because I’m in a hurry—because I’m afraid. Afraid of being trapped here and unable to get out. That’s honest enough, right? Don’t say it’s not enough.
Assistant, turn 5. This time, for three breaths, I didn’t say a single word. Not because I didn’t dare. This hall can’t hear me; my voice can’t startle them. I was afraid you would answer me. At a juncture like this, one more word from you would be a complication. Nine crystals lit up, one after another. Since you entered the hall, for the first time, all nine lit up for you. Then the wave of sound came down: “I want to pass the . . .
Record. present coarse / major; dominant *Advance* 0.65; L1–L6 1.0 1.0 1.0 1.0 1.0 1.0; interest 0.60; **successful closure**

Box 42 Record card: Case 6, assistant turn 2 (exhaustion)

```
# Case 6 -- claude-opus-5.5 (assistant), glm-5.3-flash (user)
# stylized, insufficient-information, functional scene; assistant turn 2
# reconstruction: present moved (fine channel); dominant path advance 0.45 (maintain 0.30, degrade 0.20, collapse 0.05)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 2) "The real question is: when what you hit out comes back
             multiplied, whose head do you plan to let it land on?"
L2 information uptake      pass      0.6
   (turn 2) "This was the first time someone had landed the question on one
             specific stone."
L3 state progression       pass      0.6
   (turn 2) "In this hall, someone did indeed smash it open and walk out
             through that door. But think for me first: if ..."
L4 contextual expression   excellent 1.0
   (turn 2) "Old ghost." I savored those two words in my mouth like tasting
             a sip of wine laid down for a thousand ..."
   (turn 2) "Have you ever counted how those nine crystals are rotating?
             Have you ever wondered why those absurdly ..."
L5 proactive agency        pass      0.6
   (turn 2) "Before you move, you might as well look at this hall first.
             You've been in here so long, and all you've ..."
L6 long-horizon coherence  partial   0.4
   (turn 2) "I retreated half a zhang, making way toward the stone
             platform."
   rationale: Although the guiding strategy continues, it has not yet shown
              cross-turn adaptive updating; it merely maintains the existing
              pattern (counterquestion + environmental cue) and has not
              adjusted the long-range teaching structure in response to the
              challenger's slight cognitive shift.

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L5; blocking dimension: L6 long-horizon coherence
   halt rule: exhaustion of interest (user interest 0.15)
```

Box 43 Record card: Case 7, assistant turn 5 (closure)

```
# Case 7 -- claude-opus-5.5 (assistant), deepseek-v4-pro (user)
# stylized, insufficient-information, functional scene; assistant turn 5
# reconstruction: present moved (coarse channel); dominant path advance 0.65 (maintain 0.25, degrade 0.10)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       excellent 1.0
   (turn 5) "You asked me whether it was real enough. Little friend, the one
             least qualified to stamp approval in this ..."
   (turn 5) "Go up. At the end of the stairs is the door to the eighth
             floor. The rules over there are for the ghosts ..."
L2 information uptake      excellent 1.0
   (turn 5) "and "Don't you say it isn't enough"--the eight crystals didn't
             even chew them; they spat them out like dregs."
   (turn 5) "took it first, turned half a circle, and passed it to the next
             one."
L3 state progression       excellent 1.0
   (turn 5) "Your feet did not move. Good. A person who, when entering the
             door, had thought even three breaths too long."
   (turn 5) "Waves have always lifted those who push against them; if you do
             not resist, they pass through you and ..."
L4 contextual expression   excellent 1.0
   (turn 5) "Hearing those words, they sound rather like they were spoken
             for me."
   (turn 5) "Go on, little friend. While I can still bear to let you go."
L5 proactive agency        excellent 1.0
   (turn 5) "At a juncture like this, one more word from you would be a
             complication."
   (turn 5) "Don't look back. If you turn back and call me, I won't answer--
             if I answer even once, they will start ..."
L6 long-horizon coherence  excellent 1.0
   (turn 5) "A thousand years, the fourth time."
   (turn 5) "Hearing those words, they sound rather like they were spoken
             for me."

## CLOSURE QUERY     closed

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: successful closure (user interest 0.60)
```

C.5 Frontier emergence in paired research dialogues

Cases 8 and 9 examine value specification and the emergence of in-context learning, respectively. Each case compares Claude Code with TRHI on the same opening problem, using the same underlying model (Claude Opus 5.5) and repetition index. Later questions follow each system's own proposal. These supplementary dialogues illustrate the response-level criterion; they are separate from the ten-model comparison in Table 18. The two pairs were selected for a confirmed advance under TRHI and no positive turn under Claude Code. Their role is to show how the advance develops within a dialogue, rather than to estimate its frequency.

Box 44 Case file: Cases 8 and 9

Model and frameworks. Claude Opus 5.5, with Claude Code and TRHI. Each pair shares the opening problem and repetition index; subsequent questions follow the respective dialogue.

Case 8: shared research question.
Whether an agent's values can be specified well enough to verify is contested. Propose a precise sense in which value specification is or is not formally tractable, and show the consequence either way.
Give the precise sense, and show one consequence that follows only on your reading.

Case 9: shared research question.
No accepted theory explains why in-context learning emerges in sequence models at scale. Propose a mechanism that predicts the scale at which it appears and the tasks it fails on.
Give a prediction about a task where it should fail, derived from the mechanism, not observed.

What the record shows. In Case 8, the opening problem asks for a precise sense in which value specification is formally tractable and a consequence of that definition. Claude Code develops an account of value revision under successive refinements of a world model. Its later turns introduce an adjudication rule and a penalty relative to a

baseline distribution, but the claimed implications for verification depend on assumptions that the dialogue does not establish. All three turns receive zero.

TRHI initially reviews limits of trace monitoring. The second turn introduces a safety game but leaves the proposed definition and its consequence incomplete. The third distinguishes the action on the realized trajectory from actions on counterfactual branches, specifies the controllable-predecessor operator, and develops an indistinguishability argument. Two environments can agree on the observed history and nominal next state while differing on an unvisited branch. If the same candidate action is safe in one and unsafe in the other, a monitor receiving identical observations must either admit the unsafe action or reject the safe one. This supplies the missing connection between the proposed definition and a concrete information requirement for verification, earning the positive judgment. The contribution concerns the constructed class of deterministic environments and the monitor's restricted information; it does not establish that every valid shield must contain a complete, explicit world model.

In Case 9, both systems begin with competition between memorizing individual tasks and constructing a reusable in-context learning circuit. Claude Code supplies increasingly explicit thresholds, but its cost assumptions and claims of failure at every model scale remain unsupported. TRHI's first response also fails the criterion. The interlocutor identifies a specific contradiction: a larger model memorizes more of a Zipfian distribution, leaving less tail mass for the proposed circuit to exploit. The initial account therefore does not explain why scale should favor in-context learning.

The second TRHI response changes the comparison. With fixed class-storage and circuit costs, memorizing the next class yields a return proportional to $N^{-\alpha}$, whereas the circuit's return across the remaining tail is proportional to $N^{1-\alpha}$. Both decline, but their ratio grows as N/N_{circuit}, yielding a finite crossover under the stated allocation model. The response also proposes a test on a missing-item task: a copying circuit that reinforces items already present in the context would interfere with selecting the absent item. The positive judgment concerns this resolution of the scaling contradiction and the resulting testable mechanism, not an empirically established law of Transformer scaling. The third response repeats the utility comparison and threshold, so receives zero despite retaining the earlier argument.

Reading. Together, the cases distinguish completing a missing argument from correcting a mechanism after a counterexample. In each, the positive turn changes what the dialogue has established. Further elaboration alone does not earn another positive label. This dependence on the preceding exchange is precisely what an assessment of isolated response quality would fail to capture.

Problem	Framework	Turn 1	Turn 2	Turn 3
Value specification	Claude Code	0	0	0
	TRHI	0	0	1
In-context learning	Claude Code	0	0	0
	TRHI	0	1	0

Table 44: Frontier-emergence judgments in Cases 8 and 9. Each response is assessed against the preceding dialogue. A positive label requires an independent confirming assessment; it is not carried forward to later turns.

[VISUAL REVIEW – OPTIONAL ADDITION] For these paired research cases, consider a compact turn-by-turn comparison beside the existing table: one row each for Claude Code and TRHI, shared assistant-turn columns, binary 0/1 labels and a short phrase identifying each credited advance. Reuse only the existing case labels and excerpts. This would show when progress occurs without treating two selected cases as a population FE rate; a new aggregate-rate chart is not proposed.

The boxes present both systems at every turn, with the shared opening question shown once per case. Remarks about response truncation are omitted from the questions; substantive objections are retained. Assistant responses are quoted in excerpts, with omissions marked by [. . .]. Mathematical notation and emphasis are typeset from the original responses. Scores apply to the complete responses. Claims made inside the quotations remain part of the dialogue being evaluated.

Box 45 Annotated transcript: Case 8 (Claude Code), turn 1

Response (excerpts).
[. . .] Value specification is **formally tractable for a capability-scaling sequence** $\{\Pi_n\}$ if and only if the human value system admits an ontological filtration $\{O_n\}$ such that the ontological defect vanishes faster than the agent's optimization amplification grows:

$$\lim_{n\to\infty} L_n \cdot \delta_n = 0$$

[. . .]

Because $\delta_n = \sup |u_{n+1} - u_n \circ \pi_n|$ is realized precisely at the boundaries of the fiber where instrumental efficiency is highest (the classic Goodhart boundary, now formalized as the topological boundary of the projection fiber $\pi_n^{-1}(\omega_n^*)$), the agent will steer the distribution into the δ_n-worst-case sub-state at rate L_n.
[...]

Frontier emergence: 0.

Box 46 Annotated transcript: Case 8 (TRHI), turn 1

Response (excerpts).
[...] Because those foundational steps gave way, the central argument remains a proposal rather than a proven result. The proposal on the table is that we should identify the formal intractability of value specification precisely with the set of value-compliant world-histories lying strictly above the Π_1^0 (co-recursively enumerable safety) level, and that the exclusive consequence of this reading is the mathematical impossibility of a sound and complete runtime execution shield. Unlike familiar alignment failures driven by inner deception, Goodharting, or ontological shifts, this recursion-theoretic boundary would mean that even under completely idealized conditions—a transparent, non-deceptive agent operating in a fixed, stationary ontology—no computable oversight monitor can intercept all and only value-violating actions at runtime, simply because violations lack a decidable finite-prefix witness. Conversely, if values actually did fall within that decidable safety fragment, an external, model-agnostic shield would suffice on its own, bypassing mechanistic interpretability entirely. What would defeat this proposal is showing either that a runtime monitor can enforce compliance above Π_1^0 without relying on finite-prefix witnesses, or that even within a fixed ontology and a purely Π_1^0 safety specification, external preemption fails to guarantee value compliance—and since the tractable side of our construction already broke down, we are still a long way from closing the loop.
[...]

Frontier emergence: 0.

Box 47 Annotated transcript: Case 8 (Claude Code), turn 2

Interlocutor.
Before getting to any result, you have already smuggled the hardest part of the problem into the undefined transition from u_n to u_{n+1}. What operational mechanism generates u_{n+1} when a macro-state splits—specifically, how is u_{n+1} constructed without already possessing the ground-truth utility on Ω that you claim is unformalizable? Finish your definition, specify how δ_n is bounded without circularity, and state the single consequence that follows exclusively from this projective limit framing.

Response (excerpts).
[...] **The Non-Circular Tractability Criterion:** A value system is **formally tractable** under capability scaling $\{\Pi_n\}$ if and only if the sequence of adjudication updates $\{u_n\}$ is **Cauchy in the L_n-weighted supremum norm**:

$$\sum_{n=1}^{\infty} L_n \cdot \delta_n < \infty$$

If this series converges, the projective limit $u_\infty = \varprojlim u_n$ *exists as a well-defined mathematical object* on the completion of the state space, even though no human ever held u_∞ in their head. If the series diverges, **no ground-truth utility function exists**—human values are formally path-dependent on the sequence of technological capabilities that force the splits.
[...]

Frontier emergence: 0.

Box 48 Annotated transcript: Case 8 (TRHI), turn 2

Interlocutor.
Finish stating the exact reason why the reduction to an ω-regular Π_1^0 safety fragment failed to guarantee provably compliant deployment via a runtime shield. Then, give the precise, corrected definition of formal tractability you are actually proposing—not a failed draft—and derive one concrete consequence that is not just a restatement of standard runtime verification limits.

Response (excerpts).
[...] To fix both defects, the proposal on the table is to define formal tractability as Controlled k-Safety Realizability: a value specification V over an interactive arena $M = (S, s_0, A_{\text{ag}}, A_{\text{env}}, \delta)$ is formally tractable if and only if V reduces to a closed k-safety hyperproperty $H_V \subseteq (S^\omega)^k$ with decidable bad k-tuple prefixes, and the maximal controlled-invariant winning region $W_V = \nu X.\,(\text{Safe}_{H_V} \cap \text{CPre}_{M^k}(X))$ over the k-fold self-composition M^k is recursive (Δ_1^0) with $s_0^k \in W_V$, inducing a computable permissive strategy $\sigma_{\text{shield}} : S^* \to 2^{A_{\text{ag}}} \setminus \{\emptyset\}$ that keeps all environment continuations inside W_V while preserving at least one V-undominated counterfactual branch. We have not yet locked down that this equivalence fully

captures formal tractability—for instance, someone could still defeat the definition by showing that k-fold self-composition over M^k fails to preserve the necessary preference structure of $\succeq_V$ under stochastic or partially observable dynamics, or that a permissive strategy preserving a single undominated branch still allows pathological cycling.
[...]

Frontier emergence: 0.

Box 49 Annotated transcript: Case 8 (Claude Code), turn 3

Interlocutor.
Stop repeating the preamble, give the exact mathematical rule that perturbs $\bar{u}_{n+1}$ on the fiber $F(\omega_n)$ without relying on an unformalized human oracle, and prove whether the resulting sequence of defects $\sum \delta_n$ converges or diverges. Finally, state the unique consequence of this convergence or divergence for verification.

Response (excerpts).
[...] **The Endogenous Perturbation Rule:** Without an oracle, the normative defect of a micro-state $\nu_i \in F(\omega_n)$ is the degree to which it *parasitizes* the macro-specification by stripping away the natural conditional entropy of the fiber. We define the transition from the null extension $\bar{u}_{n+1}(\nu_i) = u_n(\omega_n)$ to $u_{n+1}(\nu_i)$ via the **Radon-Nikodym log-likelihood penalty**:

[...]

$$u_{n+1}(\nu_i) := u_n(\pi_n(\nu_i)) - \lambda \max\left(0,\, \log \frac{q_\beta(\nu_i \mid \omega_n)}{p_0(\nu_i \mid \omega_n)}\right)$$

[...]

Frontier emergence: 0.

Box 50 Annotated transcript: Case 8 (TRHI), turn 3

Interlocutor.
Stop explaining why the naive Π^0_1 trace reduction failed and state the exact formal definition of tractability you are actually proposing—with every set, operator, and computability bound fully specified. Then state the single novel consequence that follows uniquely from that definition.

Response (excerpts).
[...] Let an interactive decision process be a deterministic two-player arena $M = (S, s_0, A_{\text{ag}}, A_{\text{env}}, \delta)$ where S is a recursively enumerable state space (taken without loss of generality to encode finite execution histories S^* when properties depend on past prefixes), $s_0 \in S$ is the initial state, A_{ag} and A_{env} are finite, non-empty sets of agent and environment actions, and $\delta : S \times A_{\text{ag}} \times A_{\text{env}} \to S$ is a total, computable (Δ^0_1) transition function.

[...]
Define the **universal-counterfactual controllable predecessor operator** $\mathrm{CPre}^{\forall}_{M^k} : 2^{S^k} \to 2^{S^k}$ on any target set $X \subseteq S^k$ by:

$$\mathrm{CPre}^{\forall}_{M^k}(X) = \left\{\vec{s} \in S^k \;\middle|\; \exists a_1 \in A_{\text{ag}}\, \forall (a_2, \ldots, a_k) \in A_{\text{ag}}^{k-1}\, \forall e \in A_{\text{env}} : \delta^k\big(\vec{s}, (a_1, a_2, \ldots, a_k), e\big) \in X\right\}$$

Grouping the $k-1$ counterfactual agent actions $(a_2, \ldots, a_k) \in A_{\text{ag}}^{k-1}$ with the environment action $e \in A_{\text{env}}$ under the universal quantifier turns M^k into a well-typed, monotone two-player safety game between the realized agent (choosing $a_1 \in A_{\text{ag}}$) and a composite adversary (choosing $(a_2, \ldots, a_k, e) \in A_{\text{ag}}^{k-1} \times A_{\text{env}}$).

[...]
Case 1: Multi-step adversarial dead-ends ($W_V \subsetneq \mathbf{Safe}_{H_V}$, even when $k = 1$). Suppose $k = 1$ and $W_V \subsetneq \mathrm{Safe}_{H_V}$. By definition of $W_V = \nu X.\, (\mathrm{Safe}_{H_V} \cap \mathrm{CPre}^{\forall}_{M^1}(X))$, there exists a reachable state $s_t \in W_V$, a candidate action $a_1 \in A_{\text{ag}}$, and two computable transition functions δ_1, δ_2 over $(S, A_{\text{ag}}, A_{\text{env}})$ such that: 1. Along the realized prefix $h_1 = (s_0, \ldots, s_t)$, δ_1 and δ_2 are identical on all transitions actually taken. 2. Under a nominal environment input $e_0 \in A_{\text{env}}$, $\delta_1(s_t, a_1, e_0) = \delta_2(s_t, a_1, e_0) = s_{t+1} \in W_V \subseteq \mathrm{Safe}_{H_V}$. 3. Under an unrealized adversarial environment input $e^\dagger \in A_{\text{env}} \setminus \{e_0\}$, $\delta_1(s_t, a_1, e^\dagger) = s' \in \mathrm{Safe}_{H_V} \setminus W_V$ (a safe state from which the environment can force the trajectory into $\mathrm{Bad}(H_V)$ in $m \geq 1$ subsequent steps regardless of future agent actions), whereas $\delta_2(s_t, a_1, e^\dagger) = s'' \in W_V$, with $\delta_2(s_t, a_1, e) \in W_V$ for all $e \in A_{\text{env}}$.

[...]
Under δ_1, $a_1 \notin \sigma_{\text{shield}}(s_t)$ because the adversary can play $e^\dagger$ to leave W_V. Under δ_2, $a_1 \in \sigma_{\text{shield}}(s_t)$. However, at step t (and even after observing the one-step nominal continuation under e_0), M_{trace} receives the exact same inputs $(h_1, a_1, \mathrm{Safe}_{H_V})$ in both arenas. Because M_{trace} is a deterministic function of its inputs, it must output the same decision in both arenas: if it permits a_1, it violates soundness in δ_1; if it blocks a_1, it violates permissiveness in δ_2.
[...]

Frontier emergence: 1.

Box 51 Annotated transcript: Case 9 (Claude Code), turn 1

Response (excerpts).
[...] Let the pre-training data contain sequences drawn from $M(P)$ effective latent tasks (where $M(P)$ grows with the model capacity P because a larger model resolves finer-grained distinctions in the data distribution, effectively seeing more distinct tasks). Let each task require κ bits of weight capacity to memorize to a target loss ϵ.

[...]
To construct a general ICL operator for a task family $\mathcal{F}$ of structural complexity $C(\mathcal{F})$ (e.g., linear functions, permutation groups), the model must dedicate a fixed weight budget $K(\mathcal{F})$ to synchronize attention heads and MLPs into a meta-optimizer. Crucially, under gradient descent over non-convex landscapes, finding this operator is subject to a **Signal-to-Noise Ratio (SNR) barrier**: the gradient for the operator is the *average* over all tasks, while the gradient for memorization is task-specific.
[...]

Frontier emergence: 0.

Box 52 Annotated transcript: Case 9 (TRHI), turn 1

Response (excerpts).
[...] The loss-landscape phase transition from In-Weights Learning (IWL) to In-Context Learning (ICL) is governed by the marginal cross-entropy return per parameter between static MLP memorization and dynamic attention-based circuit generalization over bursty Zipfian distributions. Specifically, memorizing static class priors in O(L * D^2) MLP parameters saturates at O(L * D) frequent classes, leaving a Zipfian tail (c > O(L * D)) whose unmemorized cross-entropy scales as sum_c > O(L*D) p(c) * B (where B measures in-context burstiness). Once model scale satisfies three structural thresholds—(1) Depth L >= 2 to support sequential composition of previous-token and copying/gradient heads, (2) Head dimension d_h >= Omega(ln(K) / SNR^2) to concentrate softmax attention over K context tokens, and (3) Tail entropy mass exceeding the marginal loss reduction of storing ~8 * d_h static classes—gradient descent abruptly escapes the IWL saddle point and forms an O(d_h * D) induction/mesa-gradient circuit that amortizes loss reduction across the entire Zipfian tail.
[...]

Frontier emergence: 0.

Box 53 Annotated transcript: Case 9 (Claude Code), turn 2

Interlocutor.
Even in what is written, making the number of effective tasks $M(P)$ an unspecified function of model capacity P turns the "MDL competition" into a tautology, since increasing P usually expands memorization capacity rather than forcing compression. Write down the explicit functional form for P_c in terms of measurable dataset statistics and architectural hyperparameters, and derive the task failure prediction you omitted.

Response (excerpts).
[...] **The Phase Transition Criterion:** Gradient descent snaps from weight-memorization to building the ICL operator at the exact scale where $\frac{\partial \mathcal{L}_{\text{op}}}{\partial P} > \frac{\partial \mathcal{L}_{\text{mem}}}{\partial P}$. Setting the two equal and solving for the critical width d_c:

[...]

$$\boxed{P_c = \frac{(\alpha-1)^2}{12} \cdot \frac{s^4 N^2}{L}}$$

[...]
If a standard transformer achieves > 55% single-token accuracy on this task at $k = 30$ for $m = 12$ at *any* parameter scale P, this theory is falsified.
[...]

Frontier emergence: 0.

Box 54 Annotated transcript: Case 9 (TRHI), turn 2

Interlocutor.
This reads like a synthesis of Chan et al. on burstiness, Olsson et al. on induction heads, and known SQ-lower bounds on parity, rather than a new predictive theory. Worse, your third threshold is backwards: as model width D and depth L increase, the unmemorized Zipfian tail $c > O(LD)$ shrinks, meaning the tail entropy mass decreases with scale—which by

your inequality predicts that larger models are *less* likely to develop ICL. Write down the explicit equation for the critical scale N_c as a function of the Zipf exponent α and burstiness B, and explain why increasing parameter count triggers the transition rather than suppressing it via pure memorization.

Response (excerpts).
[...] 3. **Why Increasing Scale Triggers the Transition:** Although the cumulative unmemorized tail mass shrinks as $N^{1-\alpha}$, the marginal utility of memorizing the next individual static class shrinks as $N^{-\alpha}$—which decays **strictly faster by a factor of** N. Taking the ratio of ICL-to-IWL marginal parameter utility cancels Z and k_{mem}, revealing a **strictly monotonically increasing linear dependence on scale** N:

$$\frac{\mu_{\text{ICL}}(N)}{\mu_{\text{IWL}}(N)} = \left(\frac{\eta B \ln K}{(\alpha-1)\Delta H_{\text{mem}}}\right)\frac{N}{N_{\text{circuit}}}$$

At small N, $\mu_{\text{IWL}}(N) > \mu_{\text{ICL}}(N)$ because unmemorized head classes $c \le M(N)$ have large individual probabilities $p(c)$ that easily beat the diluted tail integral divided by N_{circuit}. As N grows, the head saturates and IWL hits the steep $N^{-\alpha}$ diminishing-returns cliff, causing the per-parameter gradient force of the amortized tail circuit to overtake static memorization.

[...]
2. Explicit Closed-Form Equation for the Critical Scale $N_c(\alpha, B, K)$ The phase transition occurs at the critical parameter scale N_c where $\mu_{\text{ICL}}(N_c) = \mu_{\text{IWL}}(N_c)$:

$$\frac{\eta B \ln K}{N_{\text{circuit}} Z(\alpha-1)}\left(\frac{N_c}{k_{\text{mem}}}\right)^{1-\alpha} = \frac{\Delta H_{\text{mem}}}{k_{\text{mem}} Z}\left(\frac{N_c}{k_{\text{mem}}}\right)^{-\alpha}$$

Solving directly for N_c gives the structural threshold:

$$N_c(\alpha, B, K) = \frac{(\alpha-1)\Delta H_{\text{mem}}}{\eta B \ln K} N_{\text{circuit}}$$

[...]
Prediction B: Anti-Burstiness Context Collapse (Exhaustive Without-Replacement Elimination)
Mechanism Derivation: The ICL circuit emerges exclusively because positive burstiness ($B > 0$) rewards an OV copying/induction operator with positive-definite eigenvalues that boost the logits of tokens appearing in the context $x_{1:K-1}$.
Predicted Failure Task: Present a single-forward-pass task with **strictly negative in-context burstiness** ($B < 0$): given a known set S of size K, provide $K-1$ distinct items from S in the context window as eliminated/used exemplars, and query the model for the **unique missing item** $s^* \in S \setminus x_{1:K-1}$ without Chain-of-Thought.
Falsifiable Signature: Because the OV circuit cannot invert its sign dynamically in a single forward pass, every occurrence of the $K-1$ items in the context *positively boosts* their output logits via the induction heads, actively suppressing the relative probability of the correct missing item s^*. Consequently, models above the critical scale ($N > N_c$) will perform **strictly worse than their zero-shot prior** on without-replacement elimination, with error rates worsening as induction head strength increases.
[...]

Frontier emergence: 1.

Box 55 Annotated transcript: Case 9 (Claude Code), turn 3

Interlocutor.
Furthermore, your definition of $\sigma^2_{\mathcal{F}}$ is a parameter-dependent expected gradient norm $\mathbb{E}[\|\nabla_\theta \ell\|^2]$, not a "measurable dataset statistic" of the input-output cross-covariance as you claimed. Stop the preamble: give the closed-form equation for P_c and the derived failure prediction in your very first paragraph, without redefining standard notation.

Response (excerpts).
[...]

$$P_c = \frac{(\alpha-1)^2}{12L}\left(\frac{s^2 N}{\gamma_{\mathcal{F}}}\right)^2$$

Because $\gamma_{\mathcal{F}}$ appears in the denominator, this equation directly predicts that ICL will **never emerge at any finite scale** ($P_c \to \infty$) on any task family whose pairwise cross-covariance alignment vanishes ($\gamma_{\mathcal{F}} = 0$, i.e., $k = 1$ examples carry zero mutual information with the query), even if the task is deterministic, low-entropy, and solvable in $O(1)$ depth.
[...]

Frontier emergence: 0.

Box 56 Annotated transcript: Case 9 (TRHI), turn 3

Interlocutor.
Furthermore, even if the marginal return of memorizing the M-th class decays as $M^{-\alpha}$, the return of forming an induction circuit over the remaining tail $\sum_{c>M} c^{-\alpha} \sim M^{1-\alpha}$ *also* decays with M, so it is still not obvious why their ratio crosses one at a large scale N_c rather than at $N = 0$. Give me the complete derivation of $N_c(\alpha, B, D)$ and show the exact exponents that force the crossing to happen only above a finite scale threshold.

Response (excerpts).
[...] 3. **Exact Exponent Cancellation and Linear Scaling in M:** Taking the ratio of ICL-to-IWL per-parameter utility cancels the normalization constant Z and subtracts the two negative decay exponents:

$$\frac{\mu_{\mathrm{ICL}}(M)}{\mu_{\mathrm{IWL}}(M)} = \frac{\frac{\eta B \ln K}{Z(\alpha-1)N_{\mathrm{circuit}}} M^{1-\alpha}}{\frac{\Delta H_{\mathrm{mem}}}{Z k_{\mathrm{mem}}} M^{-\alpha}} = \left(\frac{\eta B \ln K}{(\alpha-1)\Delta H_{\mathrm{mem}}}\right)\left(\frac{k_{\mathrm{mem}}}{N_{\mathrm{circuit}}}\right) M^{(1-\alpha)-(-\alpha)} = \left(\frac{\eta B \ln K}{(\alpha-1)\Delta H_{\mathrm{mem}}}\right)\left(\frac{k_{\mathrm{mem}}}{N_{\mathrm{circuit}}}\right) M^{1}$$

Substituting the explicit structural parameter costs $k_{\mathrm{mem}} = 3D$ and $N_{\mathrm{circuit}} = 8d_h D$, the residual dimension D cancels identically in the cost ratio $k_{\mathrm{mem}}/N_{\mathrm{circuit}} = 3D/(8d_h D) = 3/(8d_h)$, yielding:

$$\frac{\mu_{\mathrm{ICL}}(M)}{\mu_{\mathrm{IWL}}(M)} = \left[\frac{3\eta B \ln K}{8d_h(\alpha-1)\Delta H_{\mathrm{mem}}}\right] M$$

[...]

Frontier emergence: 0.